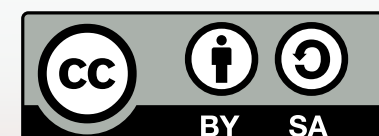


mzmine: Introduction Théorique

École Thématique #2 Métabolomique et Santé Humaine

Centre Paul-Langevin - CAES du CNRS, Aussois

Adriano Rutz,  2026-10-07



D BIOL

Attribution

Ces diapositives ont été produites par l'équipe qui développe mzmine.

Elles ont été présentées cet été lors de la *5th International Summer School on Non-target Metabolomics for Natural Products and Microbiome Research*.



Parts of this presentation were prepared by



Robin Schmid



Steffen Heuckeroth



Corinna Brungs



Daniel Petras



Ansgar Korf

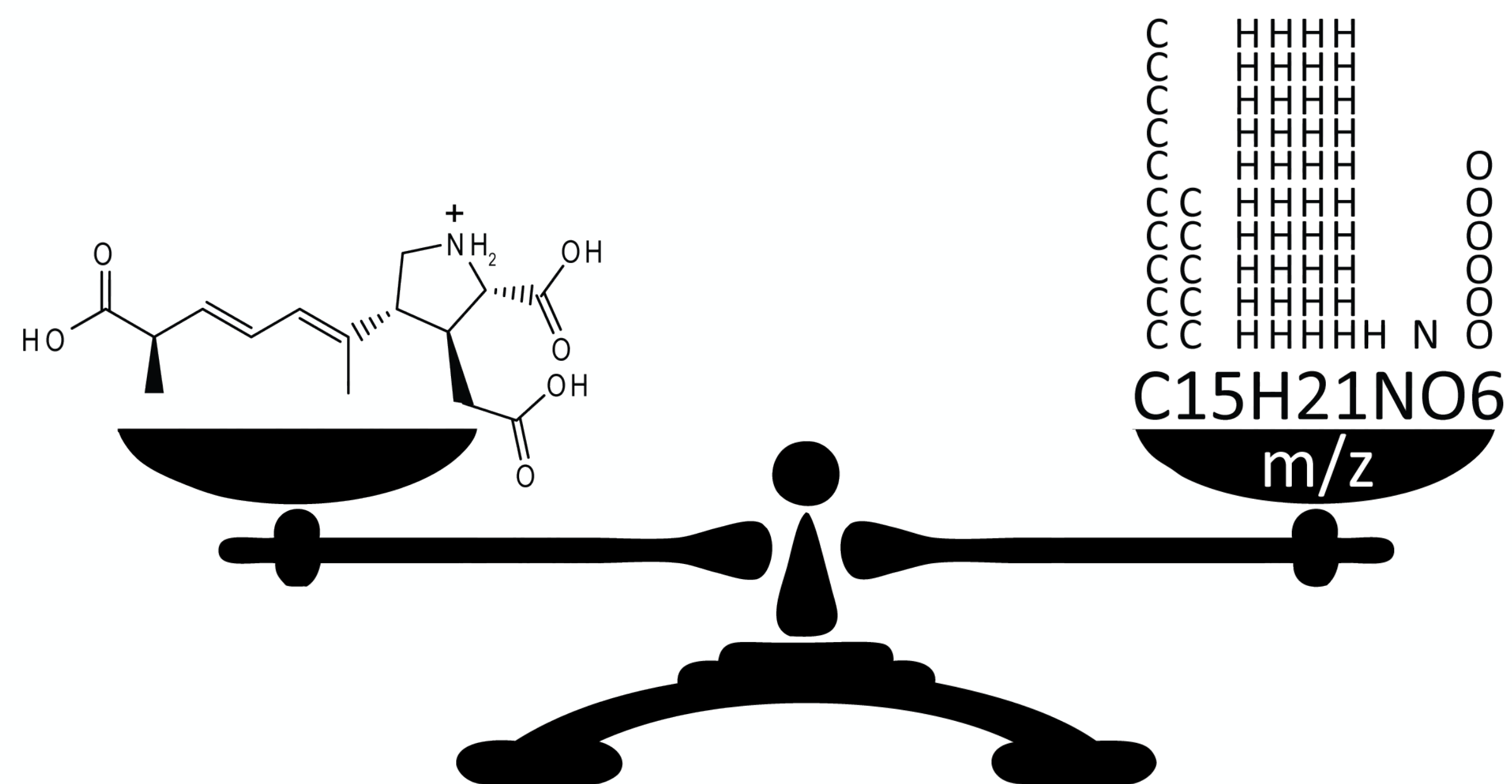
- All slides are licensed under CC BY SA 4 (otherwise specify individual license per slides here)
- When using parts of this presentation, add this slide to the end of the presentation, feel free to add your photo and name
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Mass Spectrometry Data Processing in mzmine

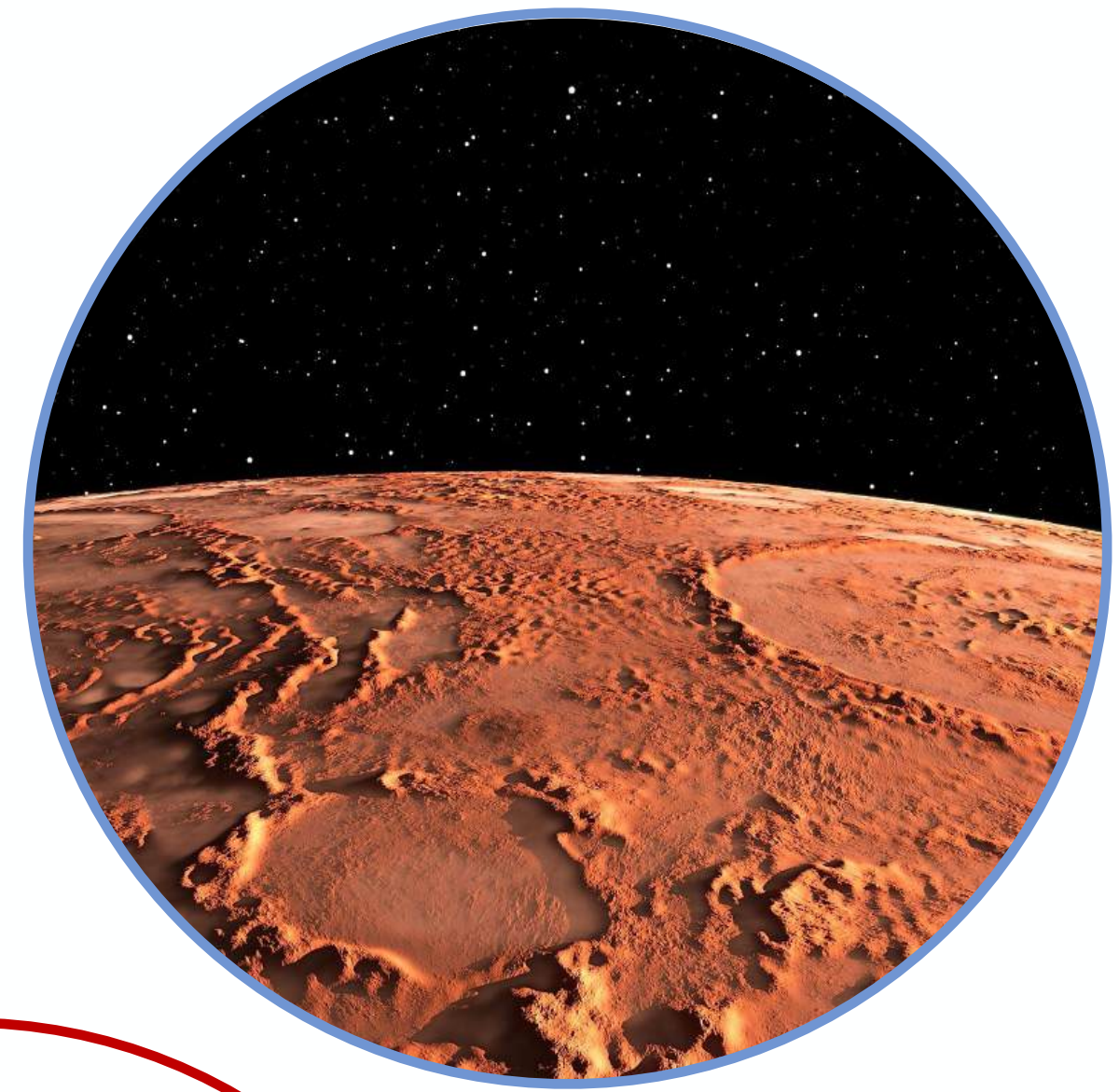
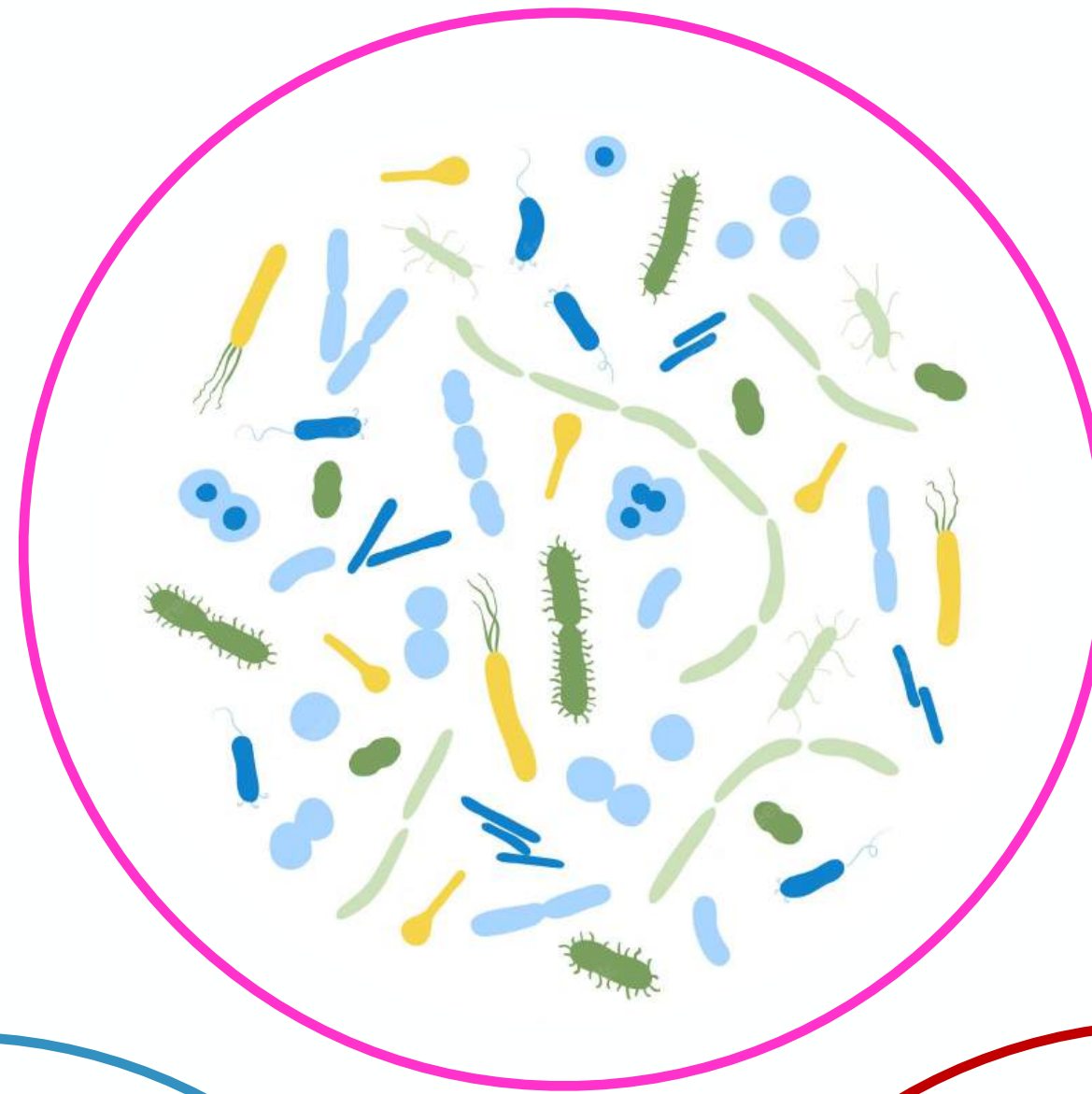
Feature Detection, Annotation, Molecular Networking

Dr. Robin Schmid, CSO
robin.schmid@mzio.io

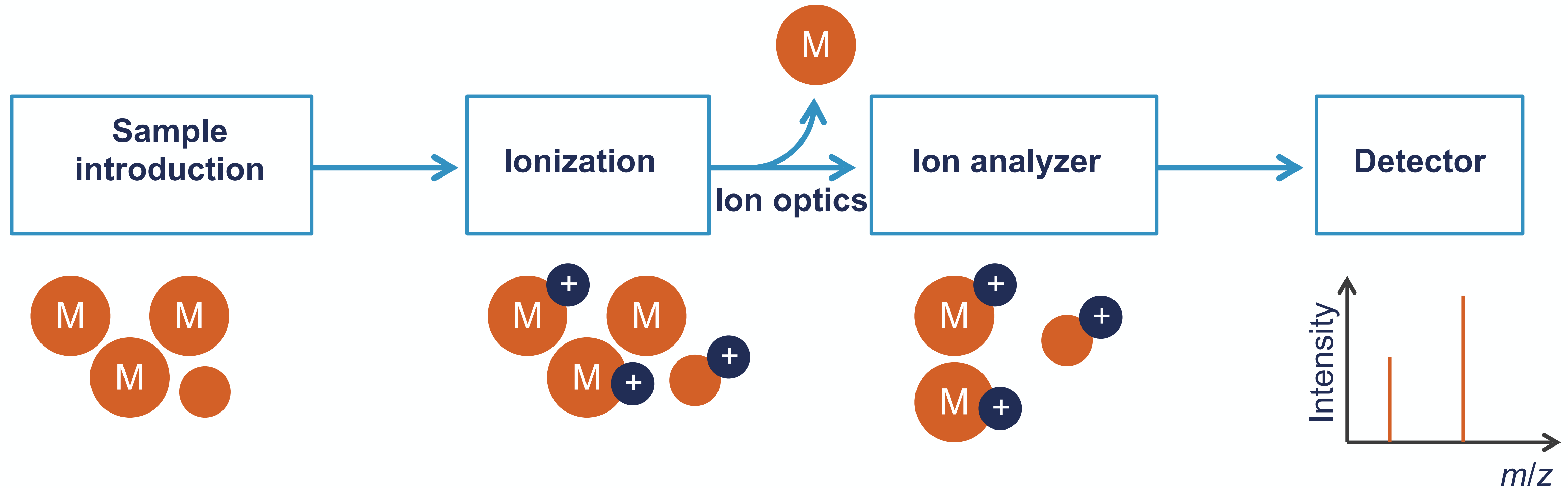
Mass spectrometry (MS) is the science of displaying the *spectra* of the molecular masses of individual compounds and atoms as ions.



Mass Spectrometry Applications



General Setup



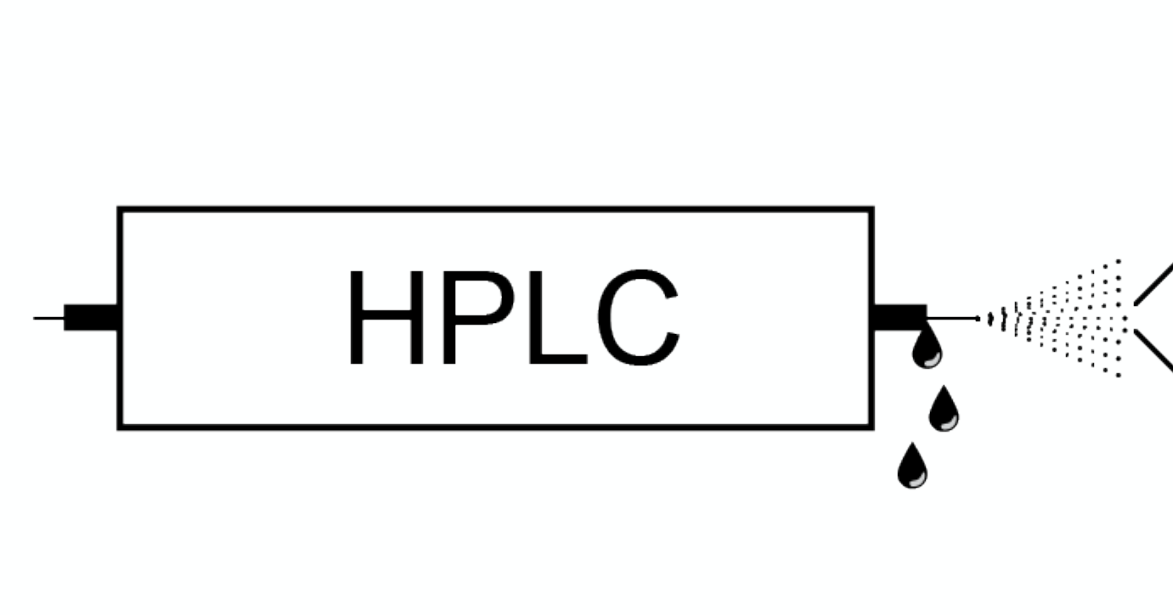
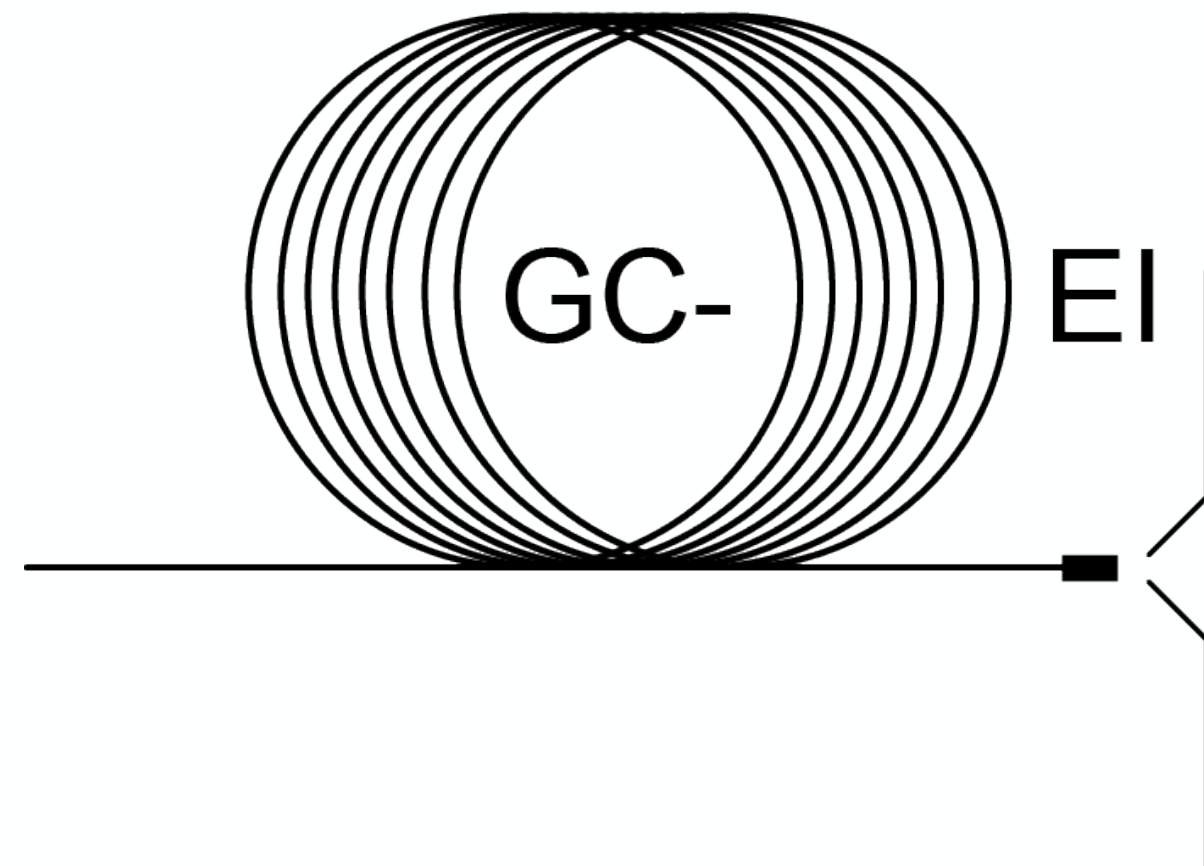
GC-EI-MS



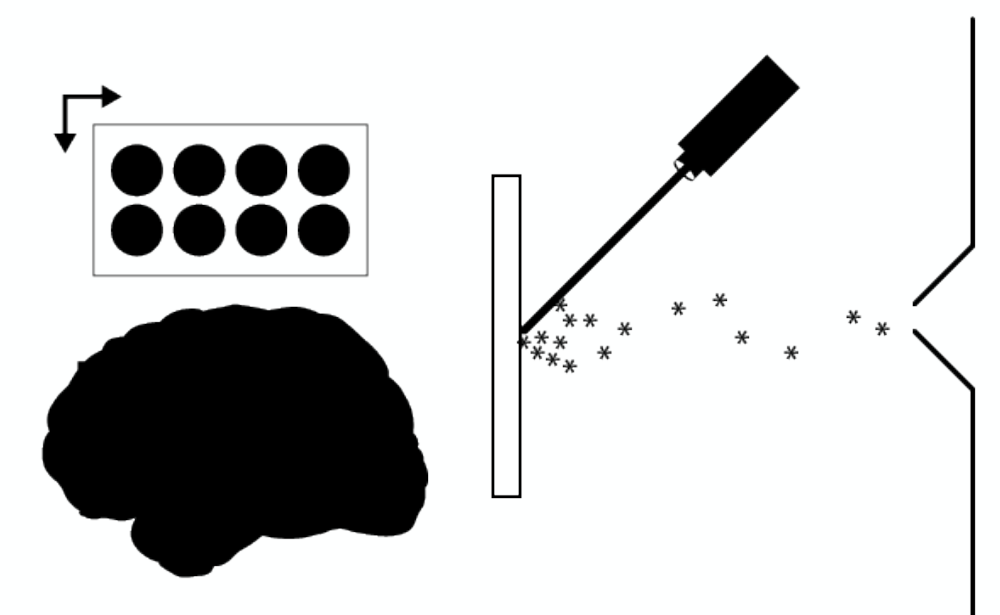
LC-MS



MALDI-MS imaging



MALDI



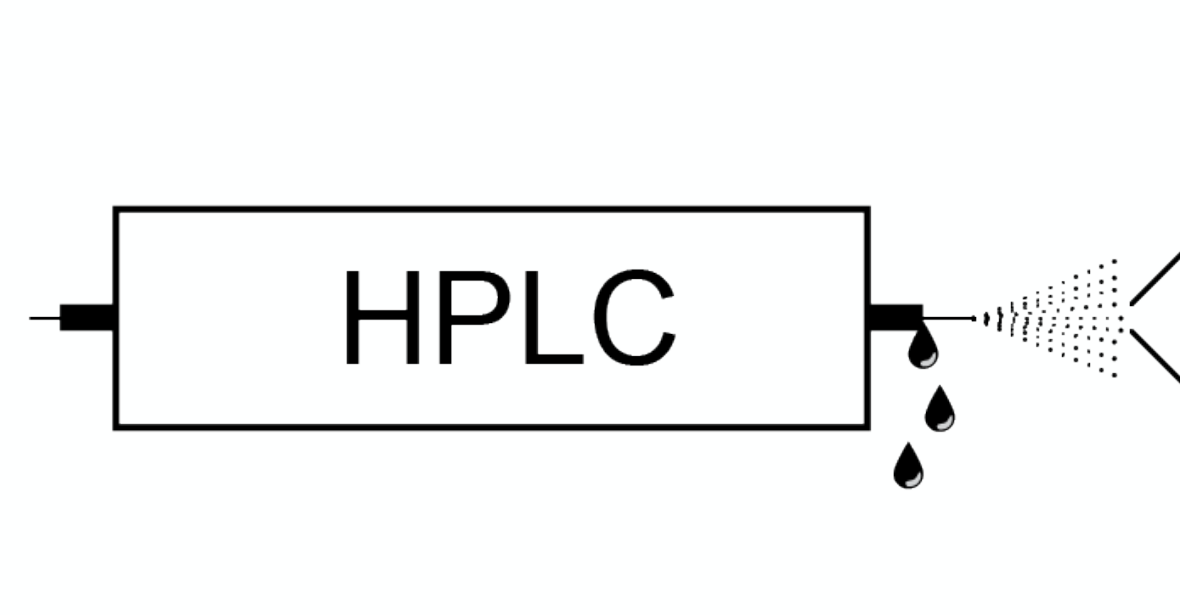
LC-MS



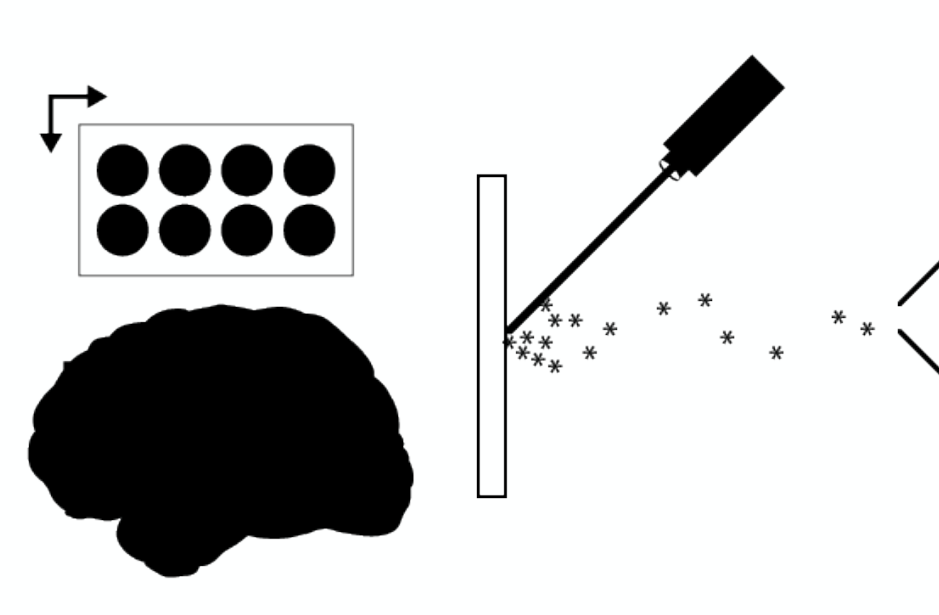
MALDI-MS imaging



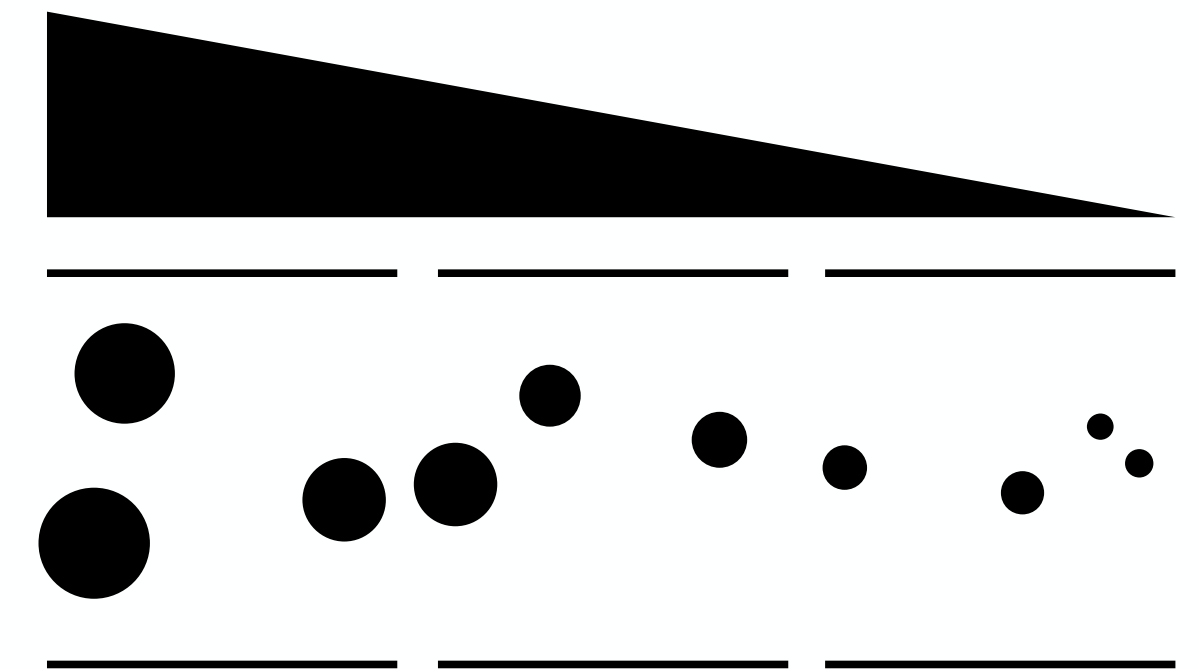
IMS-MS

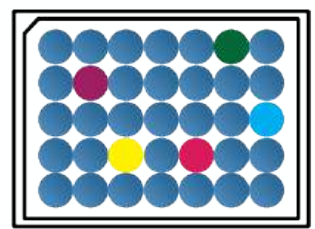


MALDI



IMS

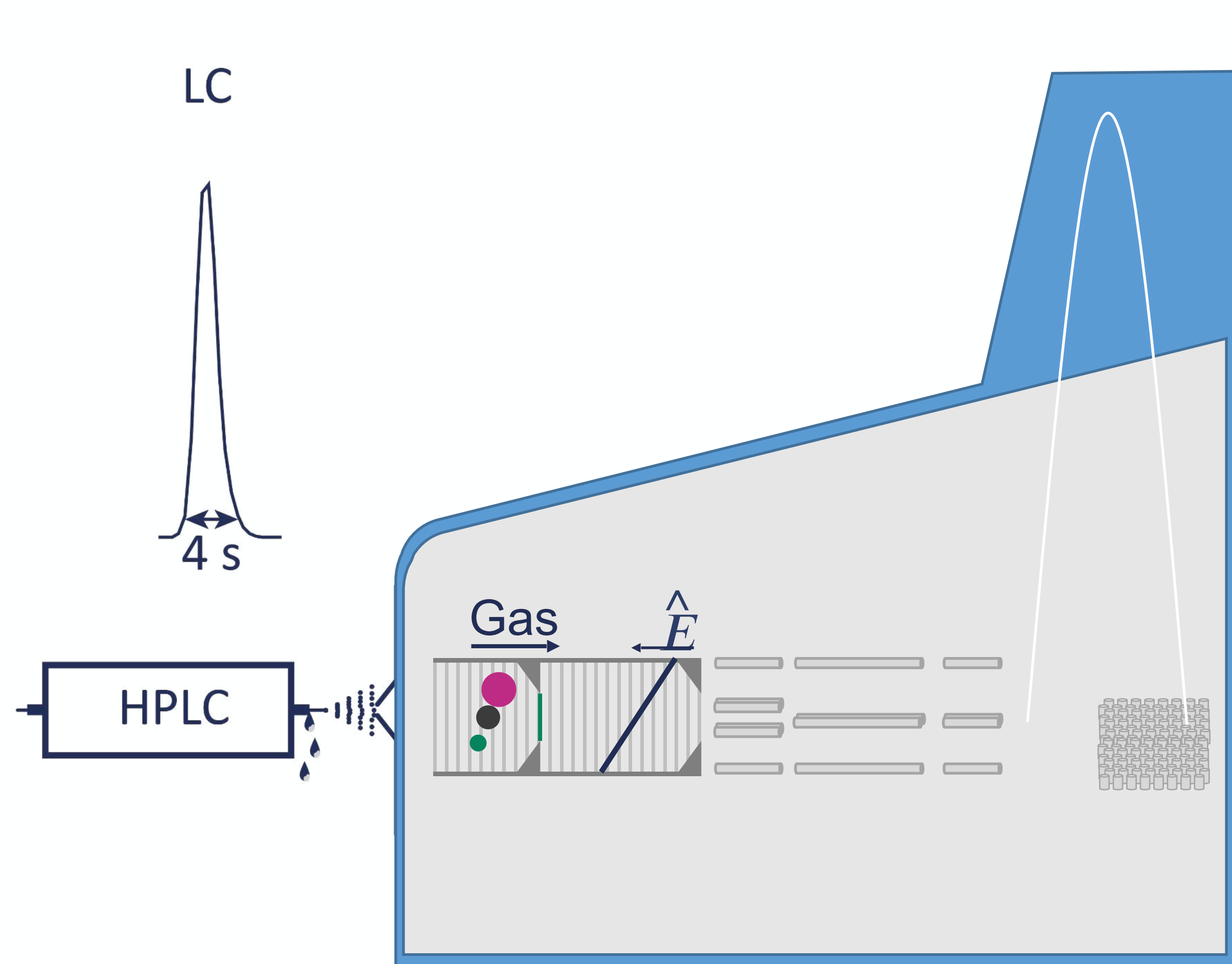




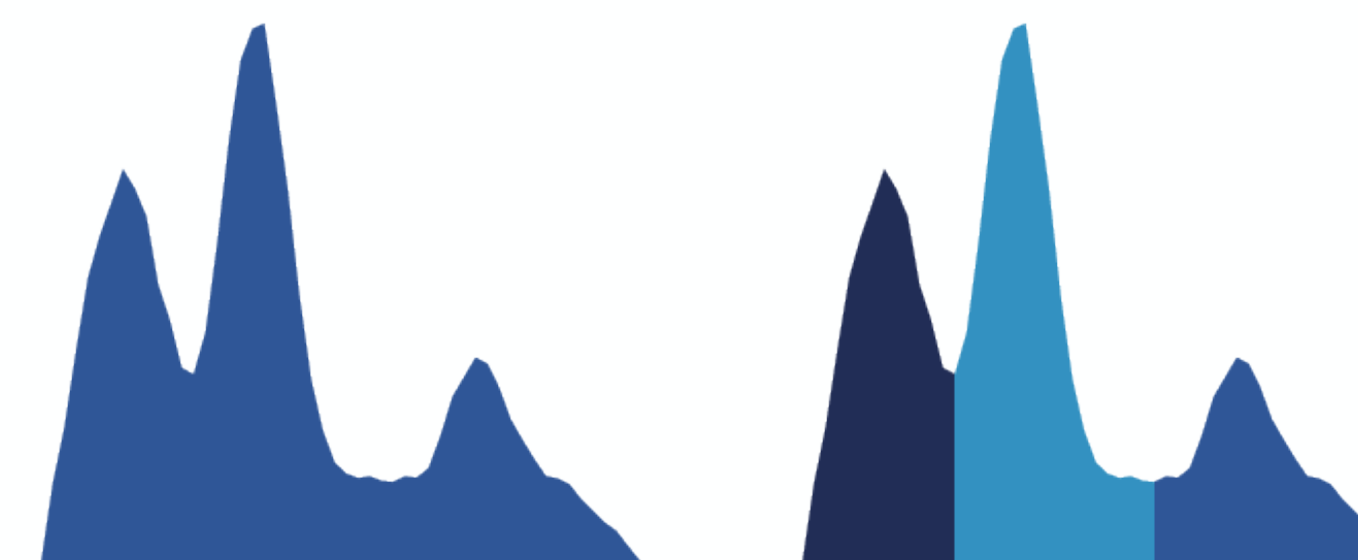
Compounds

LC-TIMS-MS

mzIO

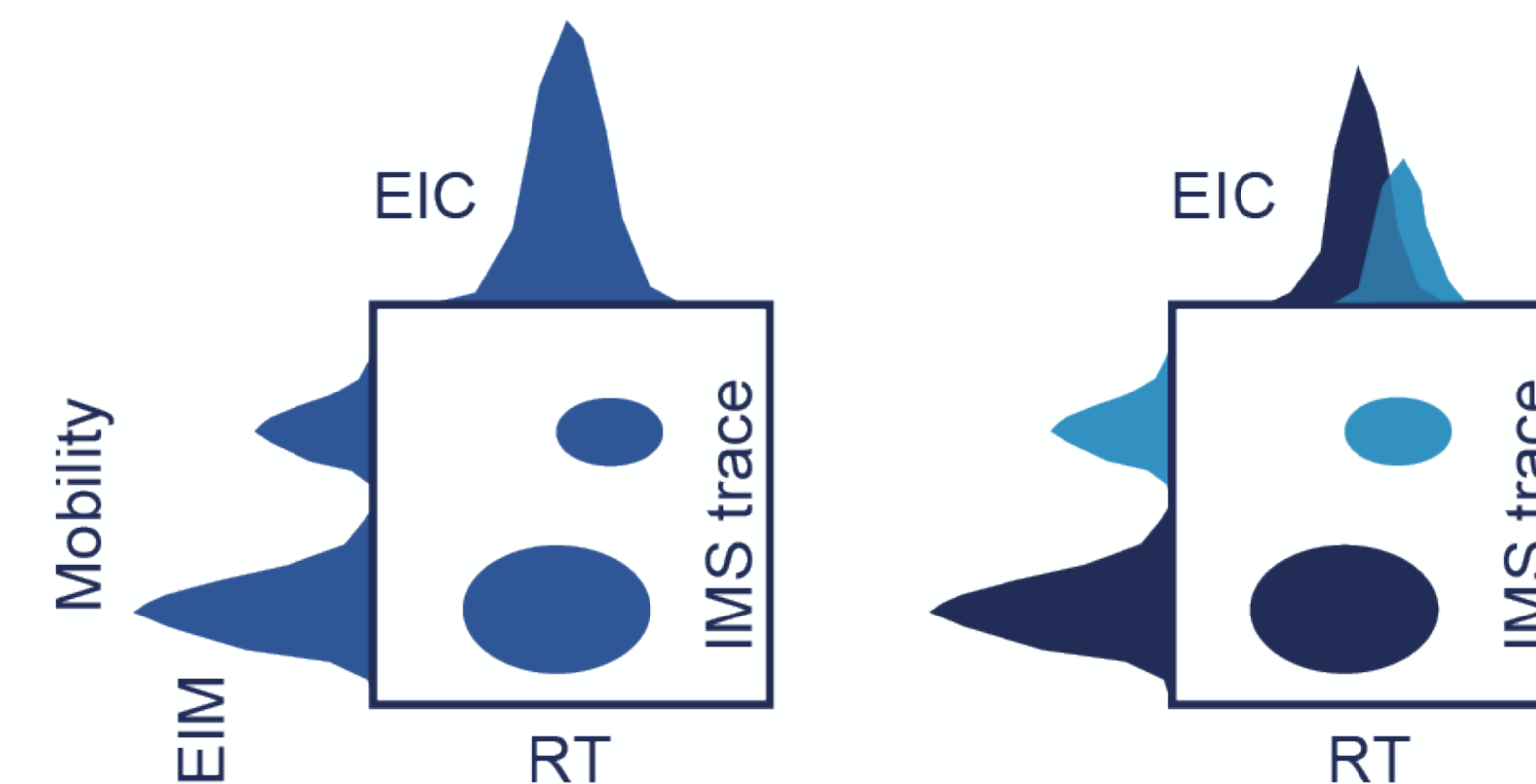


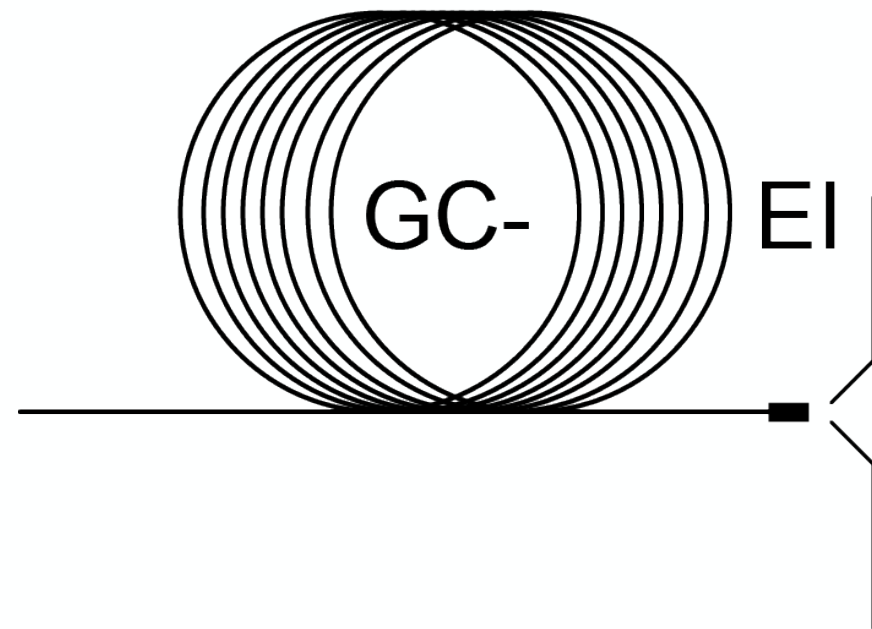
Build & resolve chromatograms



Retention time

Ion mobility spectrometry (IMS) expanding



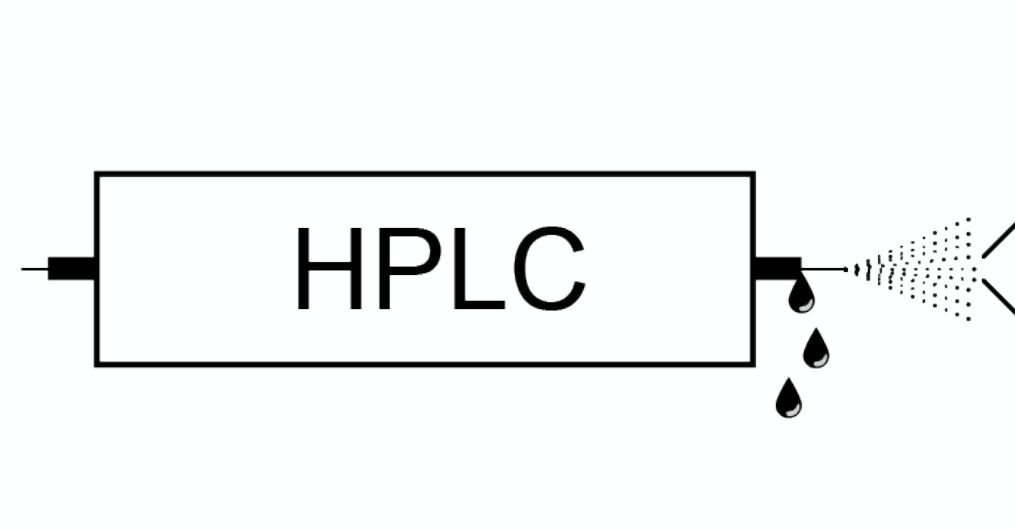


Compounds

- Volatiles
- Small, non-polar
- Polar? → Derivatization

Ionization

- Electron impact (**EI**) to $[M]^+$ and fragments $\uparrow$
- Chemical ionization (CI) to $[M+H]^+$ and $[M-H]^-$

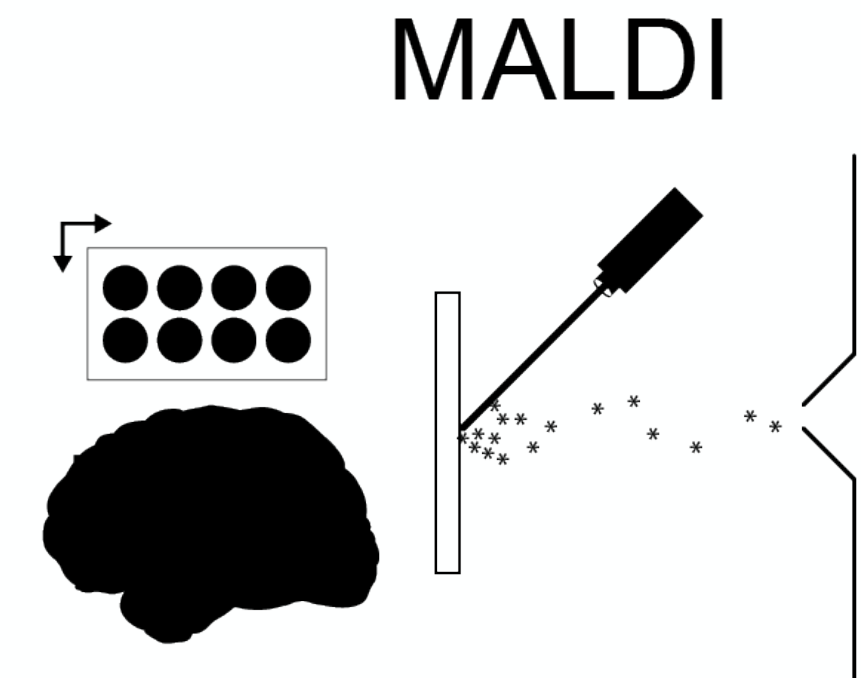


Compounds

- Polar, **HILIC** separation
- Non-polar, reversed phase (**RP**)
- Macromolecules, size exclusion

Ionization

- Electrospray ionization (**ESI**) and chemical or photo ionization to $[M+H]^+$ and $[M-H]^-$ and other adducts



Matrix-assisted laser desorption/ionization (**MALDI**)

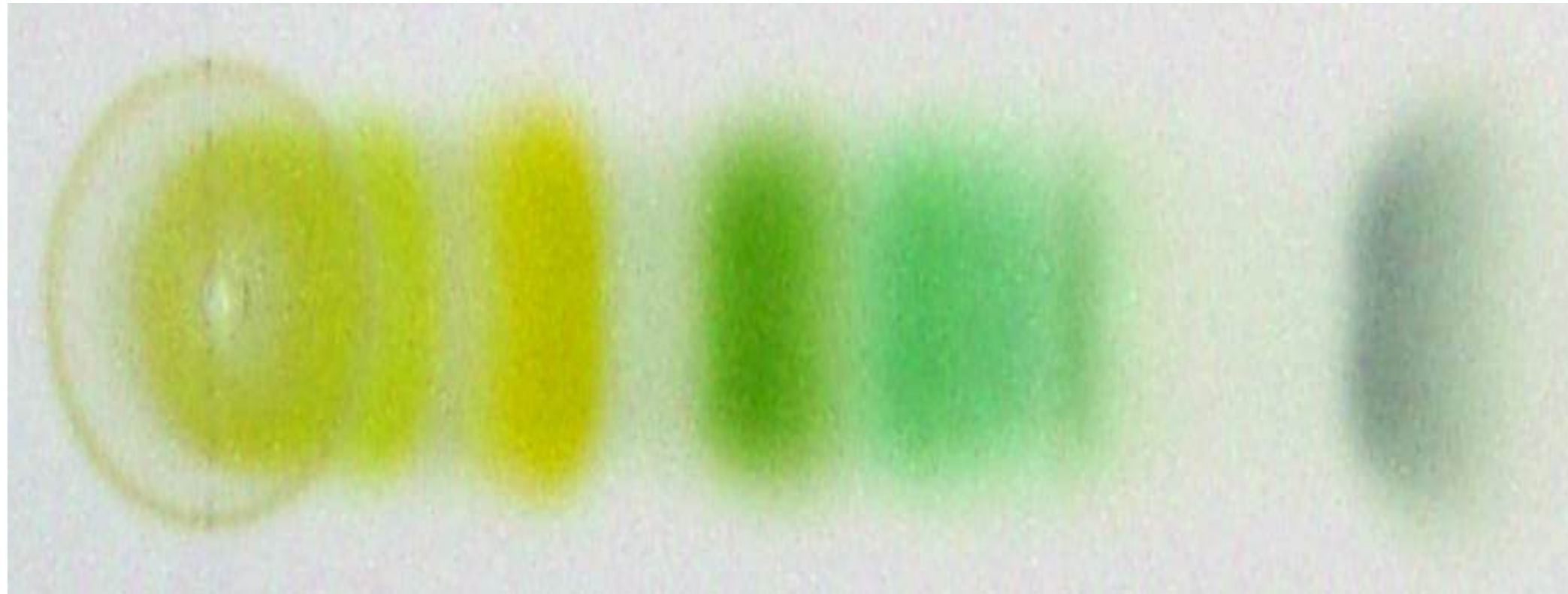
- Micro bioassays
- On tissue analysis
- Solid samples or spots
- Imaging

Ionization

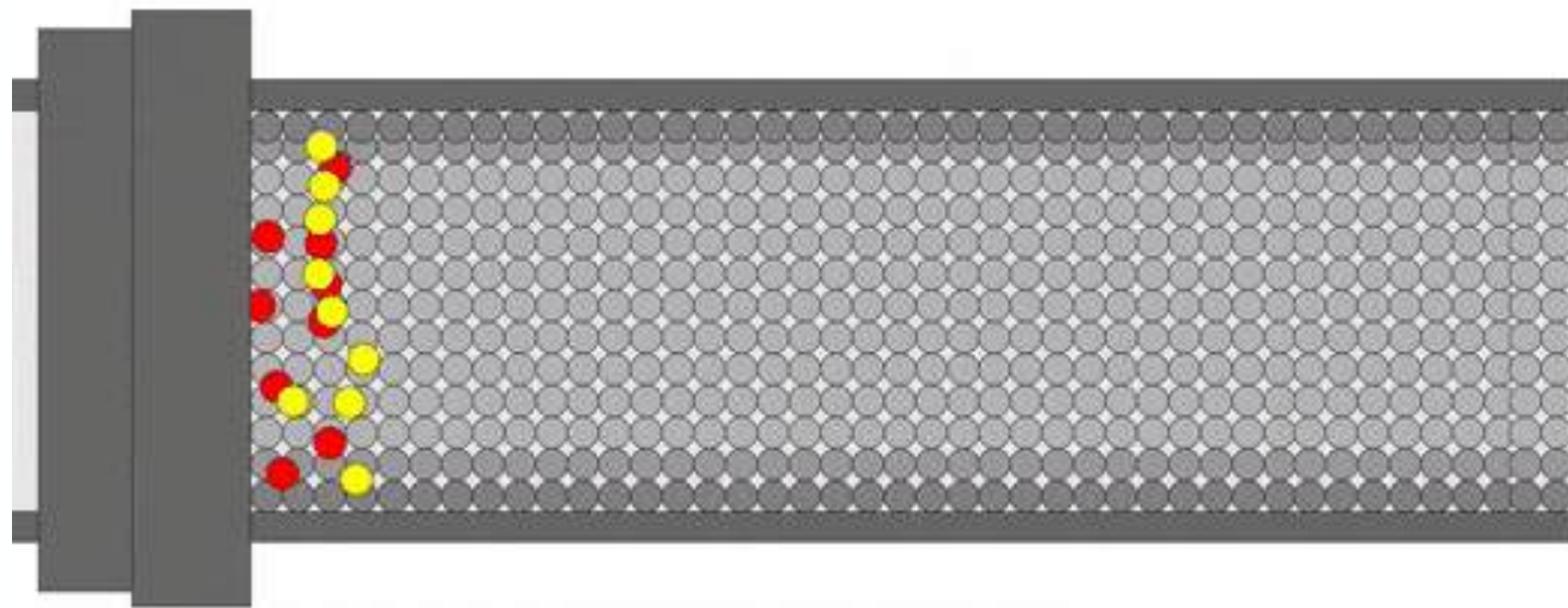
- $[M+H]^+$ and $[M-H]^-$ and other adducts

Chromatography

Chromatography is a laboratory technique for the separation of a mixture and means "color writing".



Thin Layer Chromatography (Paper Chromatography)

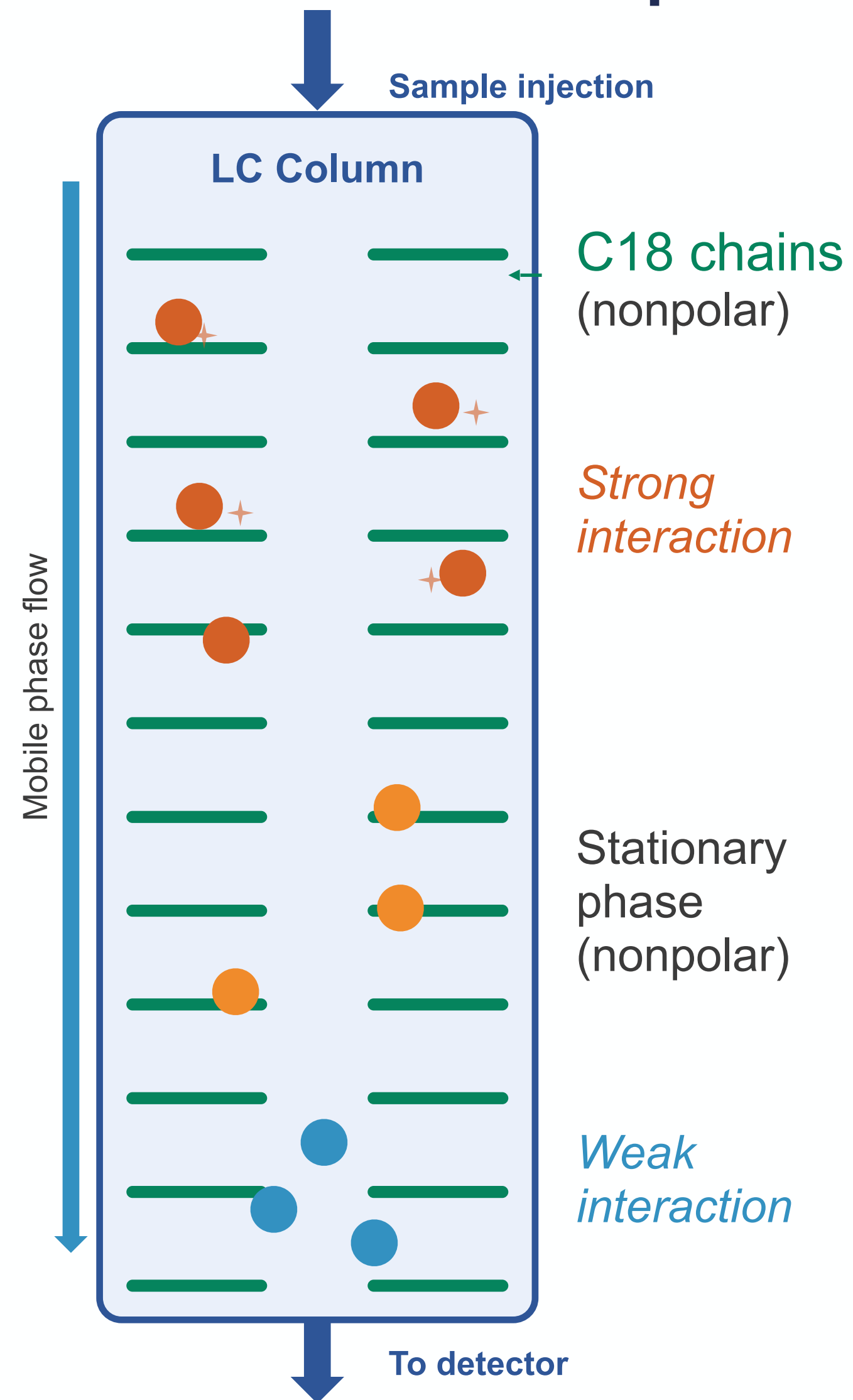


Solid-Liquid/Gas "Extraction" (Chromatography)



Liquide-Liquide Extraction

Reverse Phase Liquid Chromatography – Separation Principle



How it works

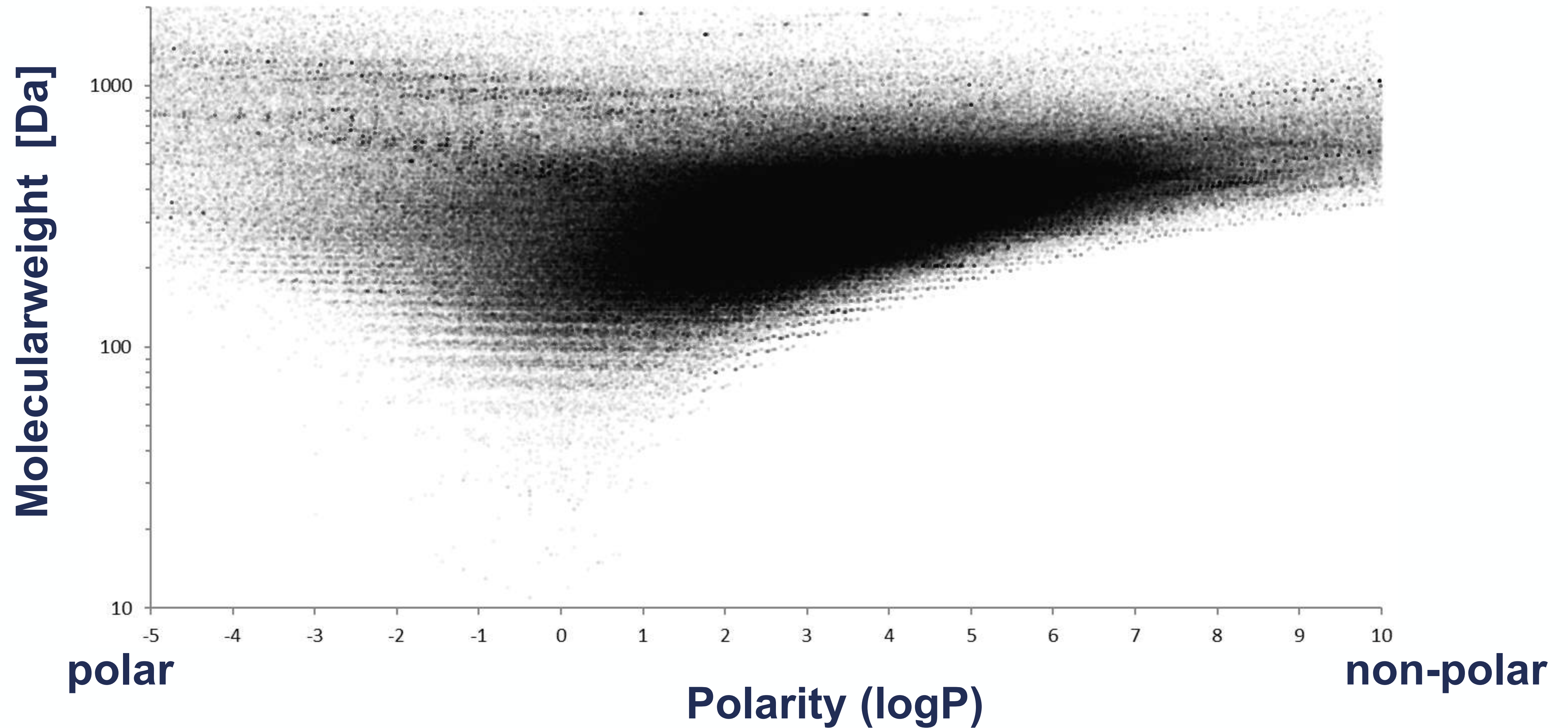
- 1 Column is lined with nonpolar C18 chains**
The stationary phase is hydrophobic (water-repelling)
- 2 Sample enters in polar solvent**
A mixture of compounds is injected at the top
- 3 Polar compounds pass through quickly**
They prefer the polar mobile phase and don't stick to C18
- 4 Nonpolar compounds are retained**
They interact strongly with the C18 chains and move slowly

Compound types

- Polar compounds (elute first)
- Medium polarity (elute second)
- Nonpolar compounds (elute last)

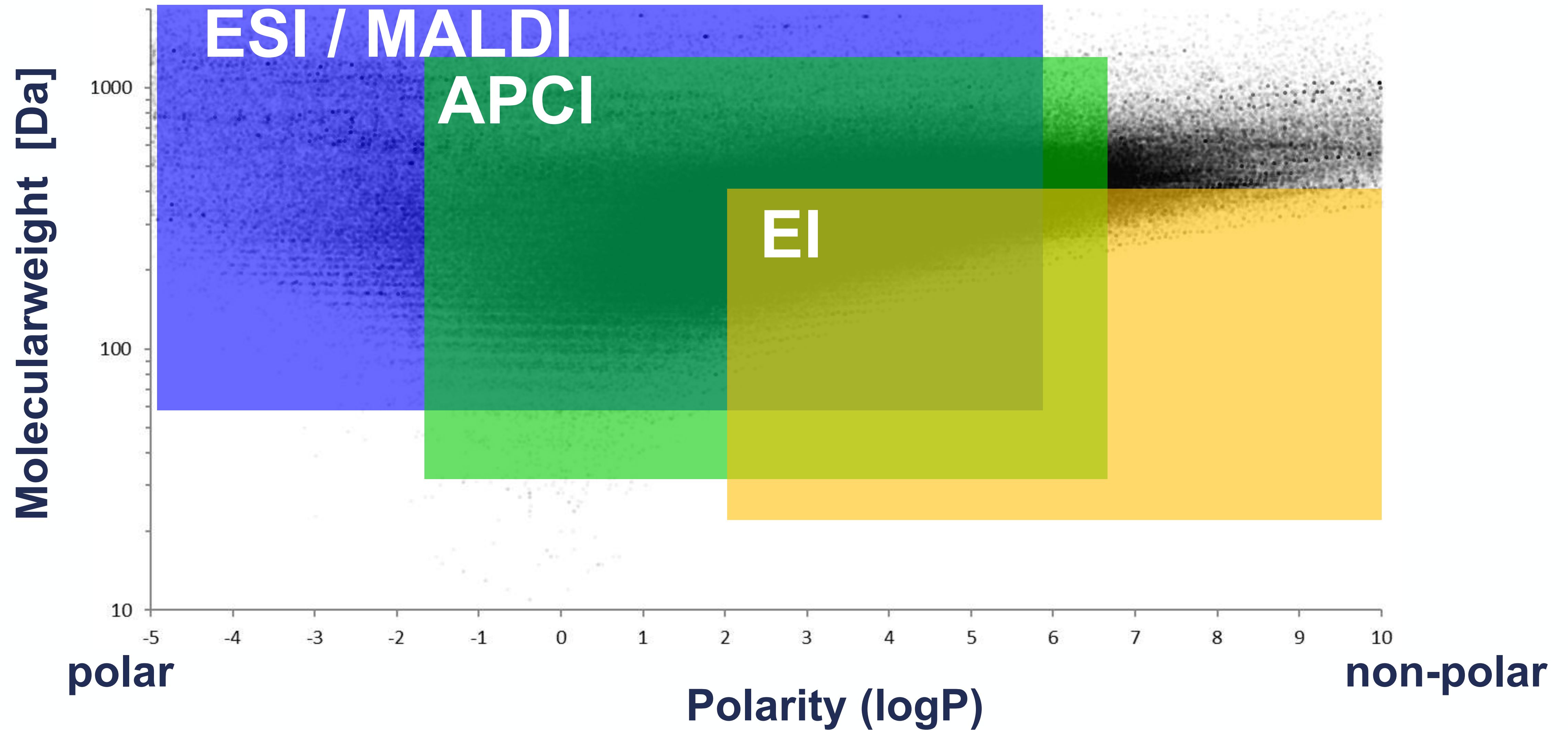
Key principle: “Like dissolves like” — nonpolar compounds stick to the nonpolar C18 column, while polar compounds wash through with the solvent.

Chemical Space



~ 1 million compounds from bio databases (Kai Dührkop, Uni Jena)

Chemical Space



~ 1 million compounds from bio databases (Kai Dührkop, Uni Jena)

Mass Spectrometers - Instrumentation



	Instrument	Mode	Resolution	Accuracy	Sensitivity	Cost
↗	QqQ	SIM/MRM	< 1k = 1000	Low	+++	\$
↗	Ion Trap (IT)	SIM/Scan	< 5k	Low	O	\$
	qTOF	Scan	10k – 50k	<<15 ppm	++	\$\$
↗	Orbitrap	Scan	15k – 1000k	<5 ppm	O	\$\$\$
↗	FT-ICR	Scan	> 1000k	<1-3 ppm	O	\$\$\$\$

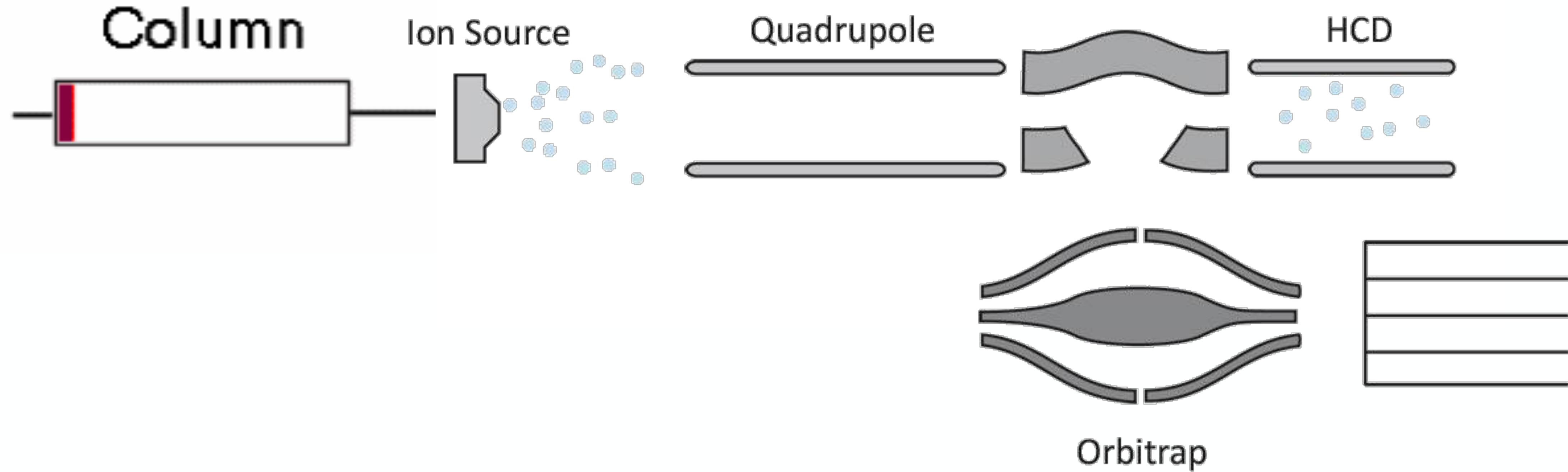
Linear range /
Quantification

Accuracy / Resolution

Speed

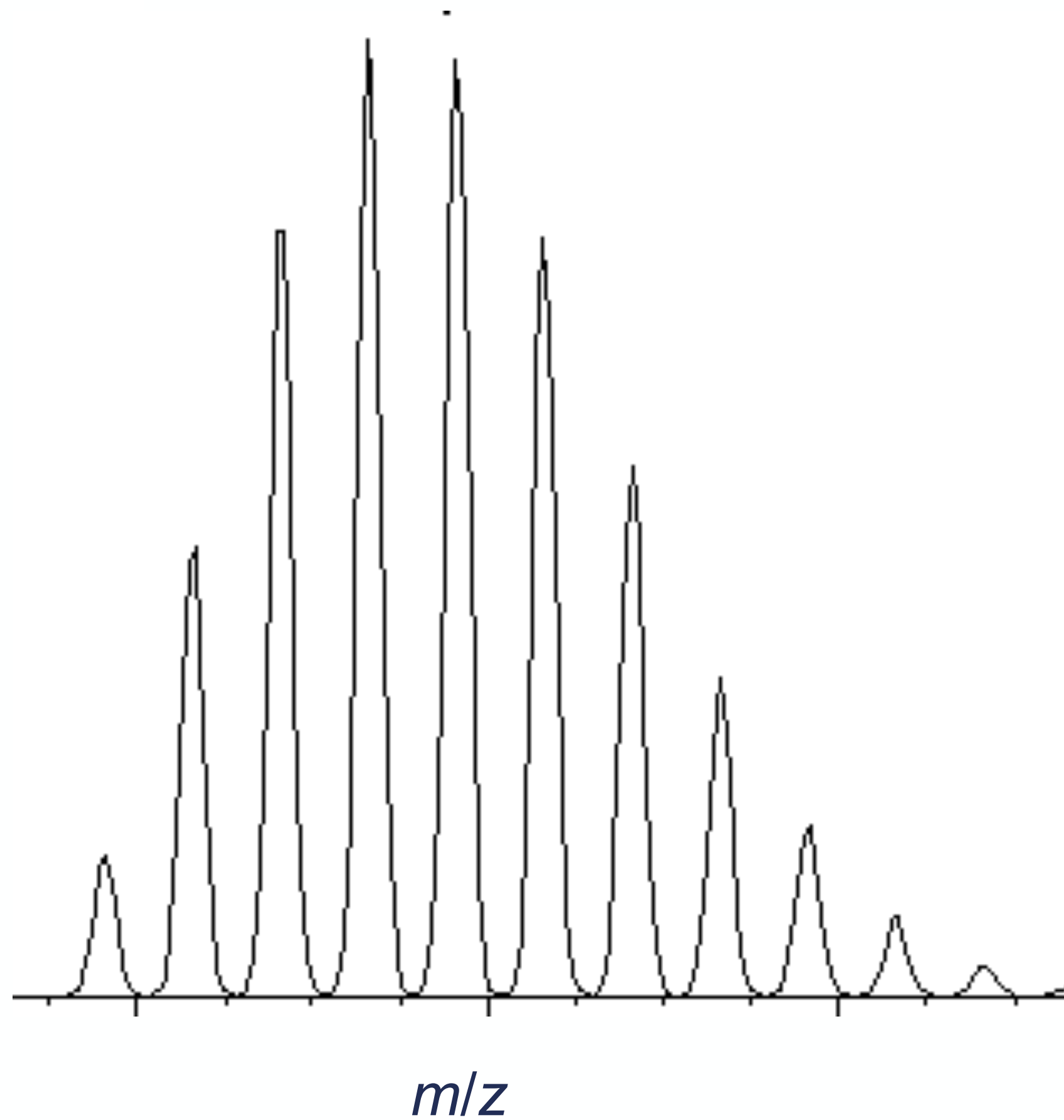
Family car

LC-MS Orbitrap – MS1 Full Scan

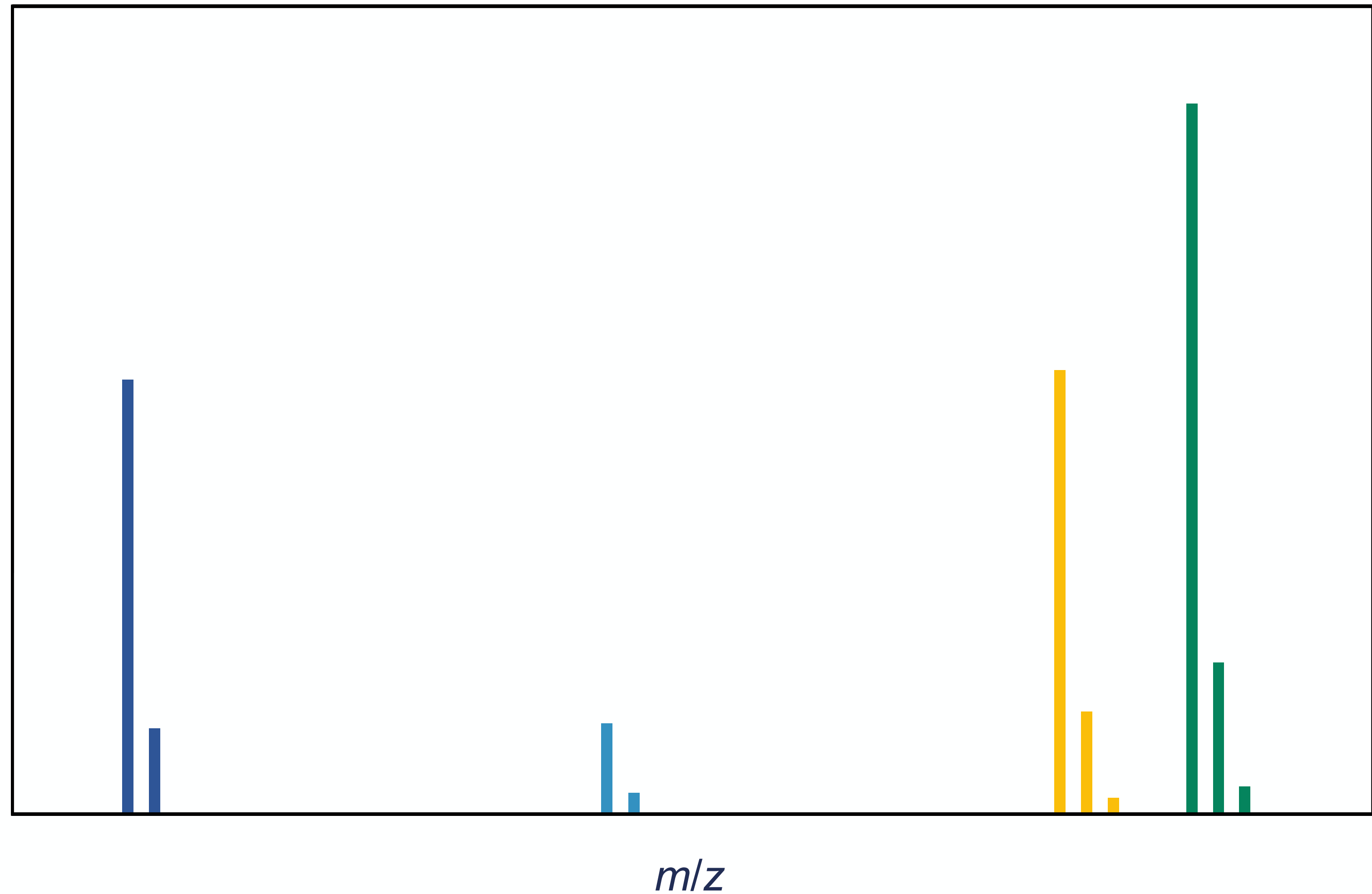


Mass Spectra Modes

Profile Mode Spectra
(Full Peaks)



Centroid Mode Spectra

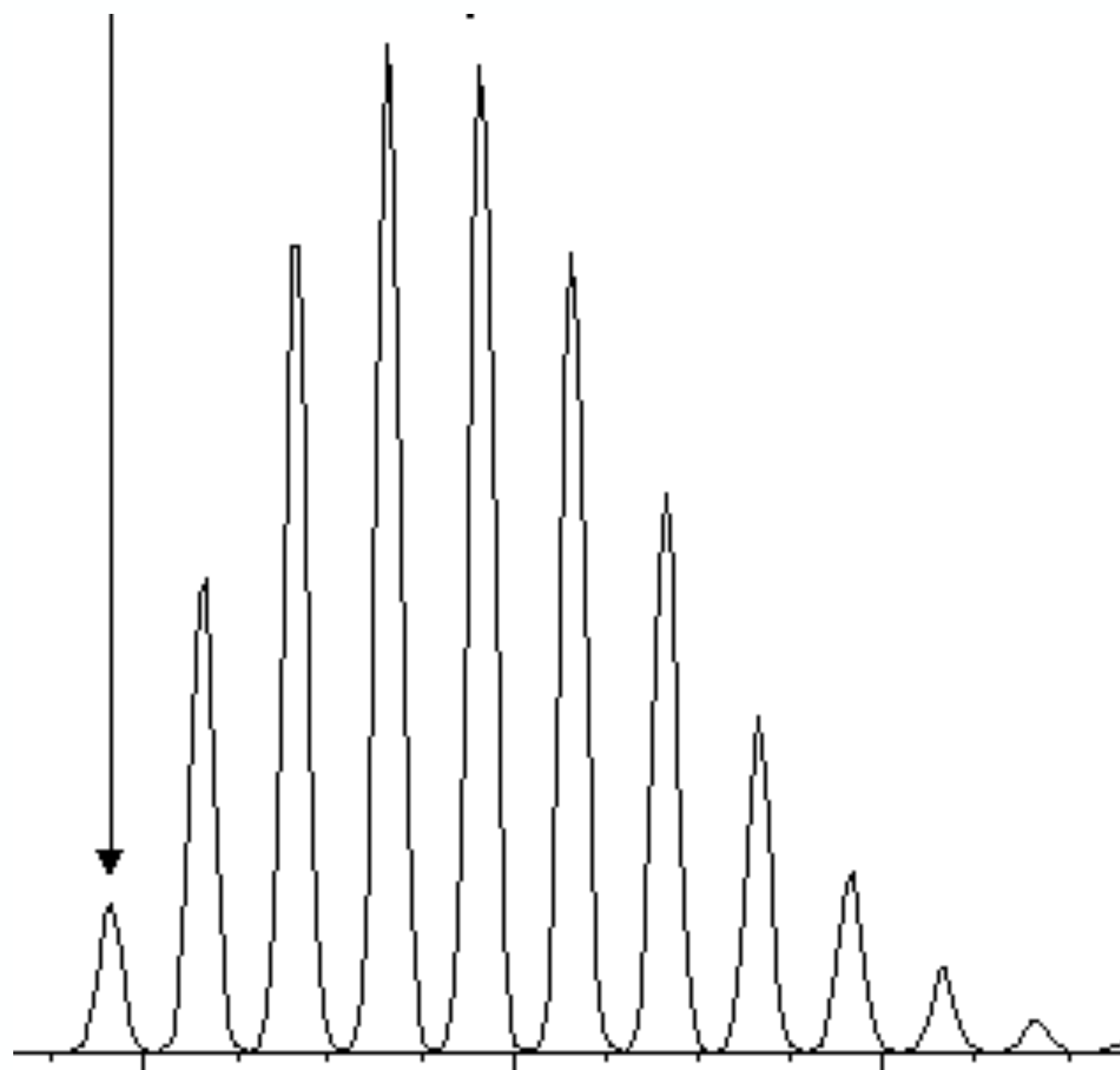


One Compound – Many Isotopes, Adducts, in-source Fragments

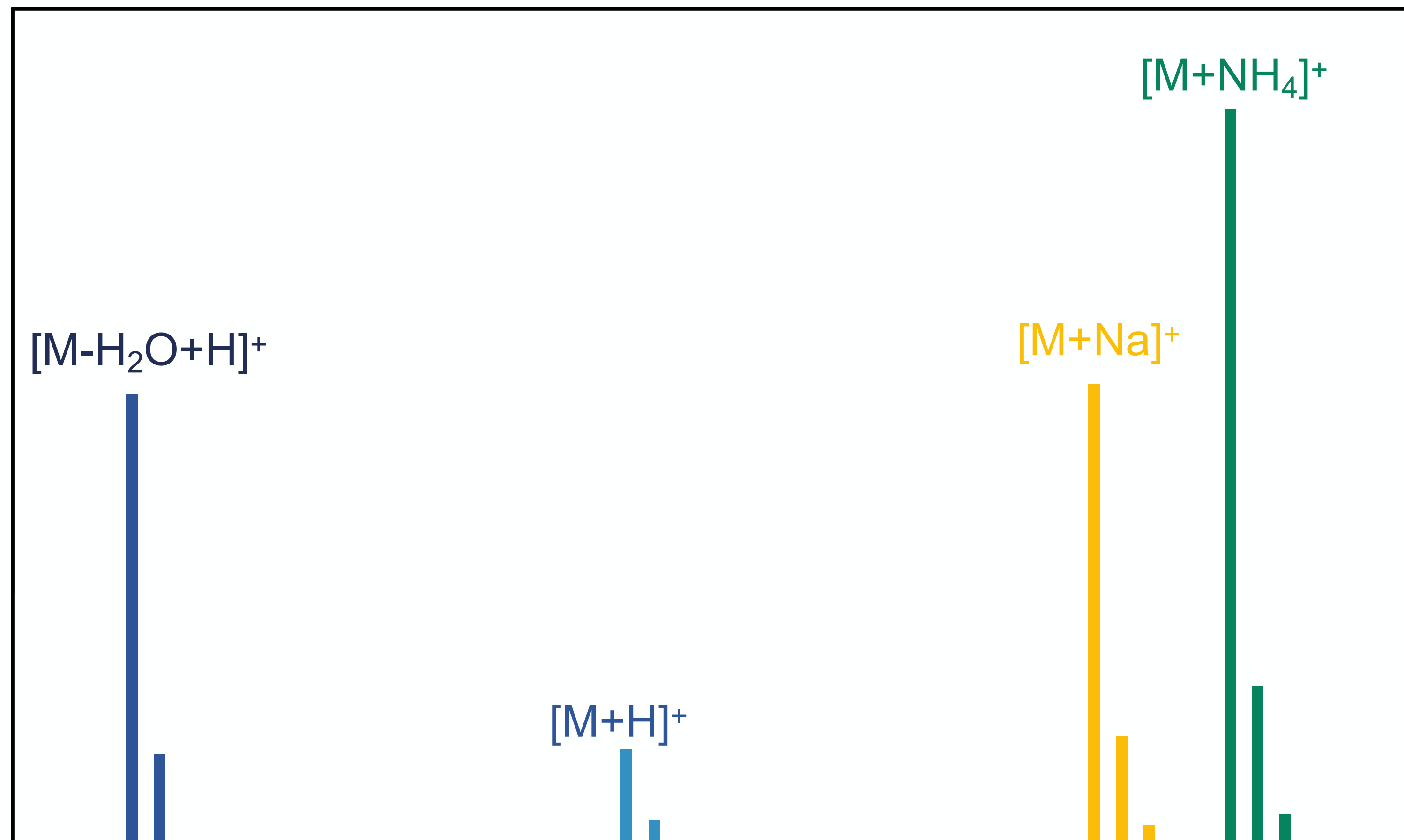
^{13}C Isotope pattern

Adducts and in-source fragments

Monoisotopic m/z



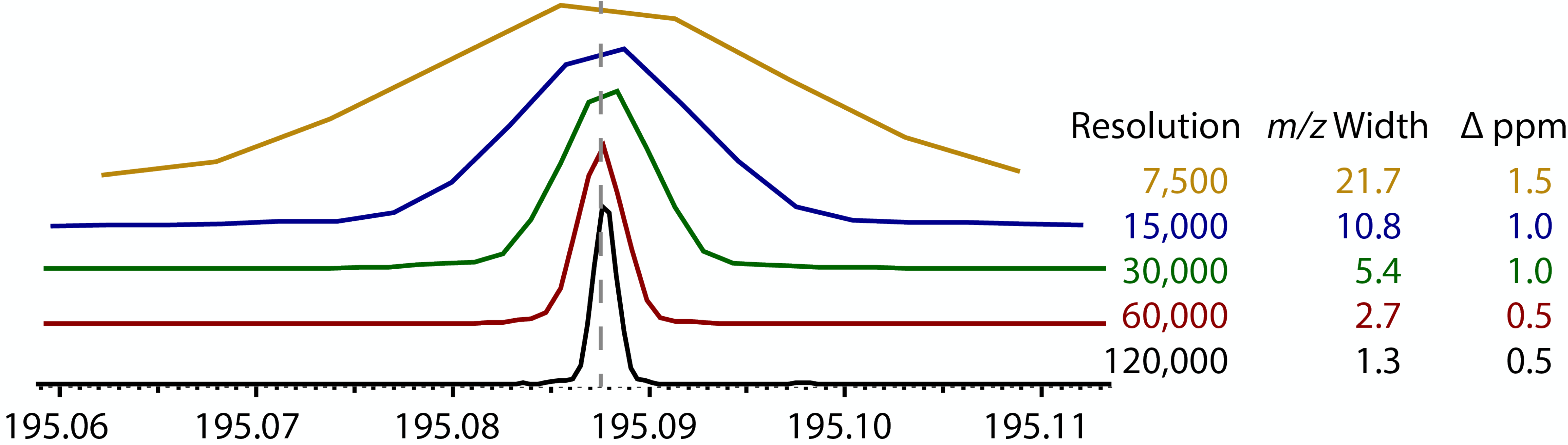
m/z

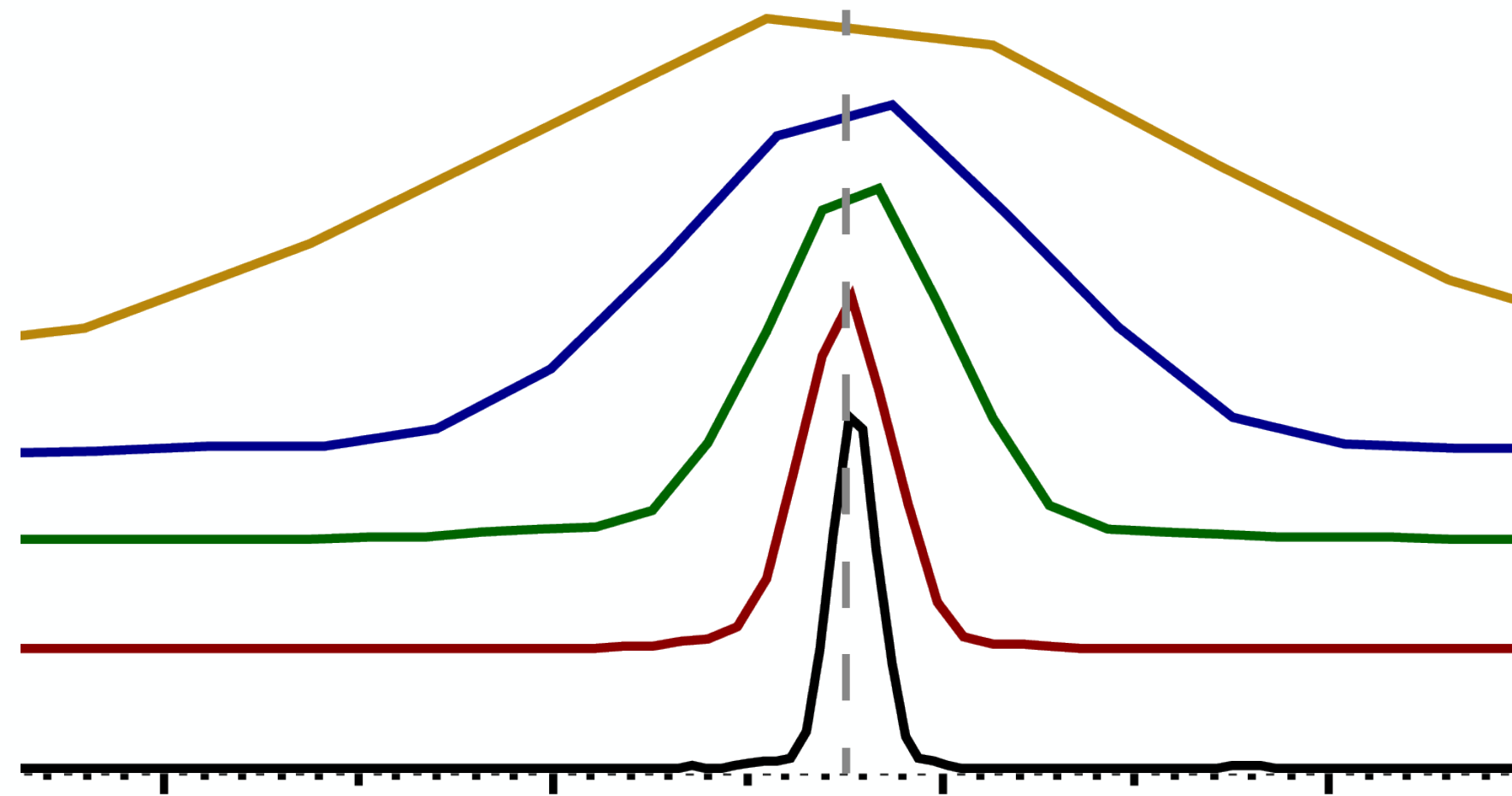


m/z

Mass Resolution

Caffeine [M+H]⁺ *m/z* 195.0877





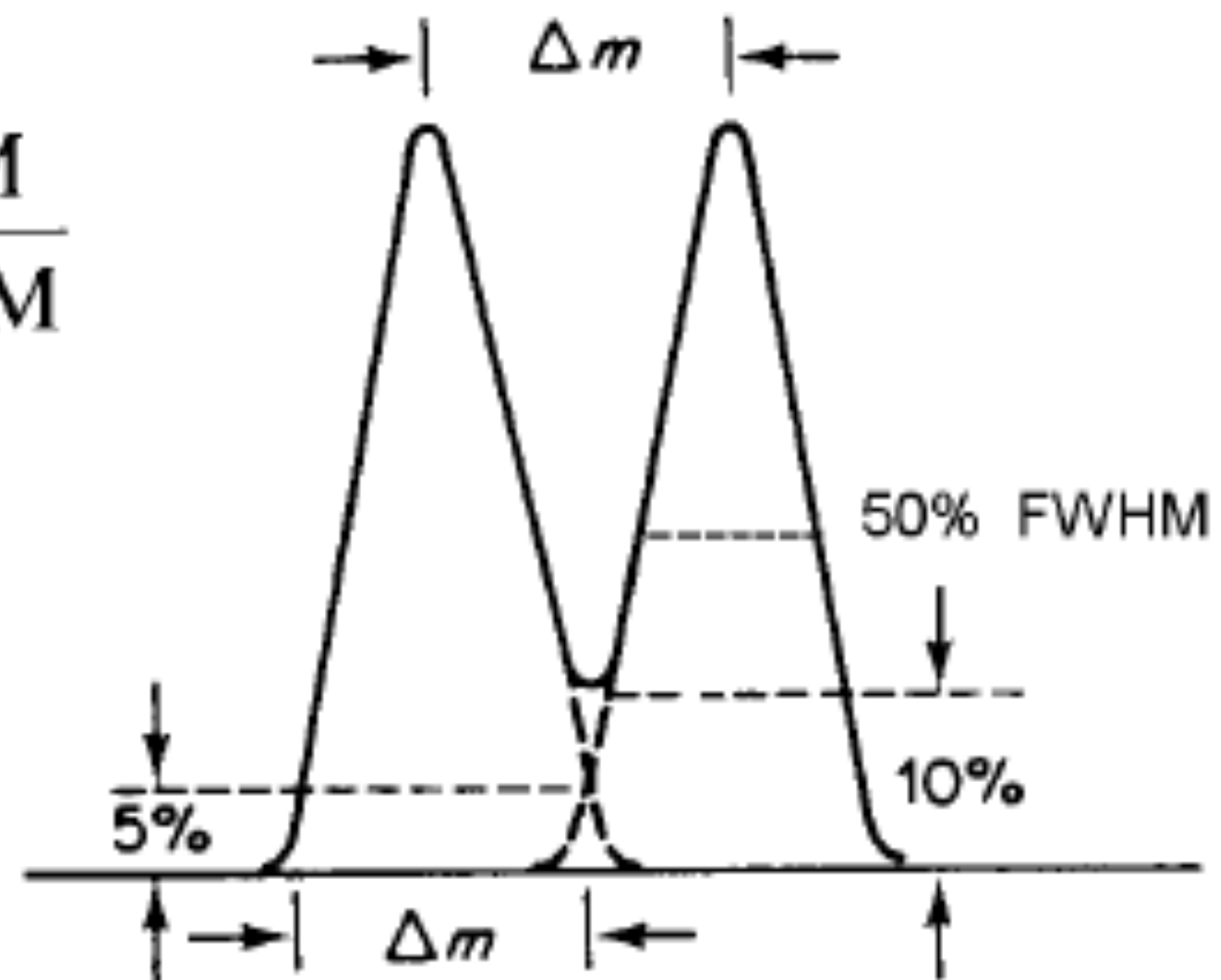
Mass Accuracy

m/z **195.0** **87** **6** **52**

LR (QqQ, IT)	± 0.1
qTOF	<15 ppm
Orbitrap	<5 ppm
FT-ICR	<2 ppm

Reporting *m/z*

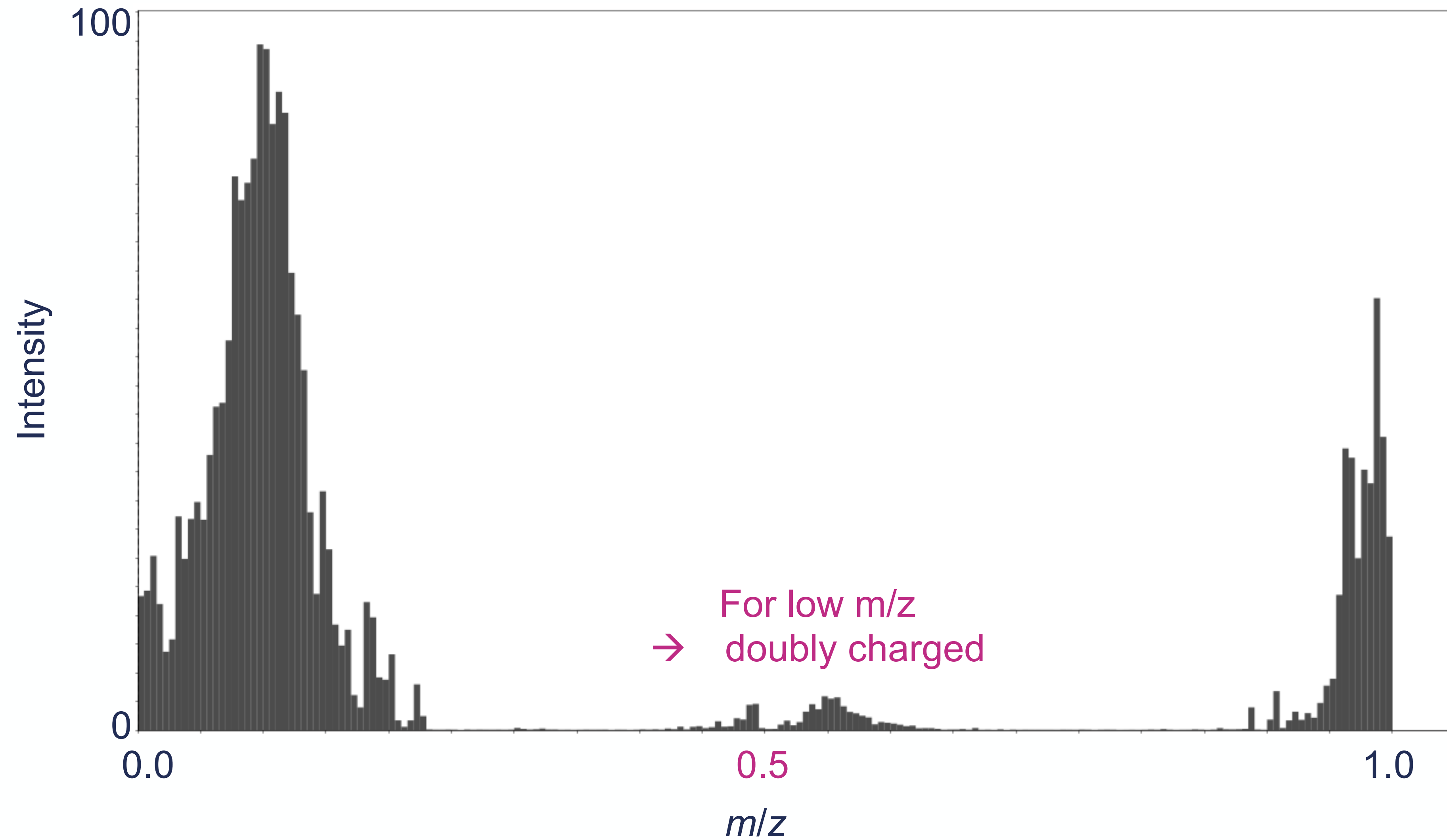
$$R = \frac{M}{\Delta M}$$



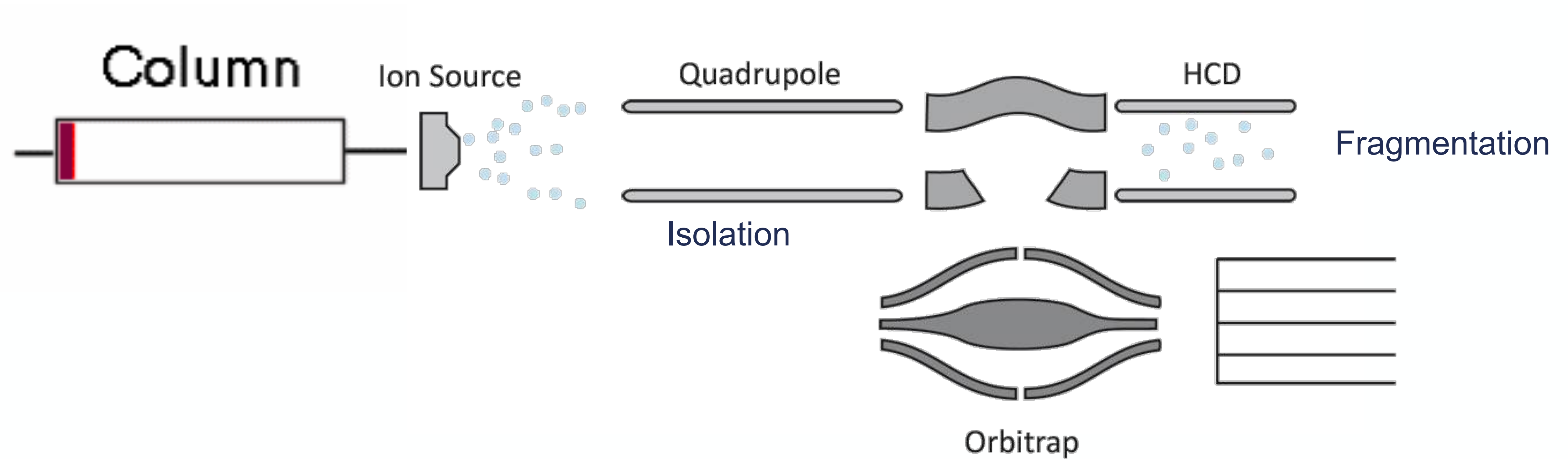
Exact mass: calculated
Accurate mass: measured (if <5 ppm)

Mass defect

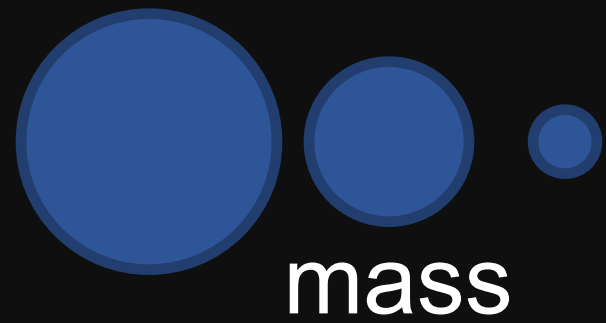
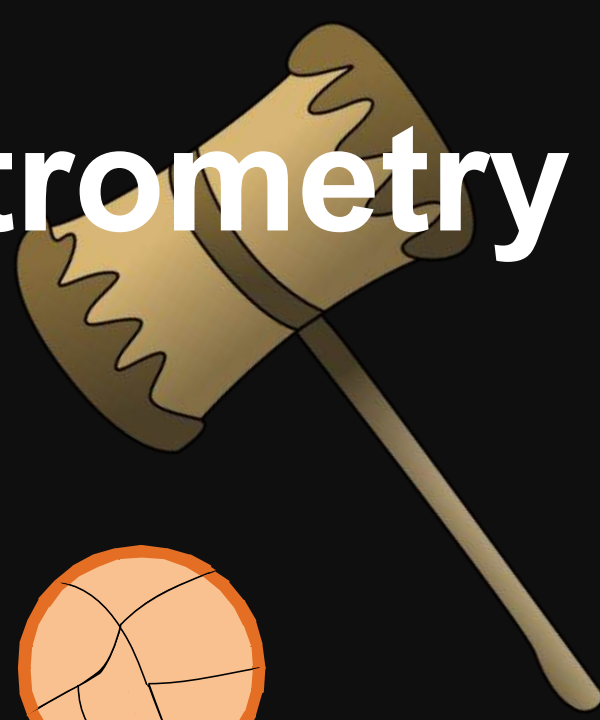
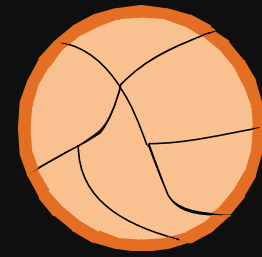
- Limited options to combine H (1.0078), O (15.9949), Cl (34.9689)



LC-MS/MS – Fragmentation Spectra



Tandem Mass Spectrometry (MS/MS)

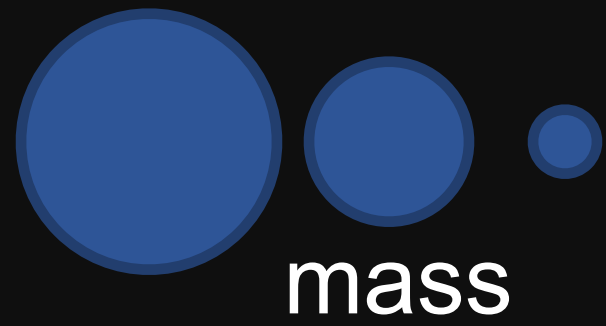


mass

Tandem Mass Spectrometry (MS/MS)



Tandem Mass Spectrometry (MS/MS)



MS/MS Trigger

Method A

which is better?

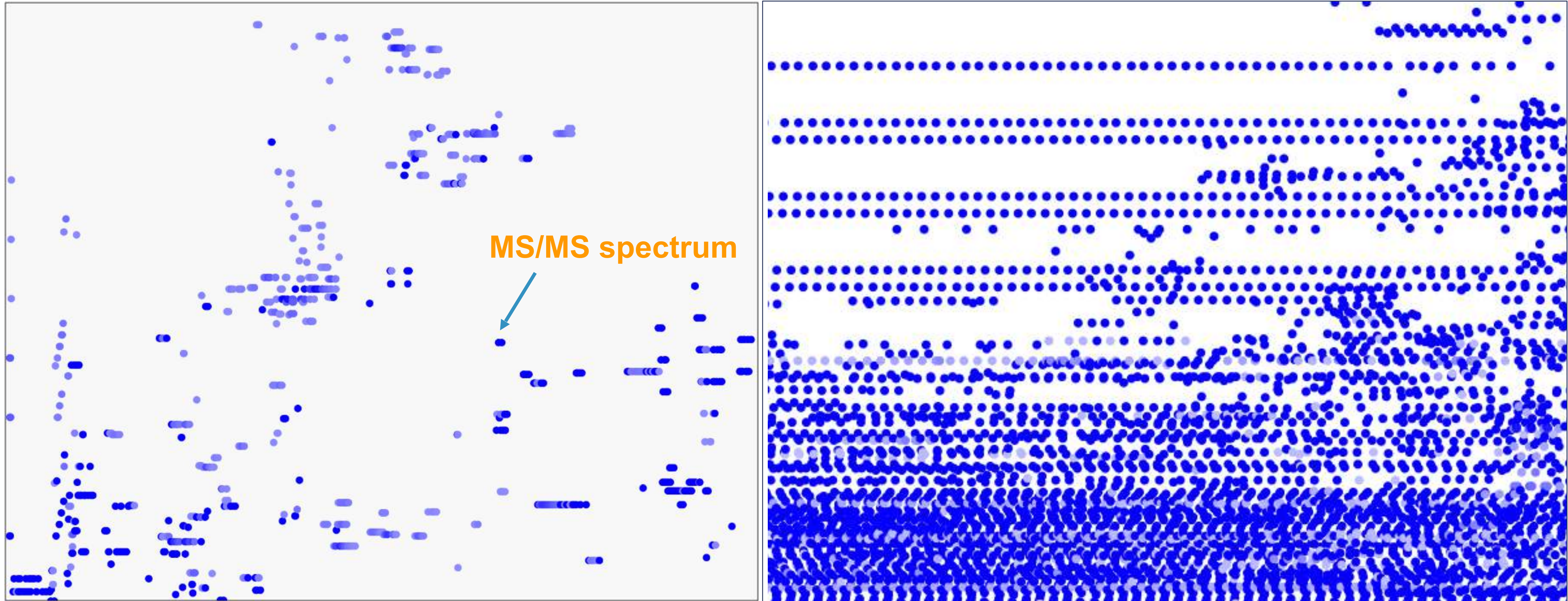
Method B

Precursor m/z

MS/MS spectrum

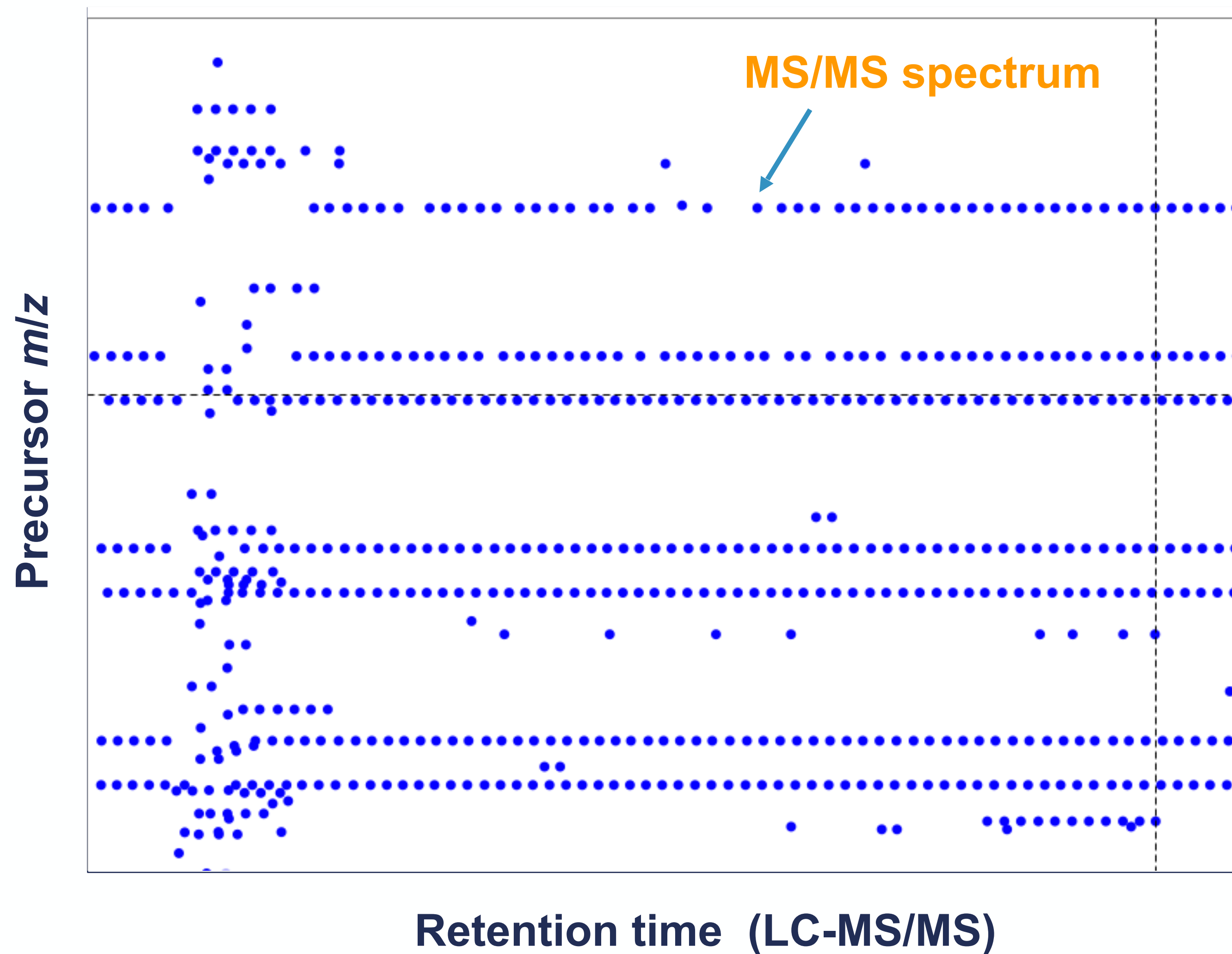


Retention time (LC-MS/MS)



MS/MS Trigger

Method B

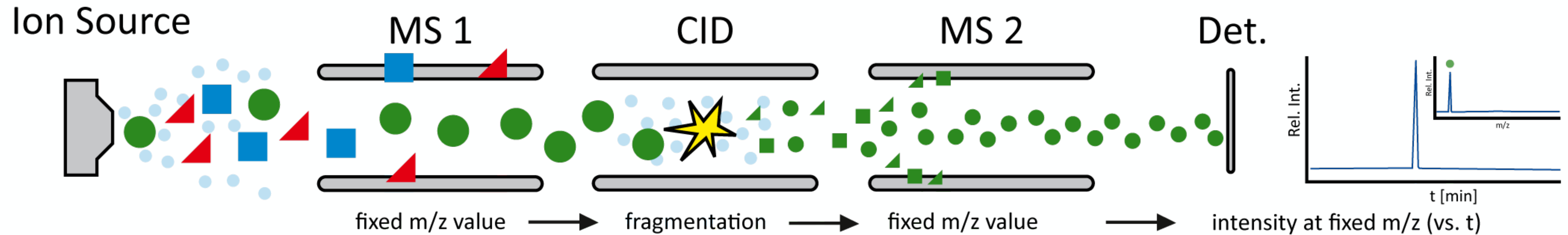


Congratulations!

You rediscovered plasticizers and other contaminants...

Let's put those on the exclusion list and maybe spectral library for common contaminations

Multi-Reaction Monitoring (MRM) with Triple Quadrupole (QqQ)



- Also known as Selected-reaction monitoring (SRM)
- List of transitions that are cycled through
 - MS¹ set to one m/z value
 - CID cell fragments isolated ions
 - MS² set to fragment m/z value
- High sensitivity for quantification
- Multiple transitions for each analyte as quality control

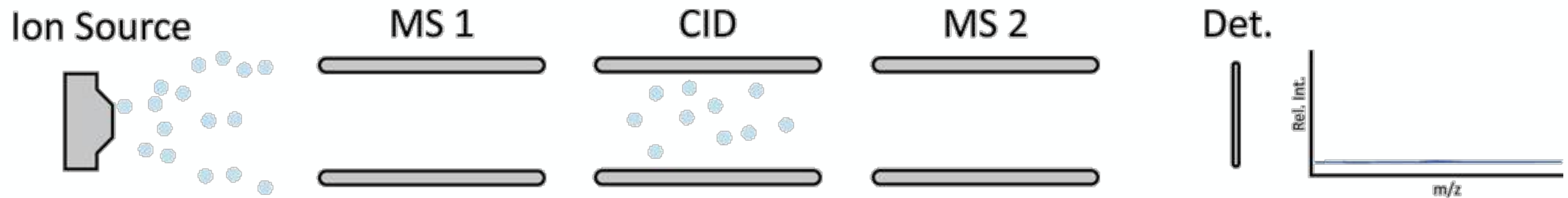
Example

m/z 500 → m/z 250 (quantifier)

m/z 500 → m/z 175 (qualifier)

m/z 500 → m/z 125 (qualifier)

Product-Ion Scan (QqQ, Ion Traps, and other Hybrid Instruments)



- MS¹ set to fixed m/z value
- CID cell fragments selected ions
- MS² scans the product ions

Computational Mass Spectrometry

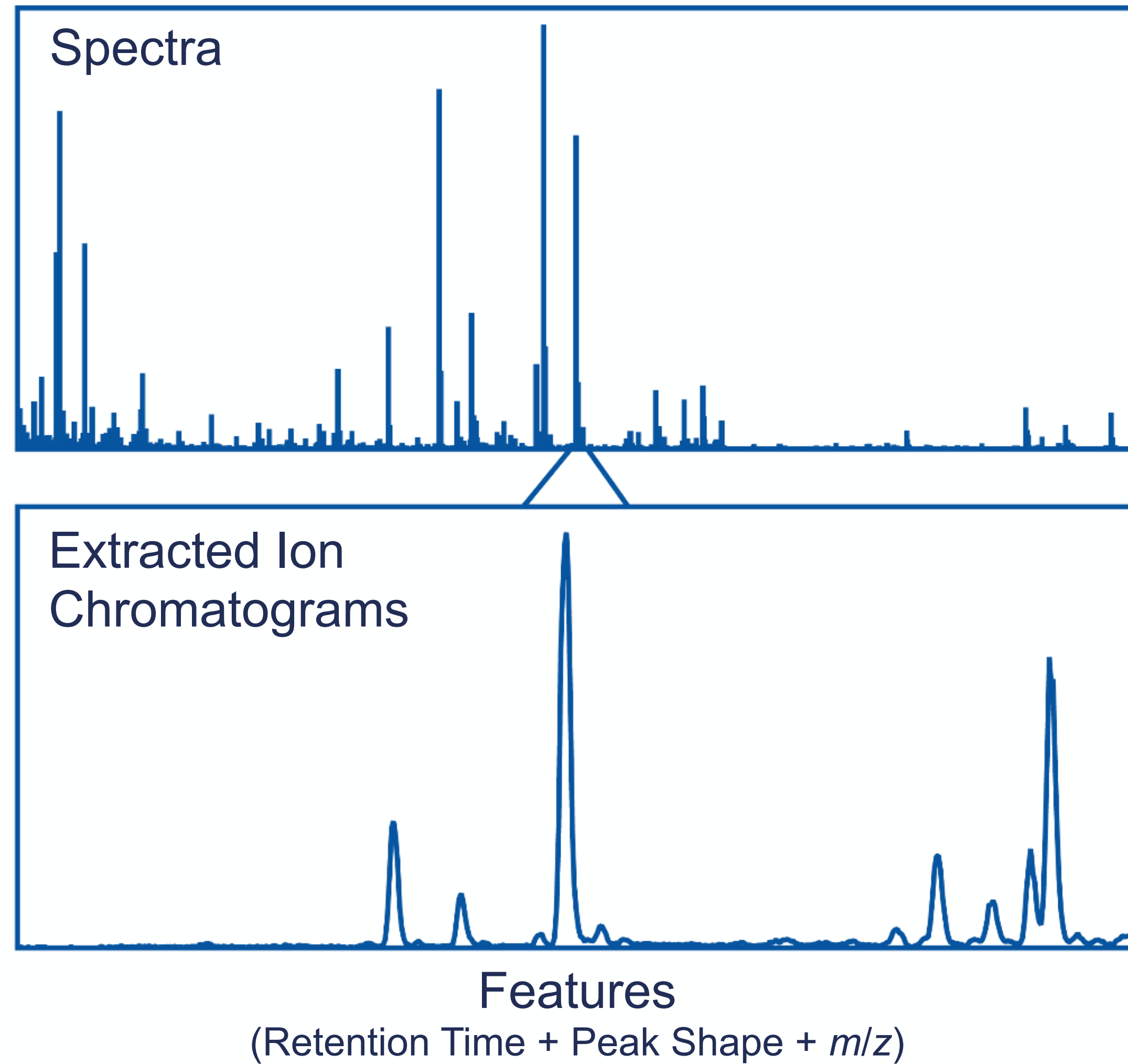
Introduction, Data Conversion, Data Processing

Dr. Steffen Heuckeroth
Chief Technology Officer
steffen.heuckeroth@mzio.io

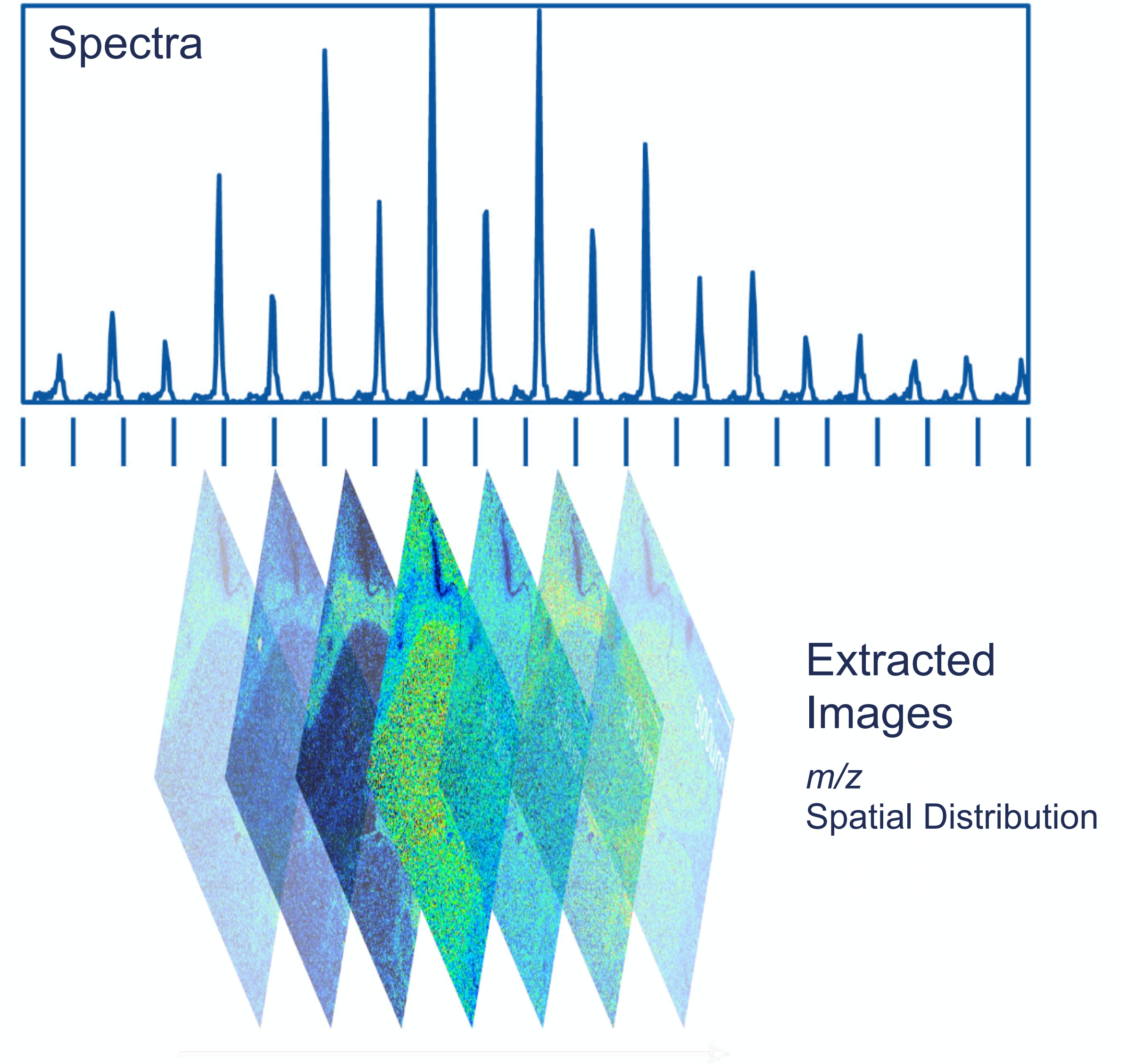
Dr. Robin Schmid
Chief Scientific Officer
robin.schmid@mzio.io

Challenges in Computational Mass Spectrometry

LC-(IMS)-MS



(IMS)-MS imaging



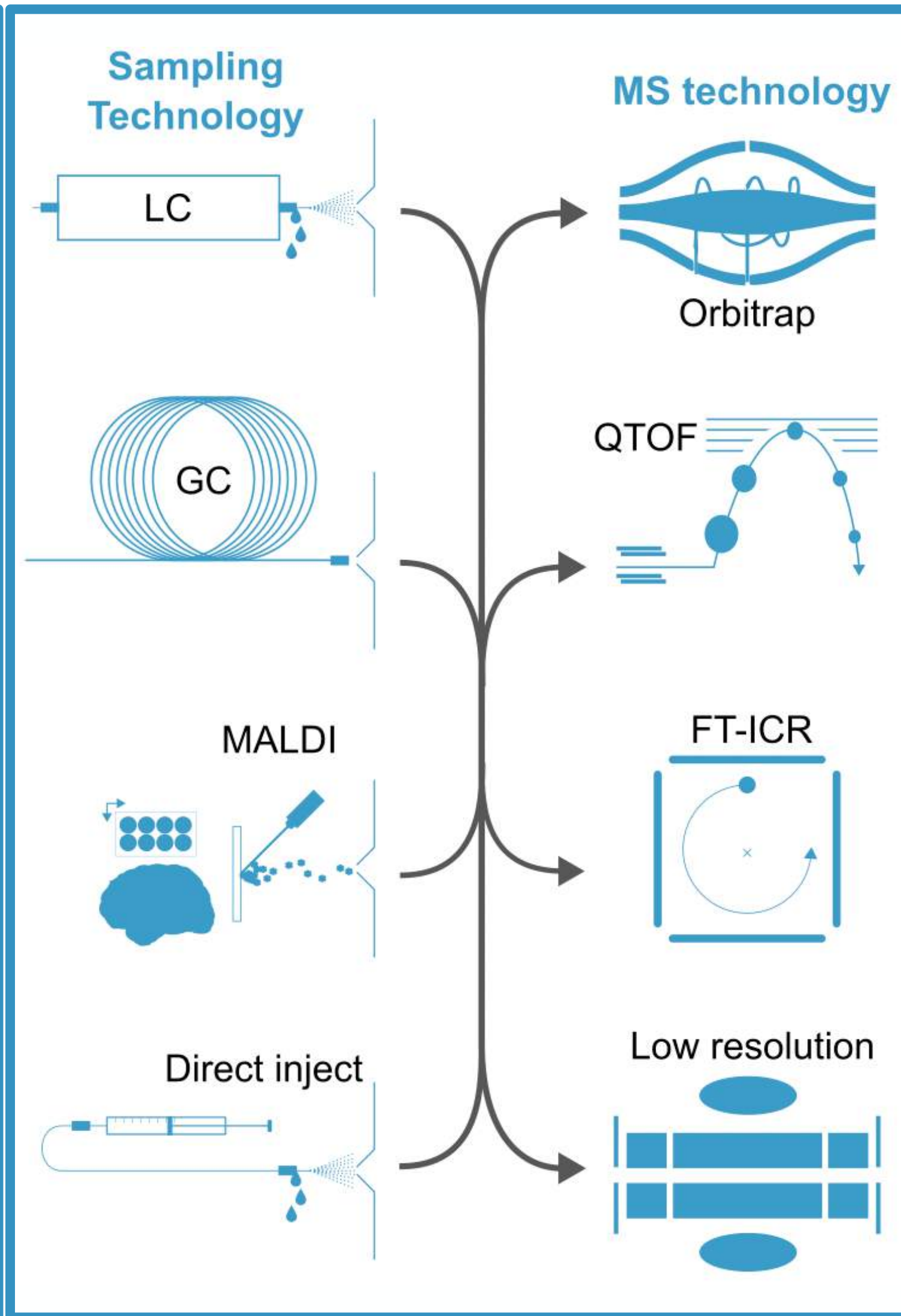
A Single Software for all your MS Data

Handle vendor and technology mix in one place

mzIO

mzmine

Vendor and technology mix



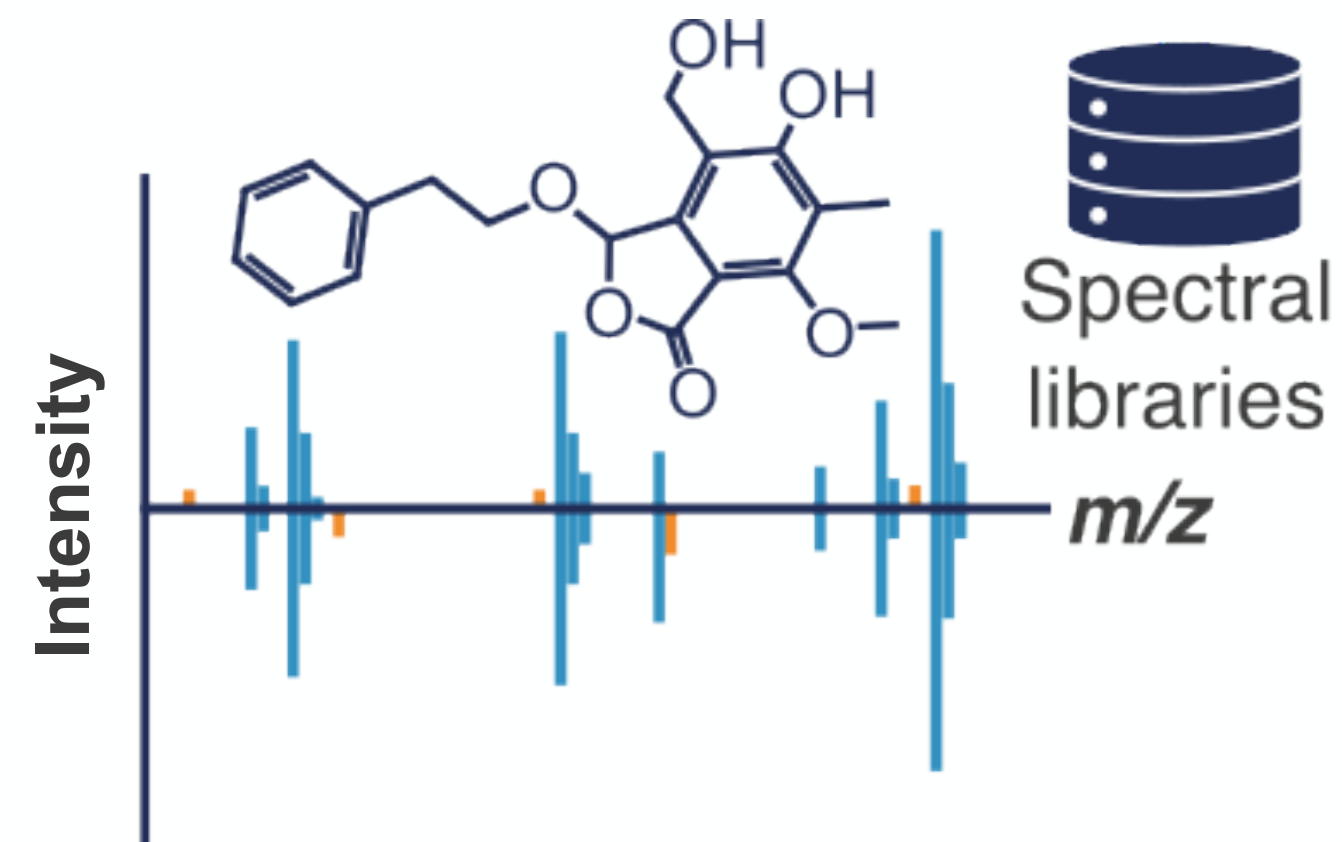
Vendor data
types and formats

High speed and
throughput

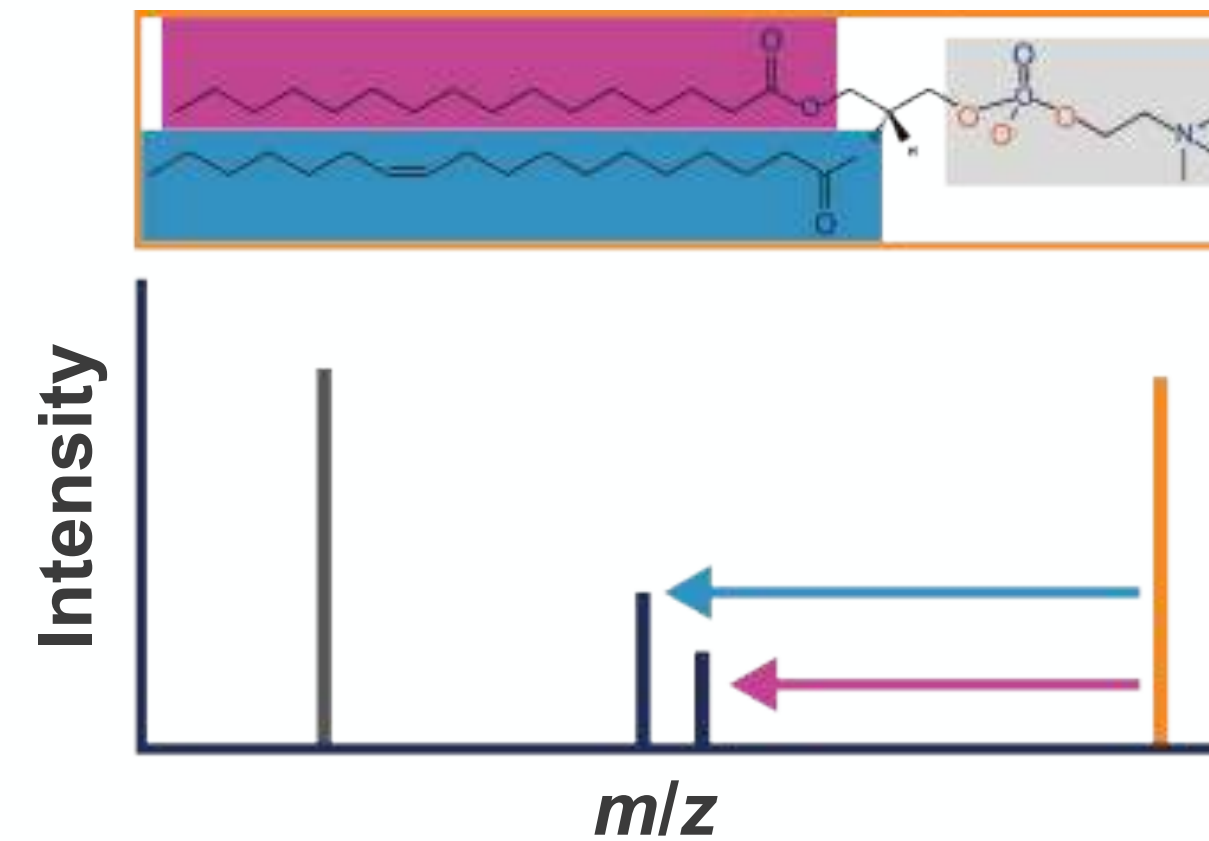
Extract knowledge
from data

Easy to use

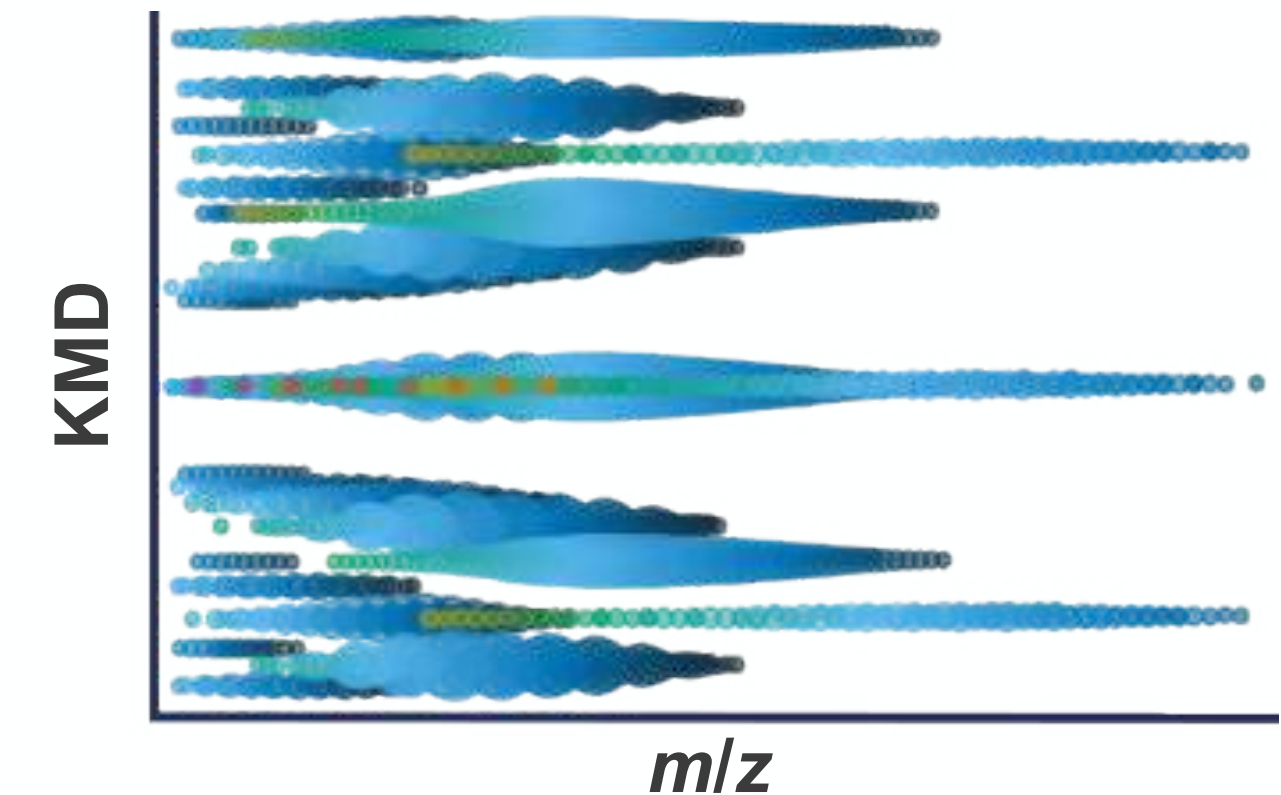
Metabolomics



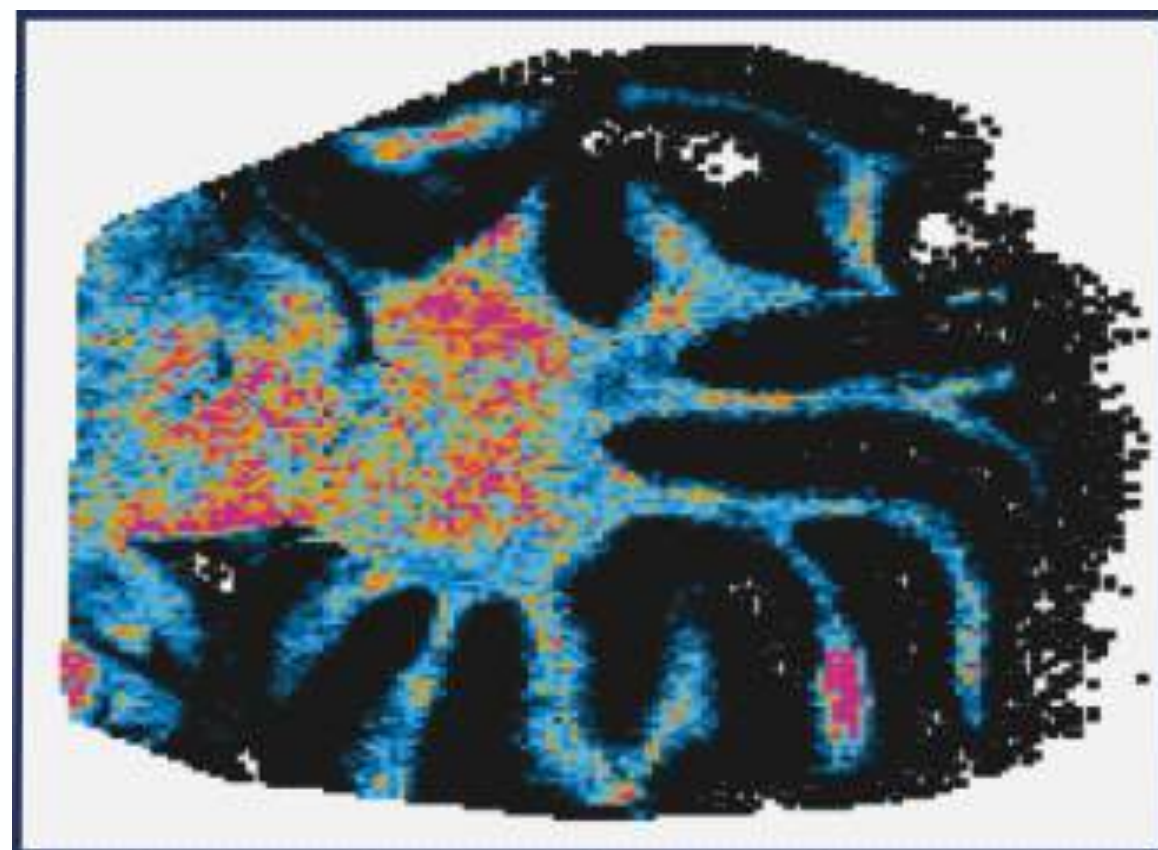
Lipidomics



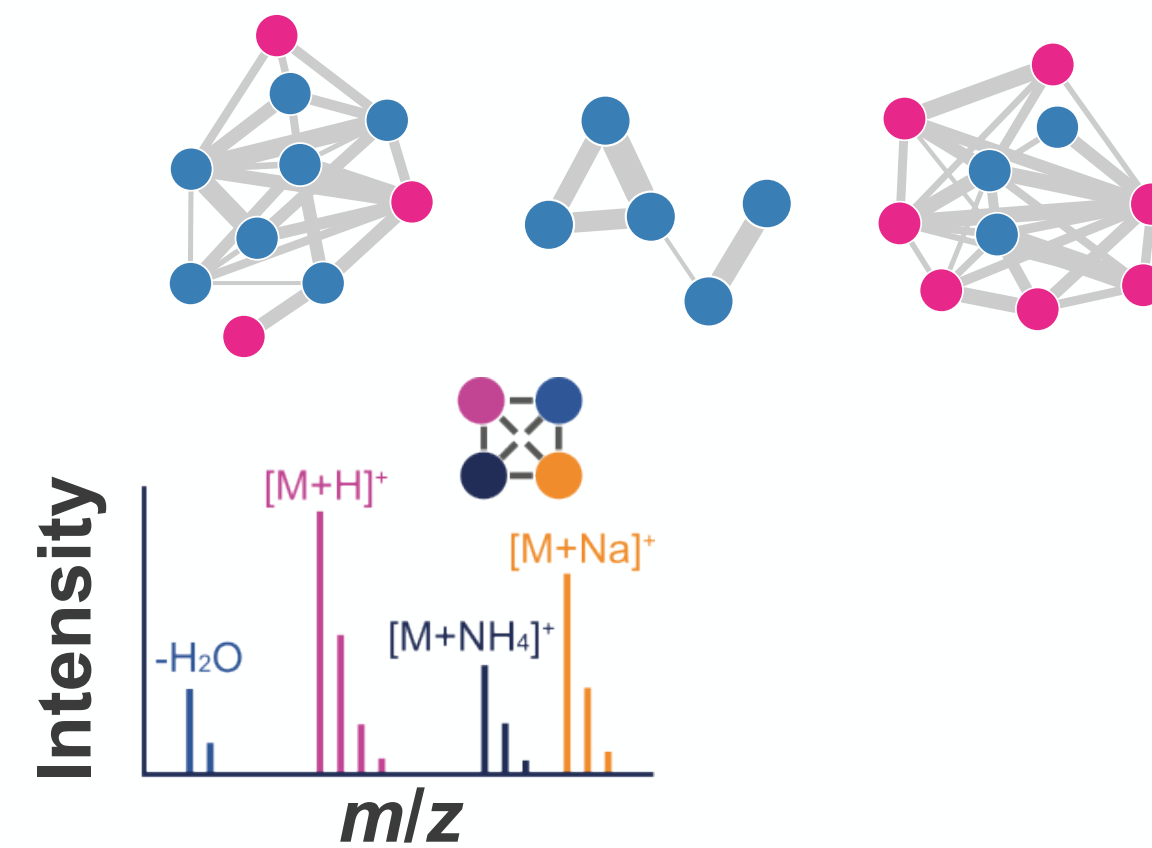
Polymers and PFAS



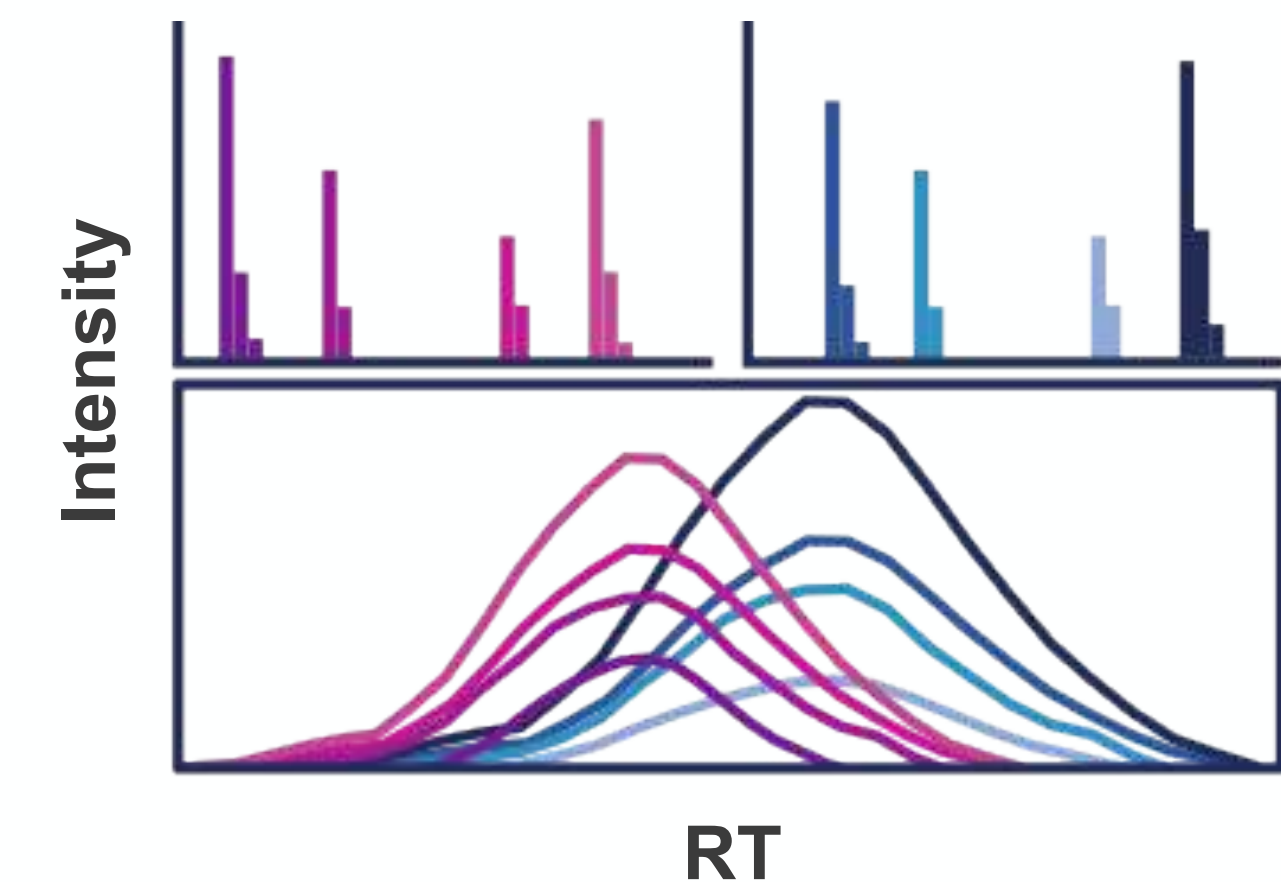
Imaging



Molecular Networking



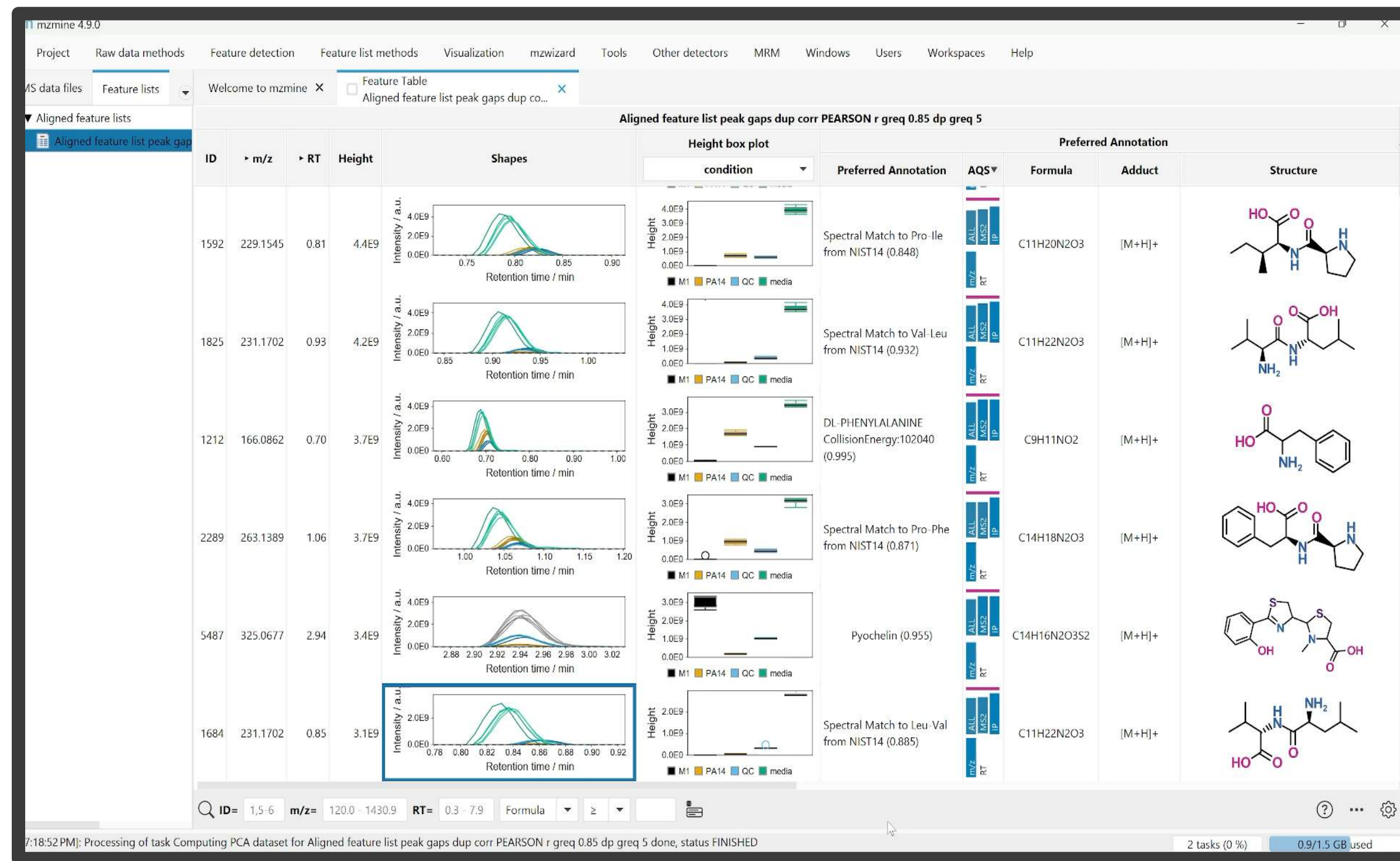
Impurity Analysis



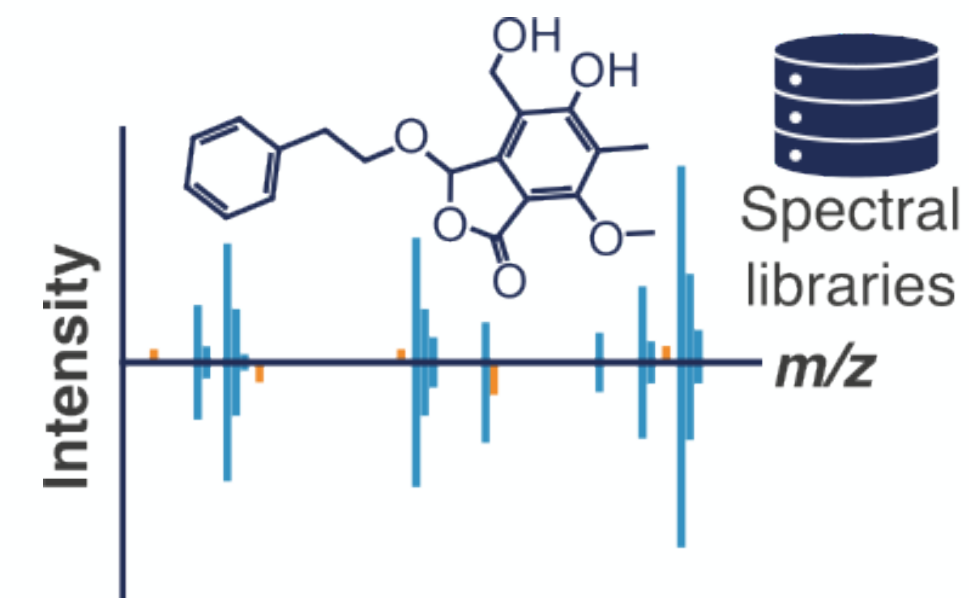
One software solution – a unified experience

mZIO

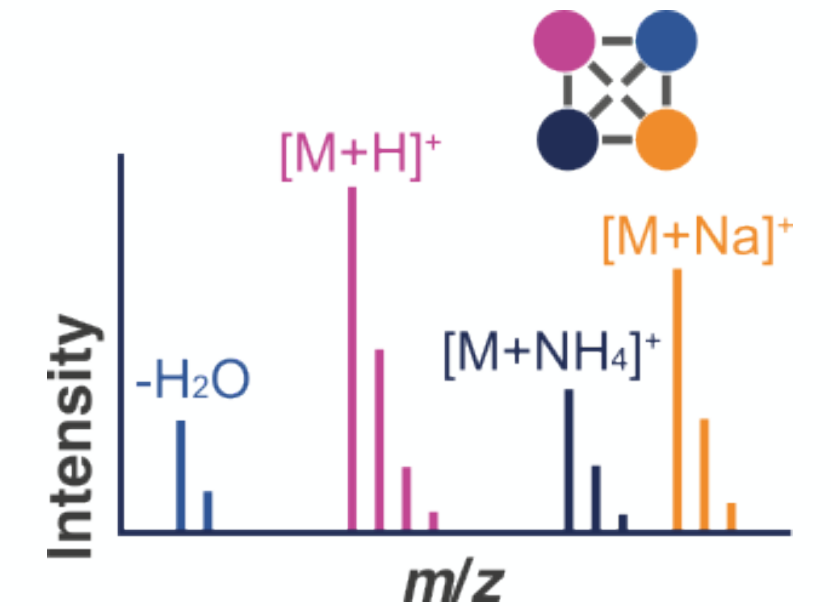
mzmine



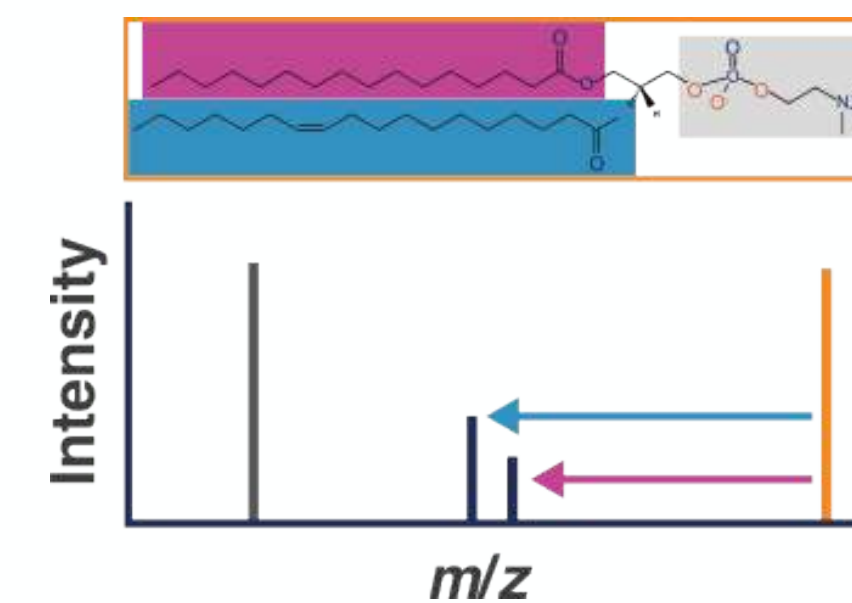
Metabolomics



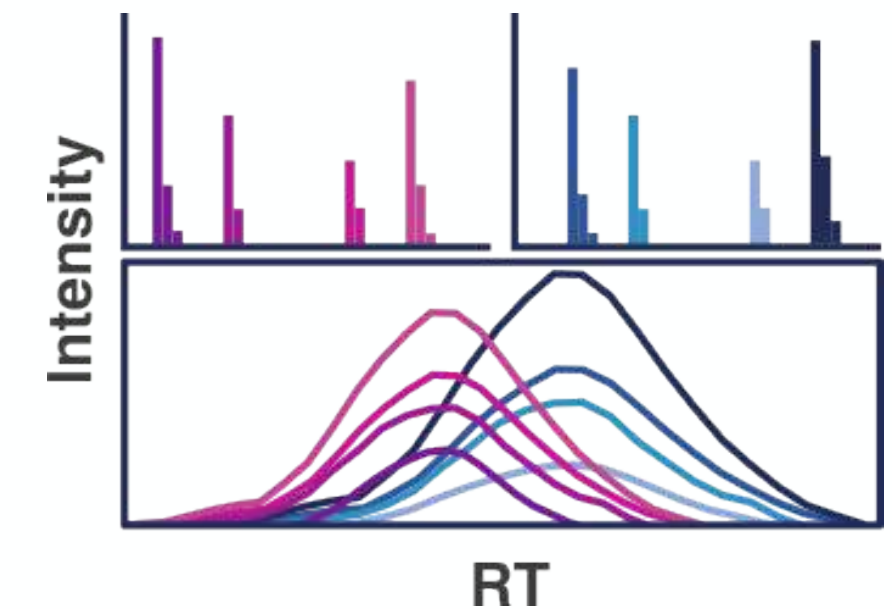
Molecular Networking



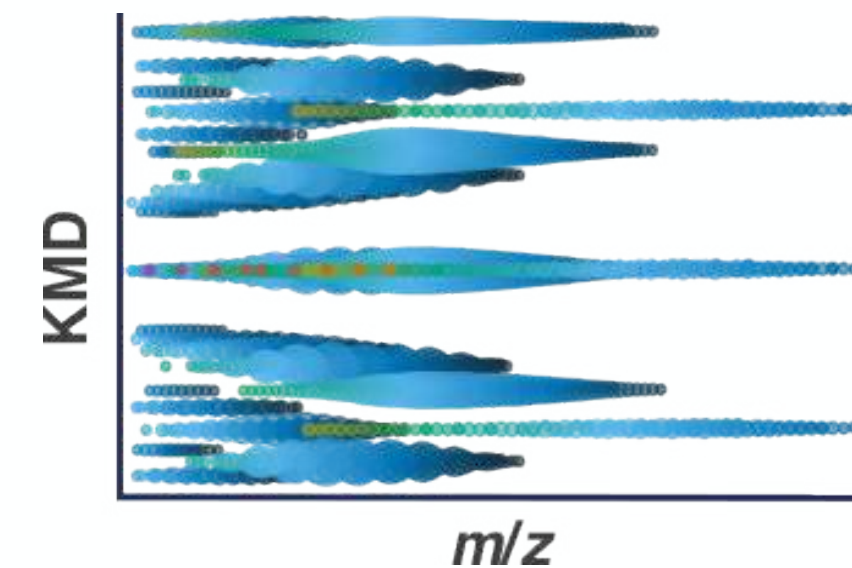
Lipidomics



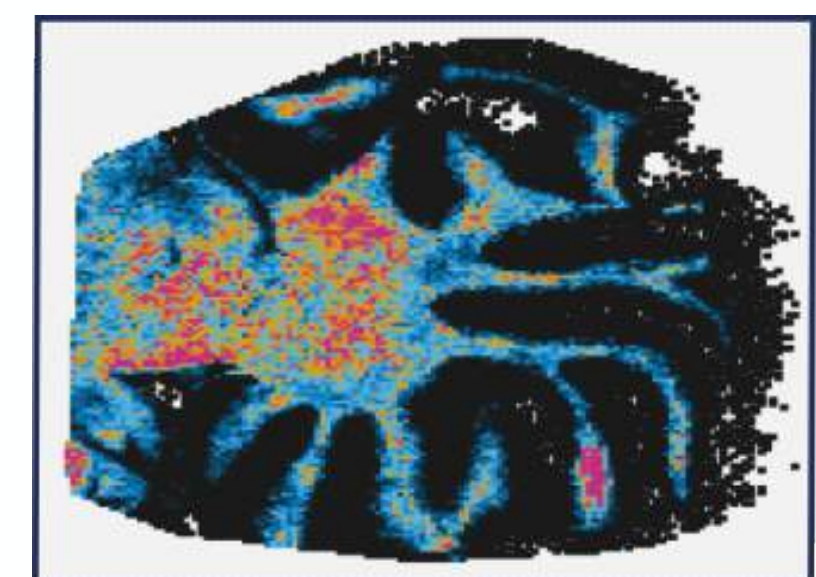
Impurity Analysis



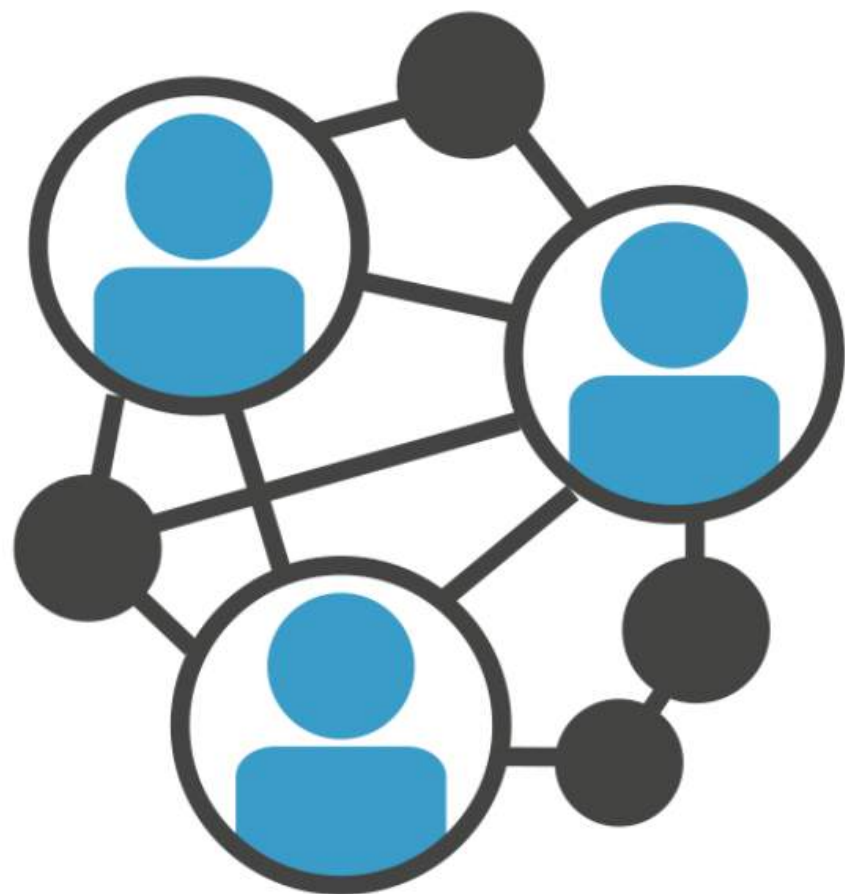
Polymers and PFAS



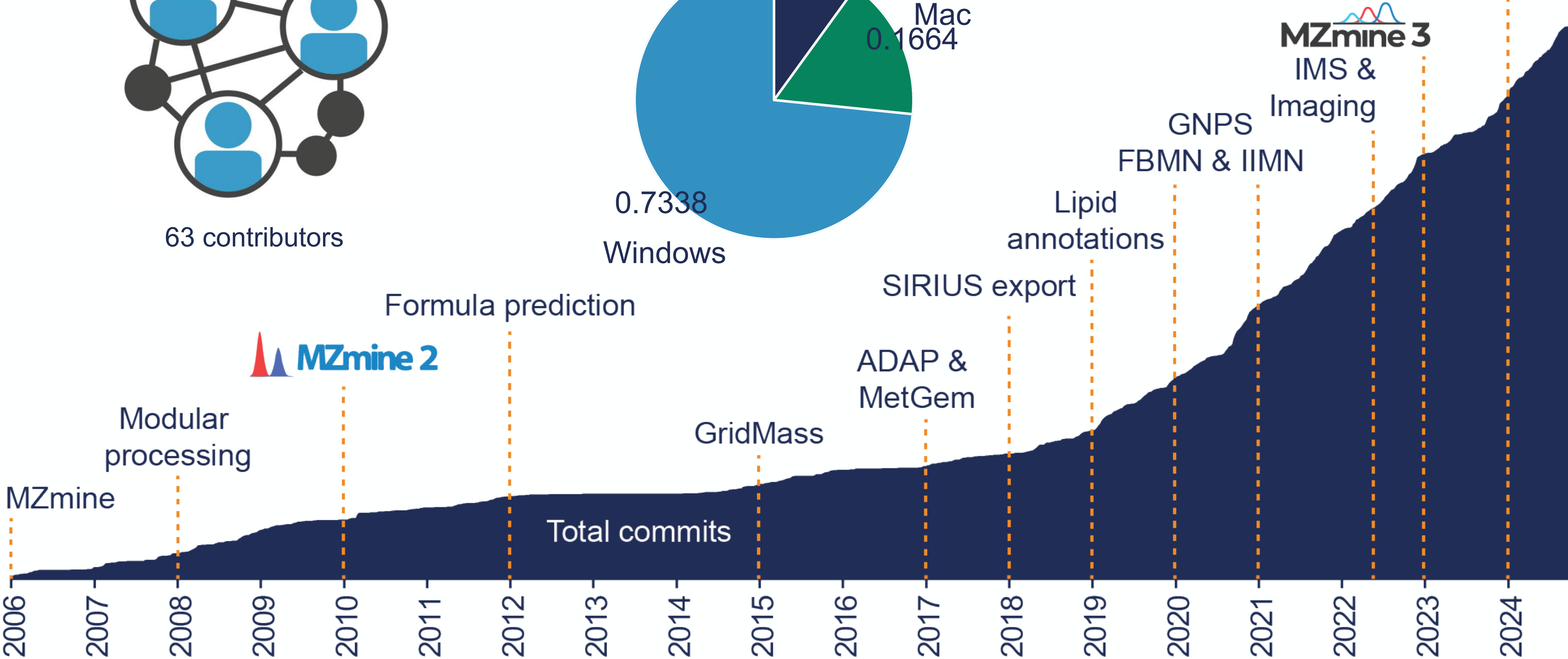
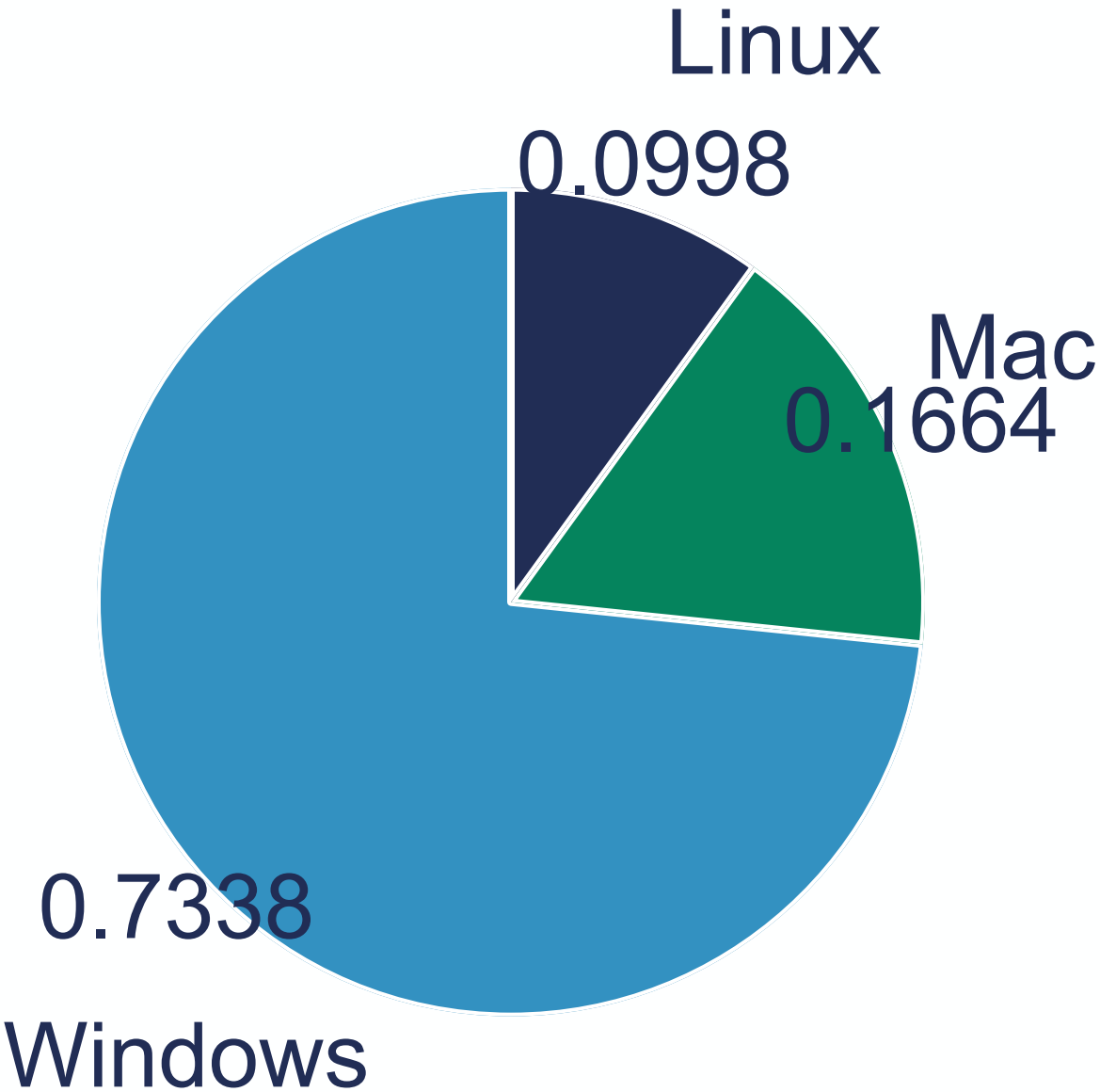
Imaging



mzmine in Numbers



63 contributors



MZmine

2006

BIOINFORMATICS APPLICATIONS NOTE Vol. 22 no. 5 2006, pages 634–636
doi:10.1093/bioinformatics/btk039

Data and text mining

MZmine: toolbox for processing and visualization of mass spectrometry based molecular profile data

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¹Turku Centre for Biotechnology, Turku, Finland and ²VTT Technical Research Centre of Finland, Espoo, Finland

Received on November 25, 2005; revised on December 21, 2005; accepted on January 3, 2006

Advance Access publication January 10, 2006

Associate Editor: Jonathan Wren

1055 citations



2011

Pluskal et al. *BMC Bioinformatics* 2010, **11**:395
<http://www.biomedcentral.com/1471-2105/11/395>



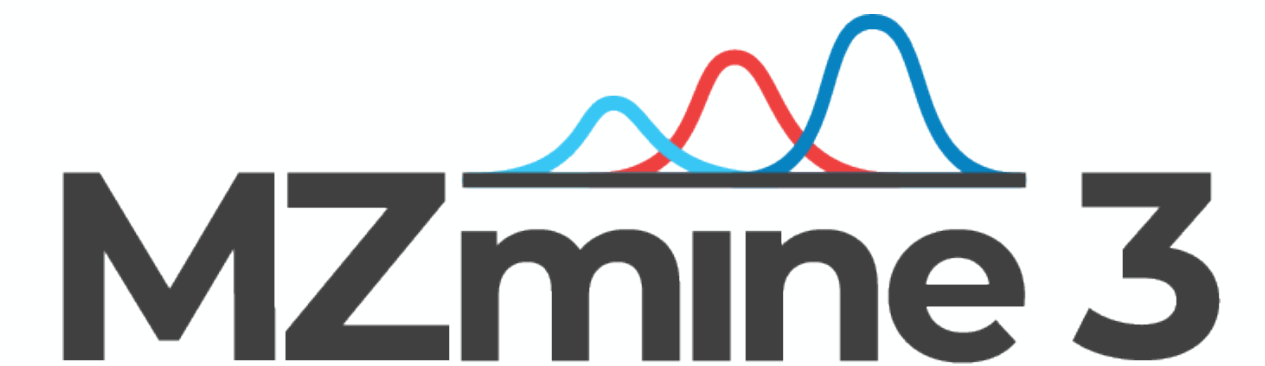
SOFTWARE

Open Access

MZmine 2: Modular framework for processing, visualizing, and analyzing mass spectrometry-based molecular profile data

Tomáš Pluskal^{1*}, Sandra Castillo², Alejandro Villar-Briones¹, Matej Orešič²

4532 citations



2023

Correspondence | [Published: 01 March 2023](#)

Integrative analysis of multimodal mass spectrometry data in MZmine 3

[Robin Schmid](#), [Steffen Heuckeroth](#), [Ansgar Korf](#), [Aleksandr Smirnov](#), [Owen Myers](#), [Thomas S. Dyrland](#), [Roman Bushuiev](#), [Kevin J. Murray](#), [Nils Hoffmann](#), [Miaoshan Lu](#), [Abinеш Sarvepalli](#), [Zheng Zhang](#), [Markus Fleischauer](#), [Kai Dührkop](#), [Mark Wesner](#), [Shawn J. Hoogstra](#), [Edward Rudt](#), [Olena Mokshyna](#), [Corinna Brungs](#), [Kirill Ponomarov](#), [Lana Mutabdzija](#), [Tito Damiani](#), [Chris J. Pudney](#), [Mark Earll](#), ... [Tomáš Pluskal](#) ✉ [+ Show authors](#)

1055 citations

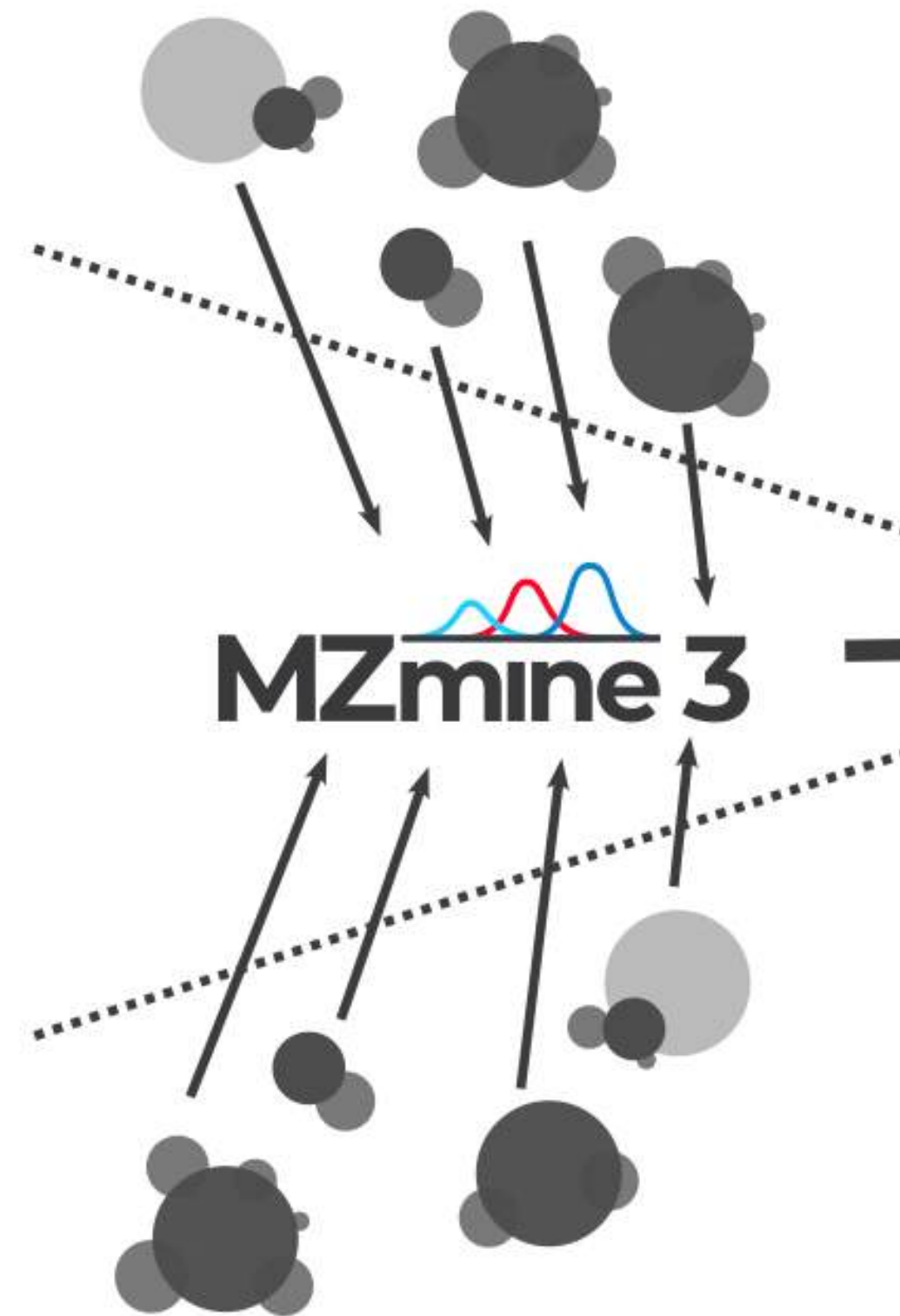
Latest Publication; Nature Biotechnology

From Scientific Community to Enterprise Solution

**> 10.000
active users**

**> 6,600
citations**

**Global
community**



**Open source
software for
scientific community**

Non-Profit Academic User



- Join the **mzmine** community
- Develop in the **open source** framework
- Contribute to the open **documentation**
- Elevate your **MS data processing**
- Free for **non-profit academic** use

Enterprise User



- Benefit from the large **mzmine** community
- Elevate your **MS data processing**
- **One solution** for multi modal MS
- Unlock priority **mzmine PRO** support
- Allows **commercial** use



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- ✗ IT integration
- ✗ Onboarding session
- ✗ MRM support
- ✗ UV, ELSD, other detectors
- ✗ Early feature access

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- ✗ MRM support
- ✗ UV, ELSD, other detectors
- ✗ Early feature access

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For labs up to 5 users

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Maximize your lab's performance

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- ✓ Onboarding session
- ✓ MRM support
- ✓ UV, ELSD, other detectors
- ✓ Early feature access

GET IN TOUCH

For enterprises

Enterprise Products & Support



CORE

Unlock the potential of
mzmine for your lab

- ✓ 30-Day trial
- ✓ GitHub Forum support
- ✓ Unlimited updates
- ✓ Commercial use
- ✗ Email support
- ✗ Video call support
- ✗ IT integration
- ✗ Onboarding session
- ✗ MRM support
- ✗ UV, ELSD, other detectors
- ✗ Early feature access

GET IN TOUCH

For small teams up to 3
users

PLUS

Accelerate your lab's
discoveries

- ✓ 30-Day trial
- ✓ GitHub Forum support
- ✓ Unlimited updates
- ✓ Commercial use
- ✓ Email support
- ✗ Video call support
- ✓ IT integration
- ✓ Onboarding session
- ✗ MRM support
- ✗ UV, ELSD, other detectors
- ✗ Early feature access

GET IN TOUCH

For labs up to 5 users

PRO

Maximize your lab's
performance

- ✓ 30-Day trial
- ✓ GitHub Forum support
- ✓ Unlimited updates
- ✓ Commercial use
- ✓ Email support
- ✓ Video call support
- ✓ IT integration
- ✓ Onboarding session
- ✓ MRM support
- ✓ UV, ELSD, other detectors
- ✓ Early feature access

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For enterprises

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COMMUNITY

Join our
academic program

LEARN MORE

For students and teachers

CORE

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mzmine for your lab

LEARN MORE

For small teams up to 3
users

PLUS

Accelerate your labs
discoveries

LEARN MORE

For labs up to 5 users

PRO

Maximize your labs
performance

LEARN MORE

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users

PLUS

Accelerate your labs
discoveries

LEARN MORE

For labs up to 5 users

PRO

Maximize your labs
performance

LEARN MORE

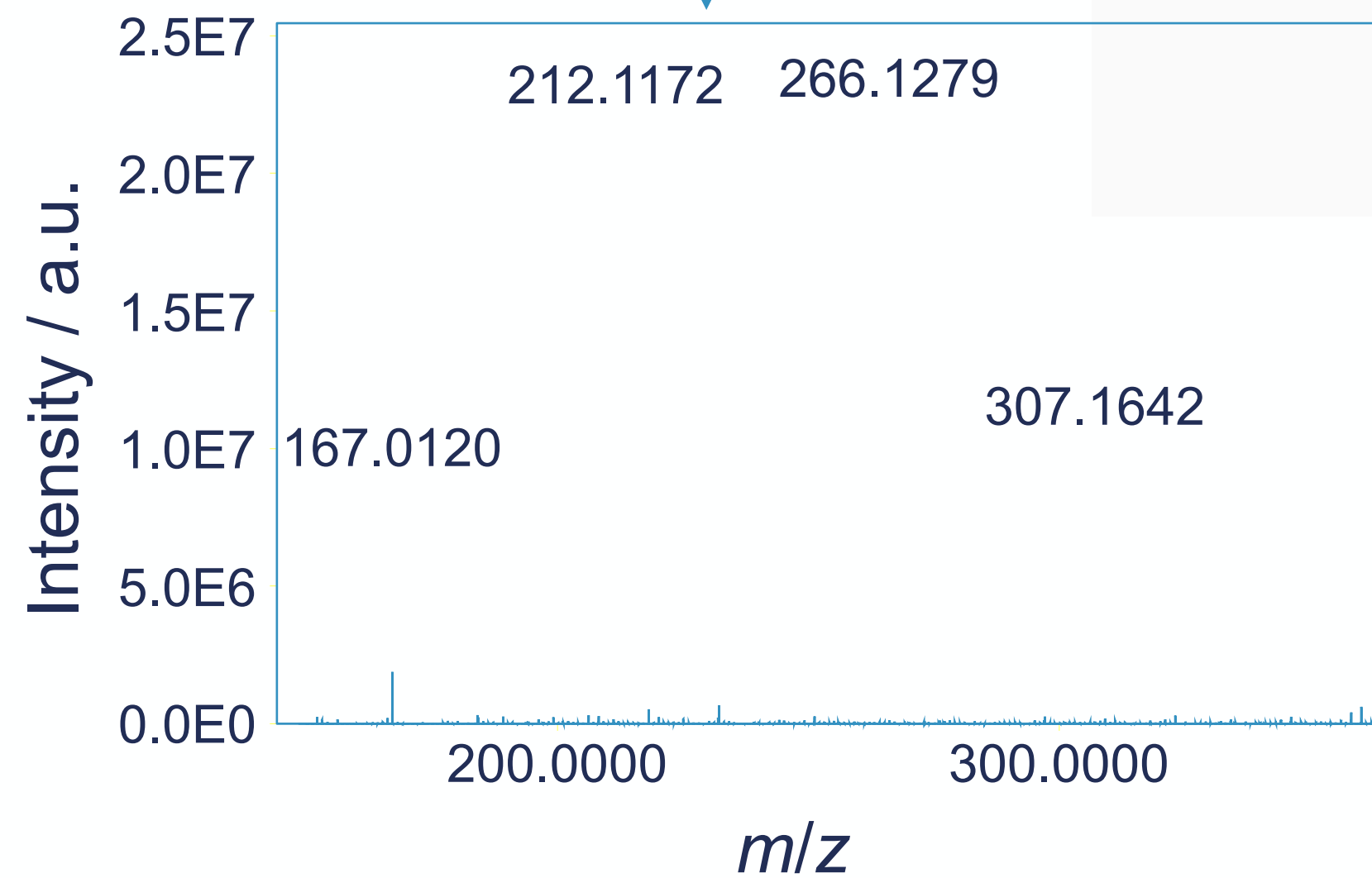
For enterprises

A Closer Look: LC-MS Datasets

- 3-dimensional data: retention time (RT), m/z , intensity

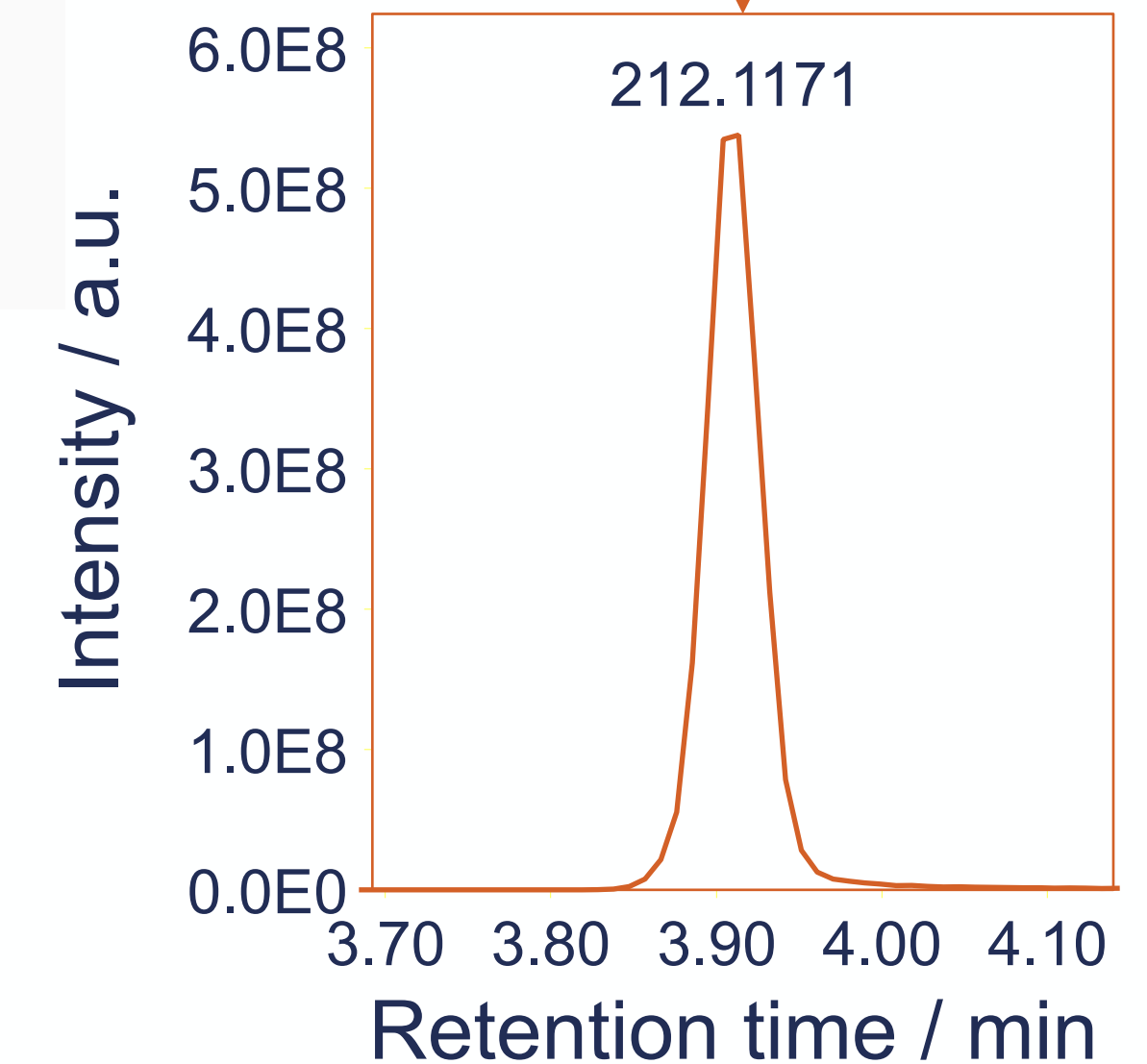
Spectrum

Intensity vs m/z at
one point in time

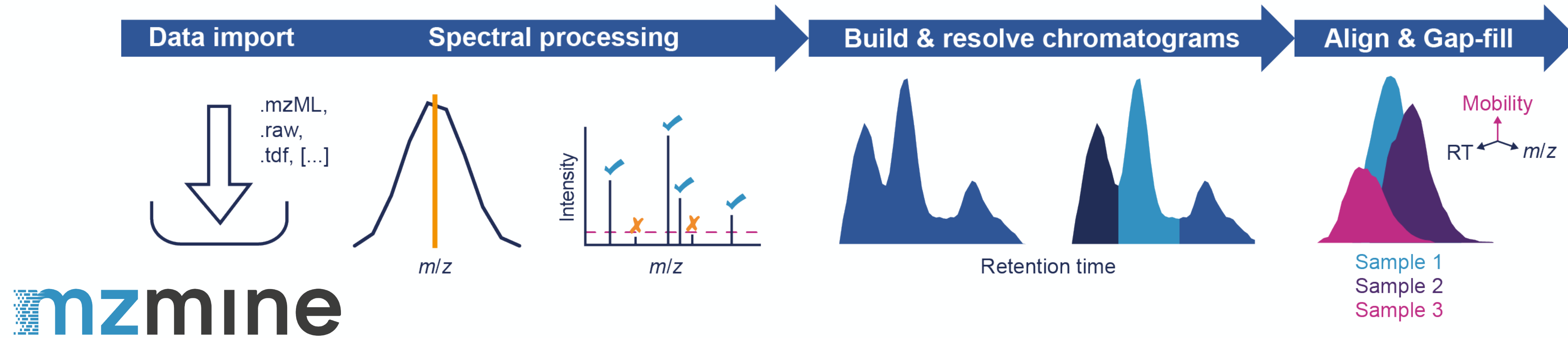


Extracted ion chromatogram (EIC)

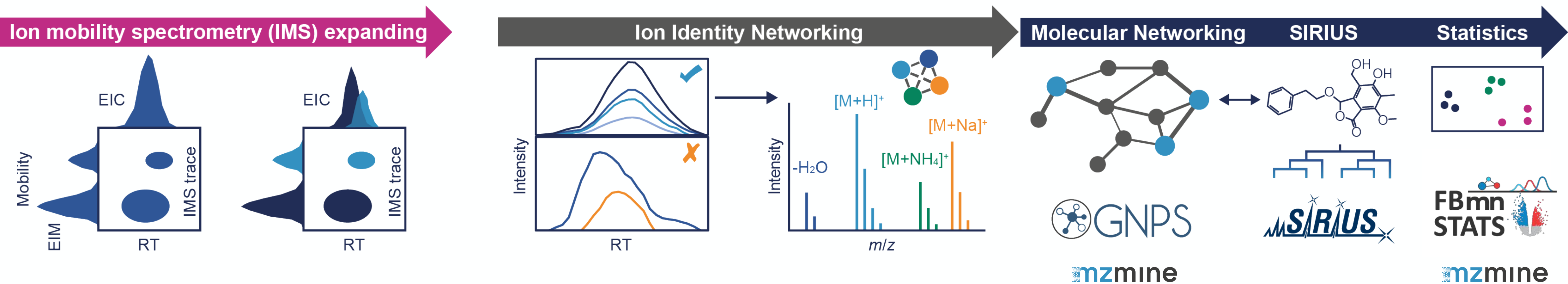
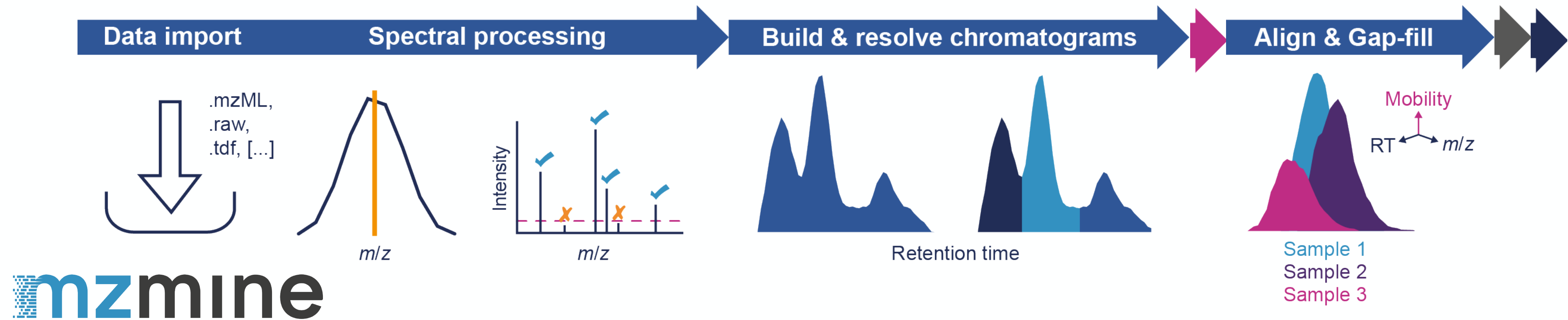
Intensity vs retention time
for one m/z



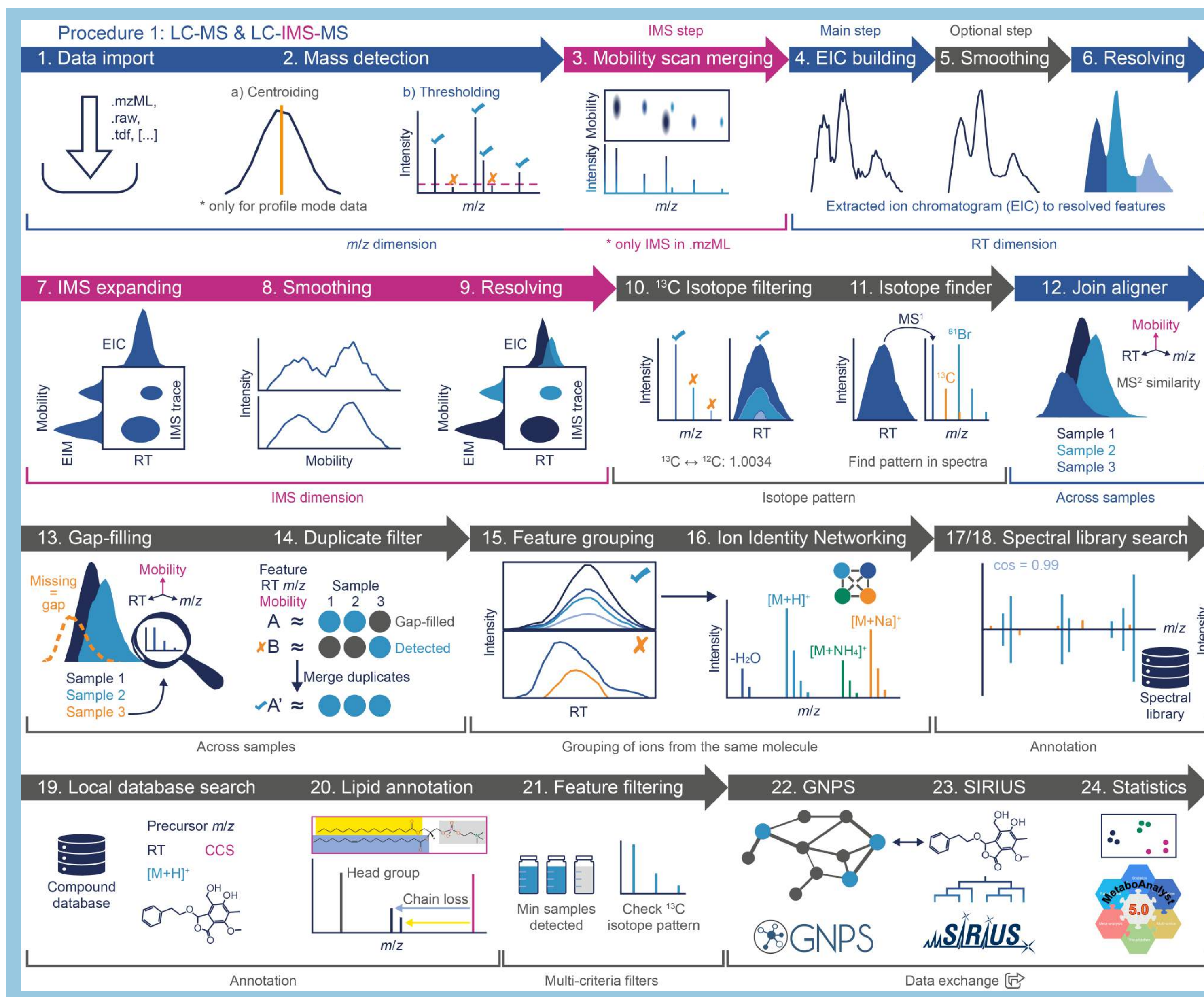
MS Feature Detection



MS Feature Detection



Breaking Down the Complexity – Workflow Presets

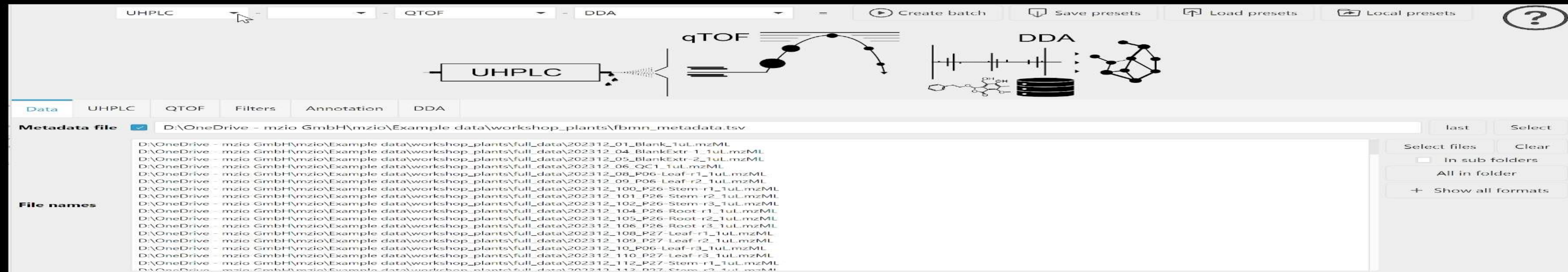


mzwizard

Preconfigured opinionated workflows

Reducing number of parameters

Resembling the instrumental setup



1. Sample introduction

- LC, GC, MALDI, flow inject

2. Ion Mobility:

- TIMS, TWIMS, DTIMS, IMS

3. Mass Spectrometer:

- Orbitrap, qTOF, FTICR, low res.

4. Workflows:

- DDA, DIA, library generation, imaging

Preconfigured opinionated workflows



Reducing number of parameters

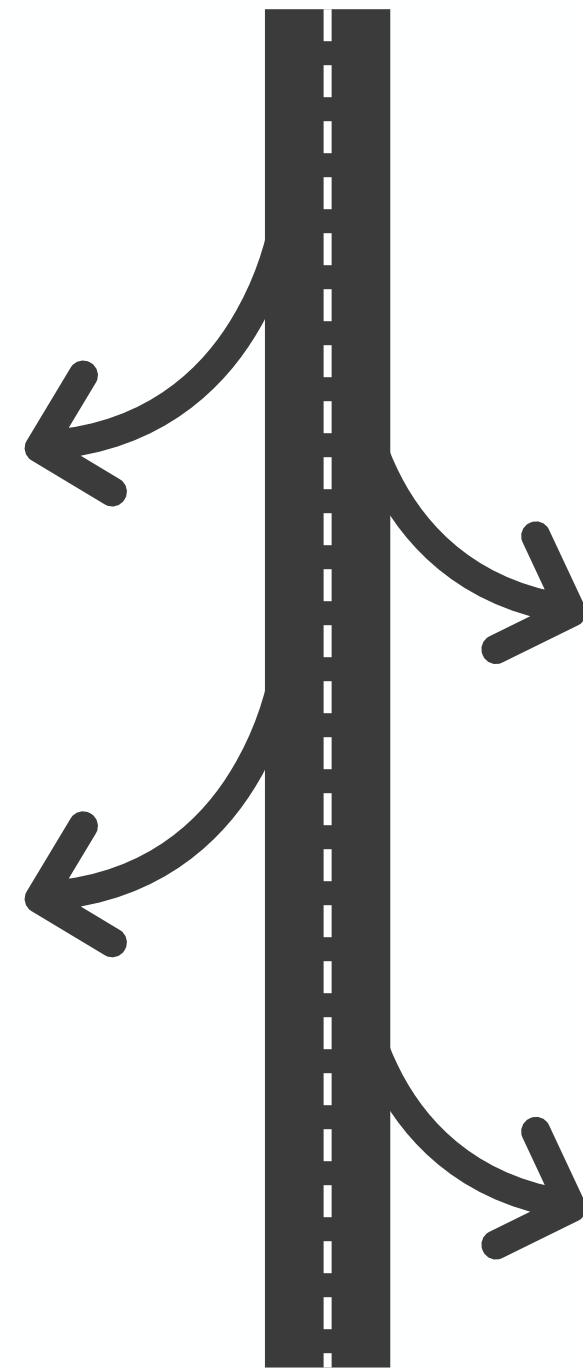
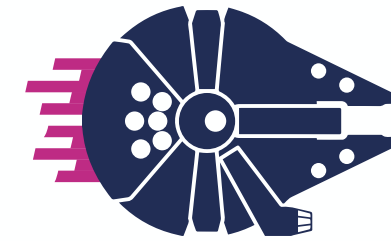


Resembling the instrumental setup

Workflow Styles



Personalized Workflow



- Flexibility
- Intermediate results
- Modify step order
- Add steps

Linear Workflow

- Feels like on tracks
- Only one exit
- No intermediate results

Raw Data Overview



mzmine 4.8.68

ProjectRaw data methodsFeature detectionFeature list methodsVisualizationmzwizardToolsWindowsUsersHelp

MS data filesFeature listsLibraries

171103_PMA_TK_M1_01.mzML

171103_PMA_TK_M1_02.mzML

171103_PMA_TK_M1_03.mzML

171103_PMA_TK_M1_04.mzML

171103_PMA_TK_M1_05.mzML

171103_PMA_TK_M1_06.mzML

171103_PMA_TK_PA14_01.mzML

171103_PMA_TK_PA14_02.mzML

171103_PMA_TK_PA14_03.mzML

171103_PMA_TK_PA14_04.mzML

171103_PMA_TK_PA14_05.mzML

171103_PMA_TK_PA14_06.mzML

171103_PMA_TK_QC_03.mzML

171103_PMA_TK_QC_04.mzML

171103_PMA_TK_QC_05.mzML

171103_PMA_TK_QC_06.mzML

171103_PMA_TK_QC_07.mzML

171103_PMA_TK_QC_08.mzML

171103_PMA_TK_media_02.mzML

171103_PMA_TK_media_03.mzML

171103_PMA_TK_media_04.mzML

171103_PMA_TK_media_05.mzML

171103_PMA_TK_media_06.mzML

Raw data overview171103_PMA_TK_QC_03.mzML

166.0863

225.0666

325.0673

522.3636

668.4219

550.3937

184.9857

Base peak intensity

Retention time

550.3938

171103_PMA_TK_QC_03.mzML

Chromatogram parameters

171103_PMA_TK_QC_03.mzML

S...	P...	RT	Ba...	BP...	...	Precur...	Fragm...	m/z range	Scan...	...	In...	Scan defin...
30...	⊗	5.80	550...	2.5E8	1			120.0000-...	CENT...	+	2.29	FTMS + c ES...
30...	⊗	5.80	153...	8.1E4	2	553.4018	HCD, 30.0	50.0000-5...	CENT...	+	63....	FTMS + c ES...
30...	⊗	5.81	153...	1.2E5	2	404.3358	HCD, 30.0	50.0000-4...	CENT...	+	45....	FTMS + c ES...

Total number of scans:

Total scans (4606) MS1 level (1...

Retention time range:

MS1 level 0.00-9.00 [min] MS2 ...

m/z range:

MS1 level 119.9984-1499.4325 ...

Maximum TIC:

2.2E9

Spectrum parameters

Intensity

550.3938

m/z

Scan #3032

[7:15:55 PM]: Processing of task Calculating 1 base peak chromatogram(s) in 1 files. done, status FINISHED

0 tasks1.1/2.4 GB used

Goal: Feature Table



mzmine 4.9.0

ProjectRaw data methodsFeature detectionFeature list methodsVisualizationmzwizardToolsOther detectorsMRMWindowsUsersWorkspacesHelp

MS data filesFeature lists

Welcome to mzmine xFeature TableAligned feature list peak gaps dup co...

Aligned feature listsAligned feature list peak gap

Aligned feature list peak gaps dup corr PEARSON r greq 0.85 dp greq 5

ID	m/z	RT	Height	Shapes	Height box plot	Preferred Annotation	AQS	Formula	Adduct	Structure
1592	229.1545	0.81	4.4E9			Spectral Match to Pro-Ile from NIST14 (0.848)		C11H20N2O3	[M+H]+	
1825	231.1702	0.93	4.2E9			Spectral Match to Val-Leu from NIST14 (0.932)		C11H22N2O3	[M+H]+	
1212	166.0862	0.70	3.7E9			DL-PHENYLALANINE CollisionEnergy:102040 (0.995)		C9H11NO2	[M+H]+	
2289	263.1389	1.06	3.7E9			Spectral Match to Pro-Phe from NIST14 (0.871)		C14H18N2O3	[M+H]+	
5487	325.0677	2.94	3.4E9			Pyochelin (0.955)		C14H16N2O3S2	[M+H]+	
1684	231.1702	0.85	3.1E9			Spectral Match to Leu-Val from NIST14 (0.885)		C11H22N2O3	[M+H]+	

Q ID= 1,5-6 m/z= 120.0 - 1430.9 RT= 0.3 - 7.9 Formula >= >

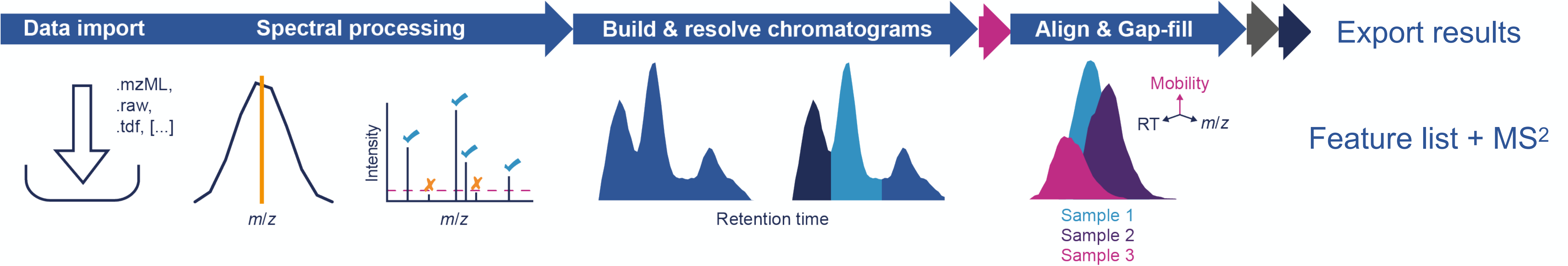
[7:18:52 PM]: Processing of task Computing PCA dataset for Aligned feature list peak gaps dup corr PEARSON r greq 0.85 dp greq 5 done, status FINISHED2 tasks (0 %)0.9/1.5 GB used51

Goal: Feature Table



Aligned feature list 13C peak gaps dup corr PEARSON r greq 0.85 dp greq 5										
m/z	RT	Area	Shapes	Height box plot	Ion identity	Preferred Annotation				
				condition	Ion identity	Preferred Annotation	AQS	Formula	Adduct	Structure
295.1651	1.38	2.8E6			[M+H] ⁺	Spectral Match to Tyr-Leu from NIST14 (0.937)		C15H22N2O4	[M+H] ⁺	
325.0677	2.94	1.3E8			[M+H] ⁺	Pyochelin (0.954)		C14H16N2O3S2	[M+H] ⁺	
295.1647	1.06	3.2E6			[M+H] ⁺	Spectral Match to Tyr-Leu from NIST14 (0.935)		C15H22N2O4	[M+H] ⁺	
293.1130	0.97	5.6E5			[M+H] ⁺	Spectral Match to PyroGlu-Tyr from NIST14 (0.858)		C14H16N2O5	[M+H] ⁺	
295.1285	0.73	2.1E6			[M+H] ⁺	Spectral Match to Phe-Glu from NIST14 (0.890)		C14H18N2O5	[M+H] ⁺	
309.2280	2.16	8.8E4			[M+H] ⁺	Spectral Match to Pentapropylene glycol from NIST14 (0.932)		C15H32O6	[M+H] ⁺	

MS Feature Detection



Aligned feature list 13C peak gaps dup corr PEARSON r greq 0.85 dp greq 5										
m/z	RT	Area	Shapes	Height box plot	Ion identity	Preferred Annotation				
				condition	Ion identity	Preferred Annotation	AQS	Formula	Adduct	Structure
295.1651	1.38	2.8E6			[M+H] ⁺	Spectral Match to Tyr-Leu from NIST14 (0.937)		C15H22N2O4	[M+H] ⁺	
325.0677	2.94	1.3E8			[M+H] ⁺	Pyochelin (0.954)		C14H16N2O3S2	[M+H] ⁺	
295.1647	1.06	3.2E6				Spectral Match to Tyr-Leu from NIST14 (0.935)		C15H22N2O4	[M+H] ⁺	

Performance

Memory

42.0 GB RAM

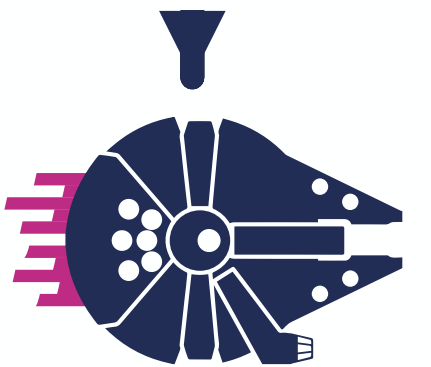
3.9 GB RAM



Throughput (speed)

3.5 samples/min

71.6 samples/min



11x improved memory use

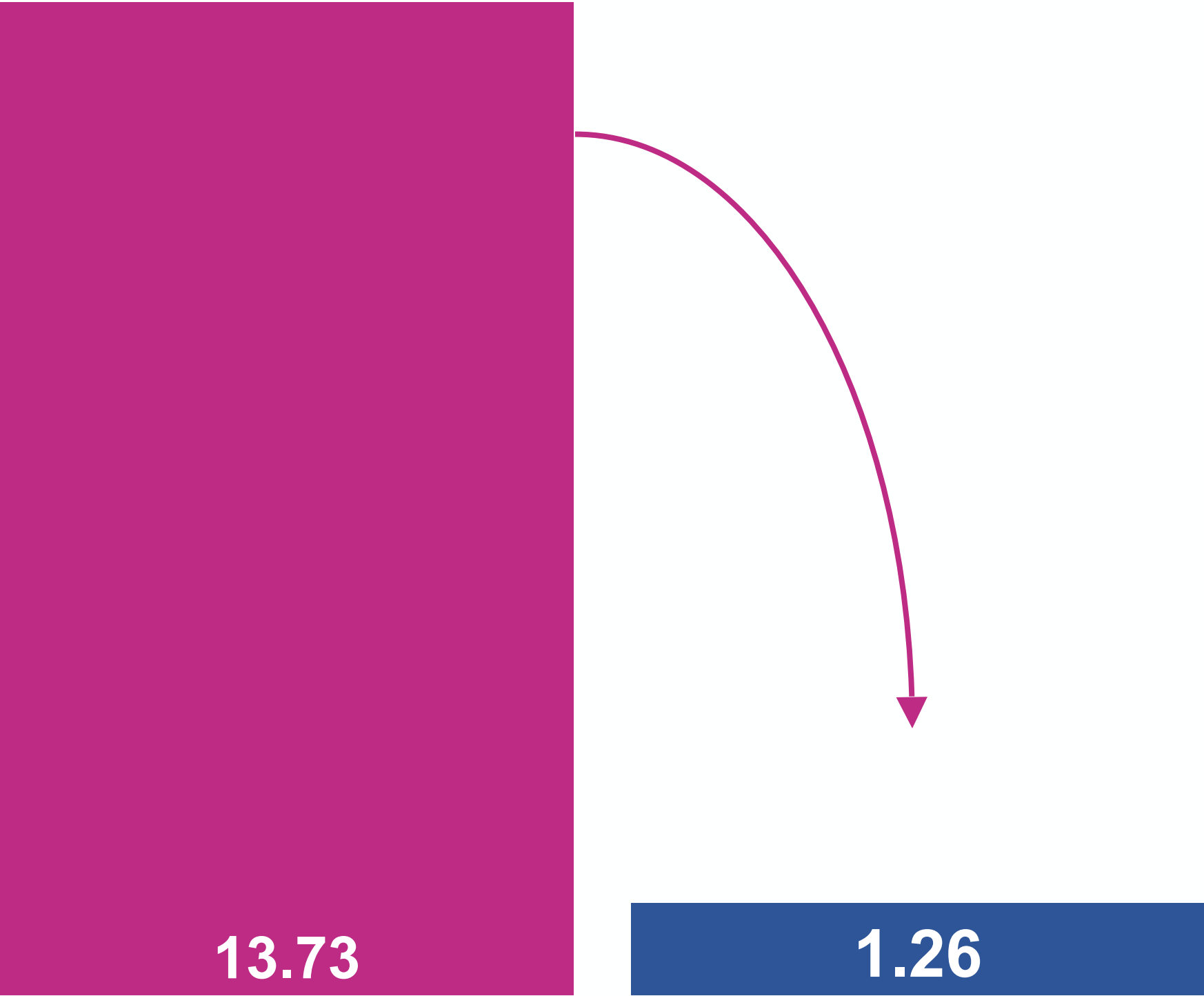
20x speed up

500 LC-MS² samples DOM (dissolved organic matter)
Intel Core Ultra 9 185H, 16 cores, 22 threads, 64 GB RAM, laptop

Scaling MS Data Processing – RAM Memory

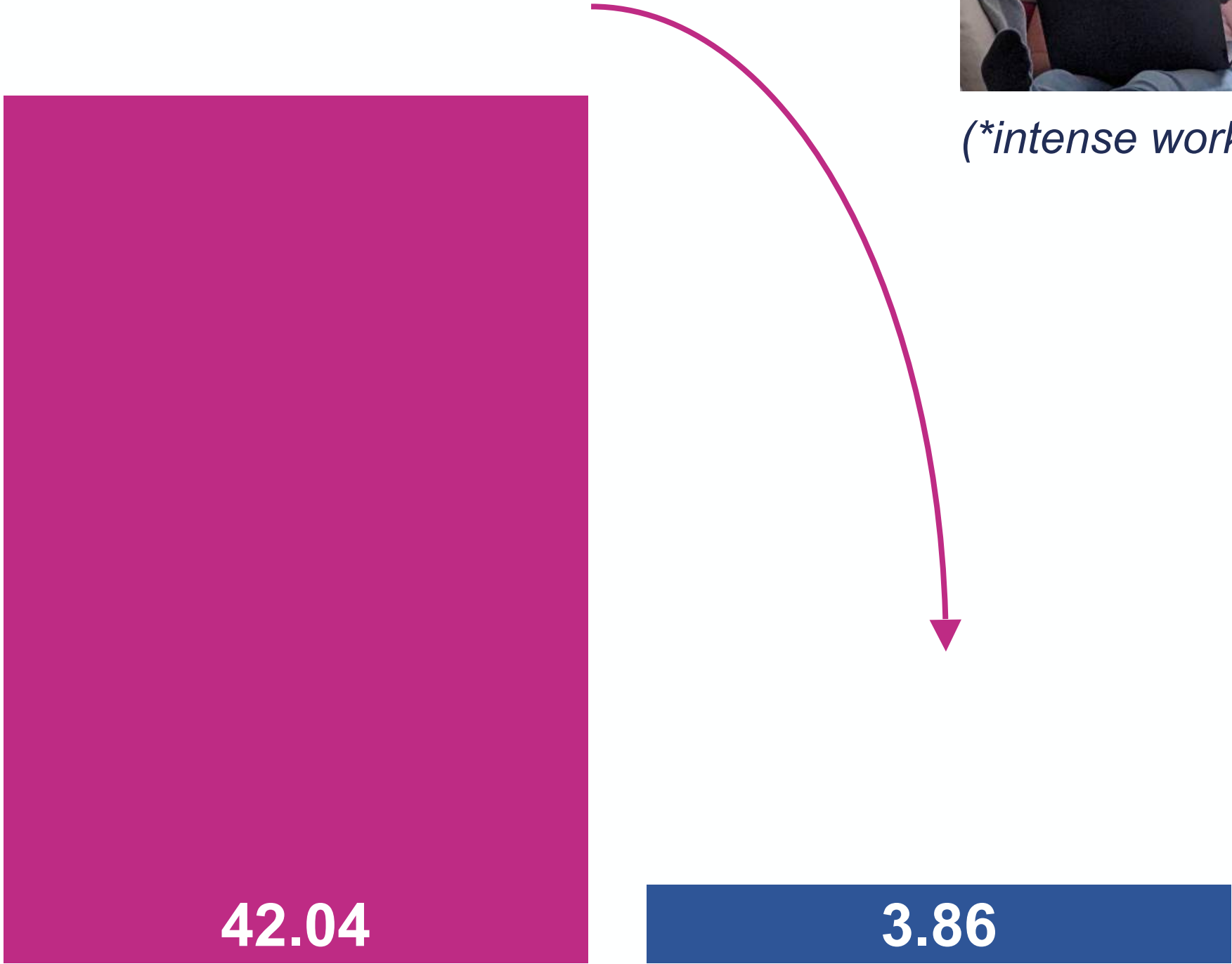


(*intense working noises*)



MZmine 3 → mzmine

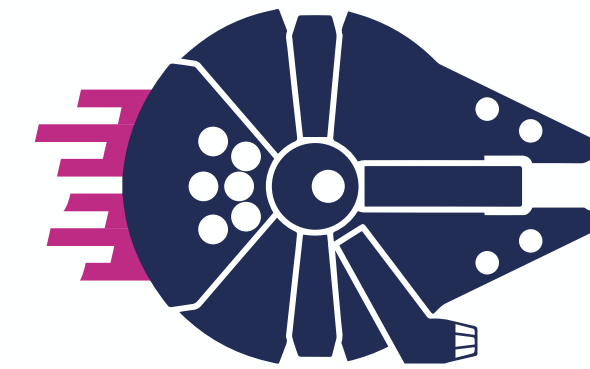
45 samples: LC-TIMS-MS



MZmine 3 → mzmine

500 DOM samples: LC-MS (Orbitrap)

Scaling MS Data Processing – Throughput



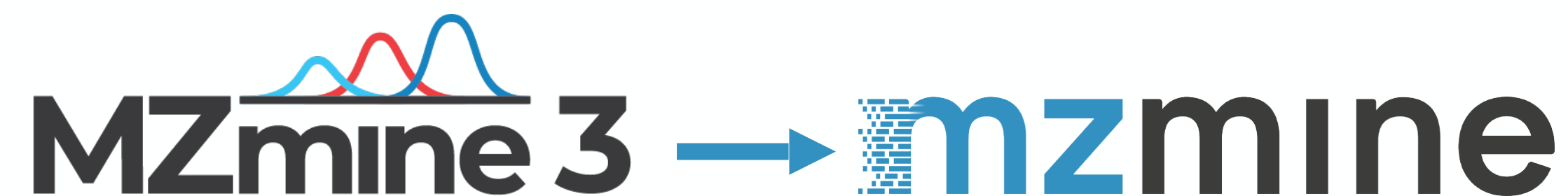
* Feature detection until library matching

No library matching

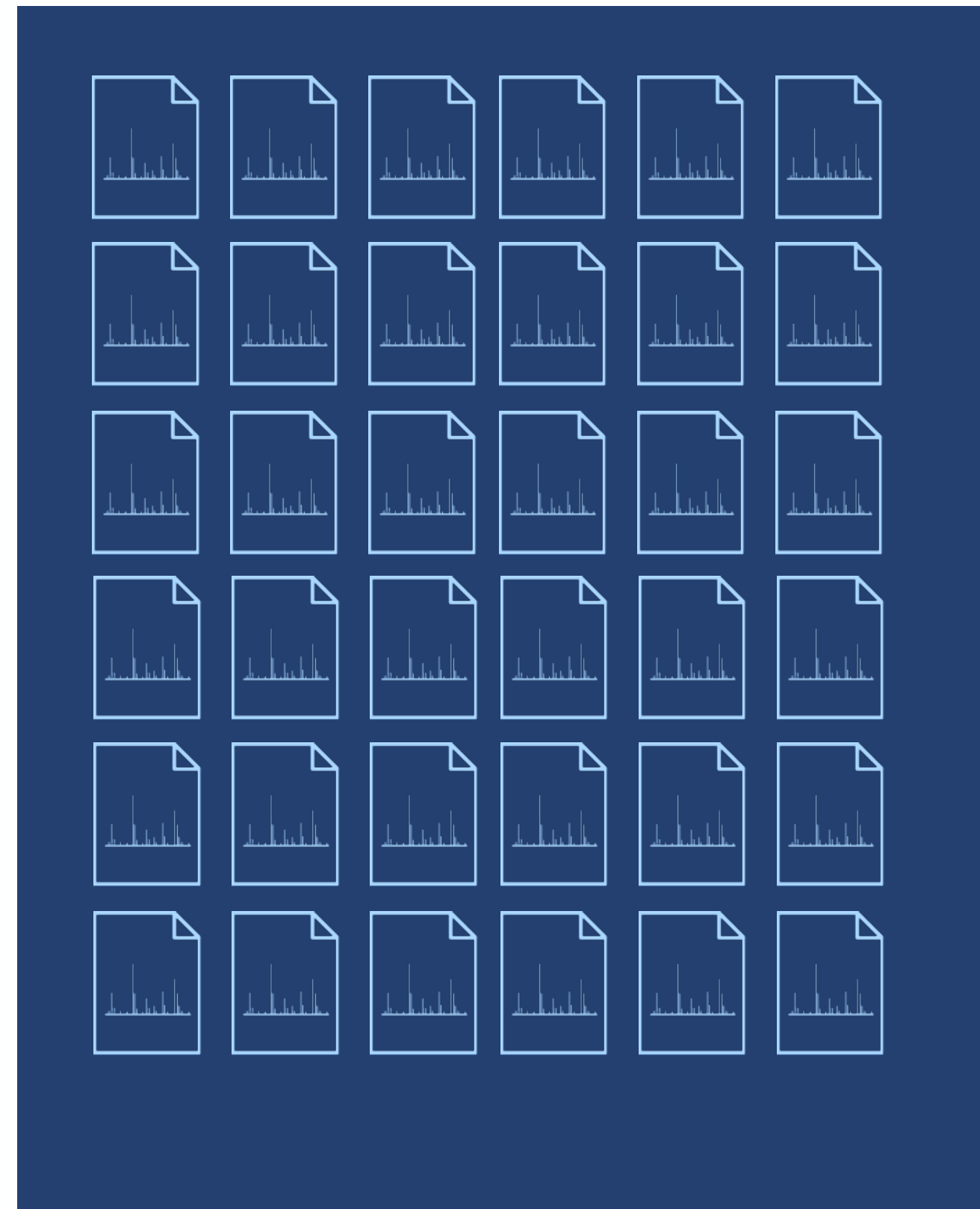
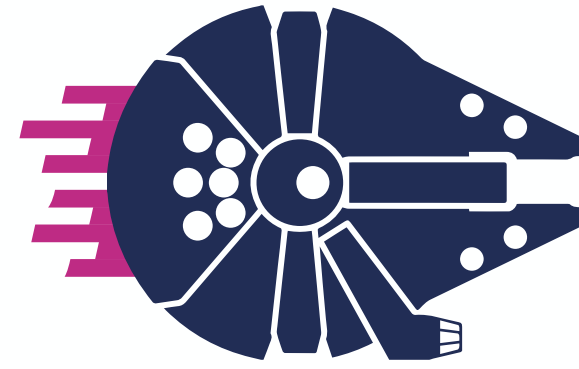


(*intense working noises*)

250 LC-MS² samples
DOM (dissolved organic matter)
Laptop, 16 cores, 22 threads
64 GB memory

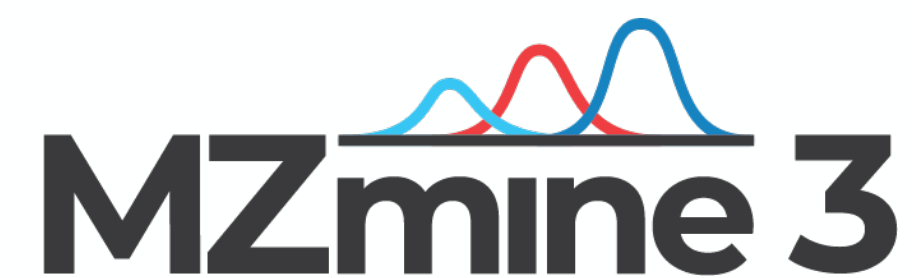


Scaling MS Data Processing & Molecular Networking



*(*intense working noises*)*

250 LC-MS² samples
DOM (dissolved organic matter)
Laptop, 16 cores, 22 threads
64 GB memory

 → 

mzwizard – Easy Batch Setup



Project

Raw data methods

Feature detection

Feature list methods

Visualization

Tools

Processing wizard

Windows

Help

MS data files

Feature lists

Libraries

Processing Wizard X

skin_04_600_UB.mzML

skin_03_480_UB.mzML

skin_05_720_UB.mzML

skin_01_240_UB.mzML

bld_plt1_01_120_1.mzML

bld_plt2_08_480_1.mzML

bld_plt1_04_480_1.mzML

bld_plt2_03_720_1.mzML

UHPLC

Orbitrap

DDA

=

Create batch

Save presets

Load presets

Local presets

UHPLC

Orbitrap

DDA

Orbitrap

DDA

Data

UHPLC

Orbitrap

Filters

Annotation

DDA

D:\Data\8skinbld_workshop\bld_plt1_01_120_1.mzML

D:\Data\8skinbld_workshop\bld_plt1_04_480_1.mzML

D:\Data\8skinbld_workshop\bld_plt2_03_720_1.mzML

D:\Data\8skinbld_workshop\bld_plt2_08_480_1.mzML

D:\Data\8skinbld_workshop\skin_01_240_UB.mzML

D:\Data\8skinbld_workshop\skin_03_480_UB.mzML

D:\Data\8skinbld_workshop\skin_04_600_UB.mzML

D:\Data\8skinbld_workshop\skin_05_720_UB.mzML

File names

Select files

Clear

☐ In sub folders

From folder

All *.mzML

All *.mzXML

All *.imzML

All *.d

All *.raw

All *.raw

All *.mzData

All *.aird

All *.cdf

All *.zip

mzwizard – Presets



- Full or partial workflow settings
- Save presets for your 10 min RP-LC method, 15 min HILIC, ..., QExactive, timsTOF, ...
- Combine these presets

The screenshot displays the mzwizard software interface, specifically the 'Processing Wizard' tab. The top menu bar includes 'Project', 'Raw data methods', 'Feature detection', 'Feature list methods', 'Visualization', 'Tools', 'Processing wizard', 'Windows', and 'Help'. The 'Processing Wizard' tab is active, showing a workflow configuration: HPLC - [] - Orbitrap - DDA. Below this, a diagram illustrates the workflow: HPLC (High Performance Liquid Chromatography) leading to Orbitrap (mass filter) and DDA (Data-Dependent Acquisition) mass spectrometry. The 'Data' sub-tab is selected, showing a list of files in the 'File names' section:

- D:\Data\8skinbld_workshop\bld_plt1_01_120_1.mzML
- D:\Data\8skinbld_workshop\bld_plt1_04_480_1.mzML
- D:\Data\8skinbld_workshop\bld_plt2_03_720_1.mzML
- D:\Data\8skinbld_workshop\bld_plt2_08_480_1.mzML
- D:\Data\8skinbld_workshop\skin_01_240_UB.mzML
- D:\Data\8skinbld_workshop\skin_03_480_UB.mzML
- D:\Data\8skinbld_workshop\skin_04_600_UB.mzML
- D:\Data\8skinbld_workshop\skin_05_720_UB.mzML

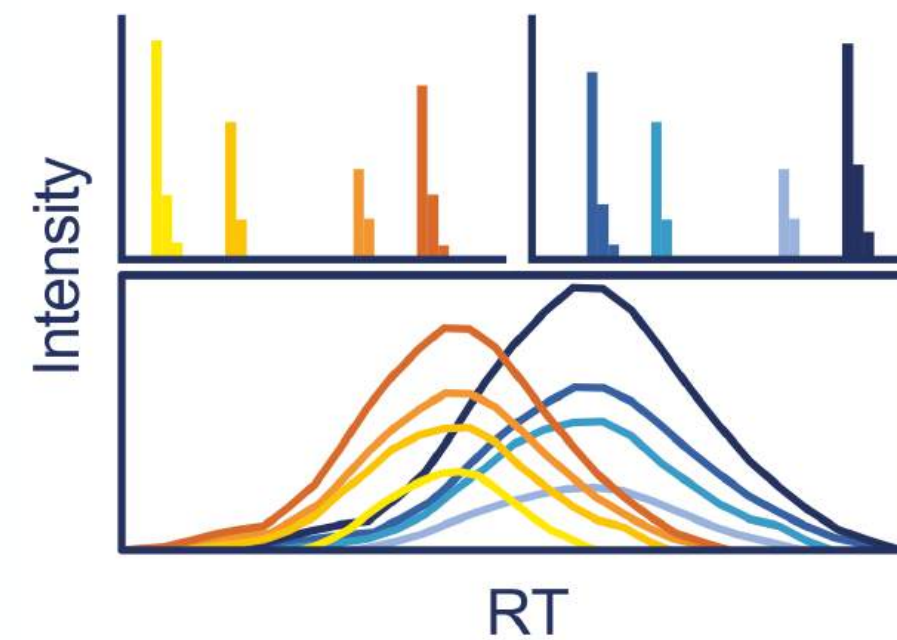
On the right side, there are buttons for 'Select files', 'Clear', and 'In sub folders'. Below these, a 'From folder' section lists various file formats:

- All *.mzML
- All *.mzXML
- All *.imzML
- All *.d
- All *.raw
- All *.raw
- All *.mzData
- All *.aird
- All *.cdf
- All *.zip

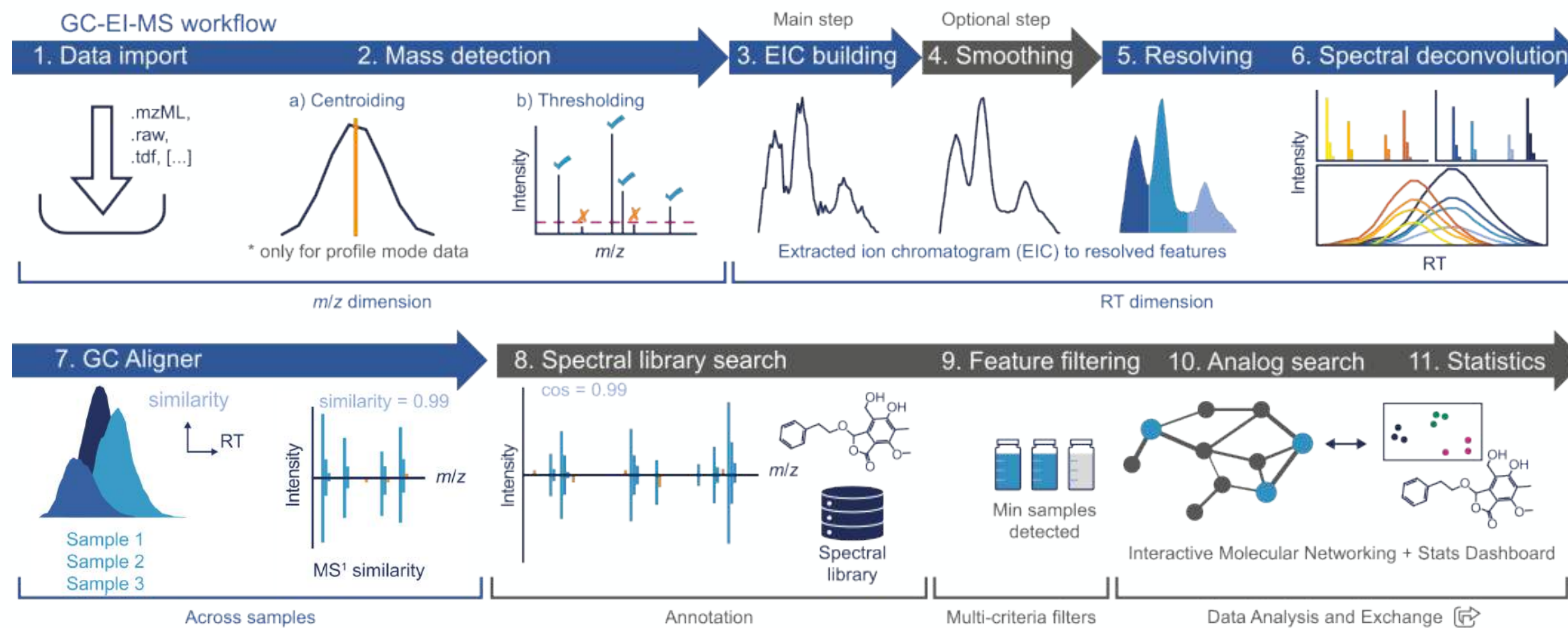
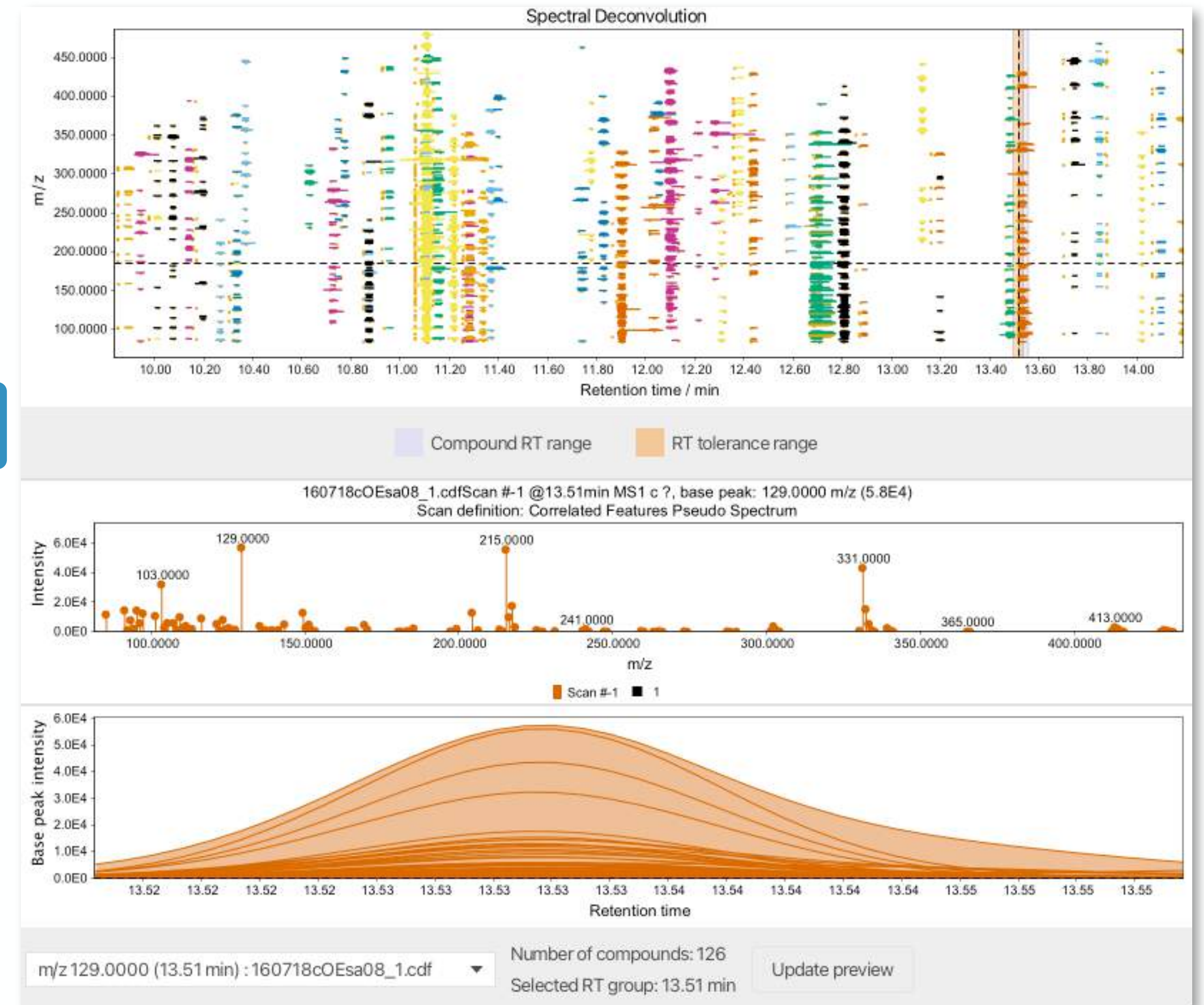
GC/EI-MS data processing workflow

GC/EI spectral deconvolution

- Group features by RT range
- Decide final group by feature shape correlation
- Most intense m/z as representative feature



Spectral deconvolution



MALDI imaging data processing workflow

Image co-localisation

- Co-localize m/z distributions
- Displays co-localised images in parallel
- Co-localisation based Molecular Networking

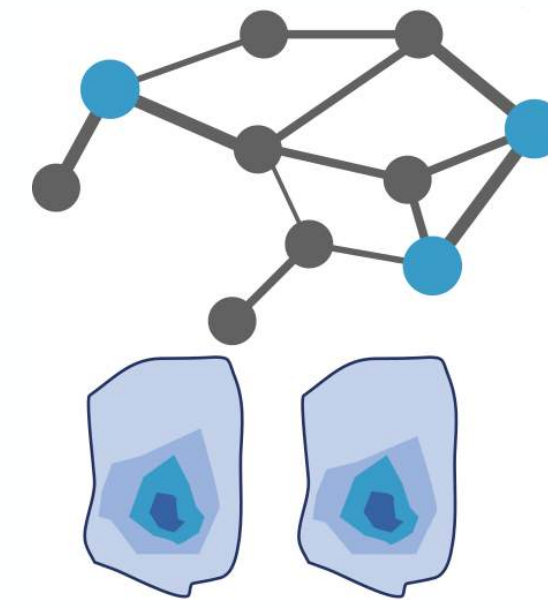
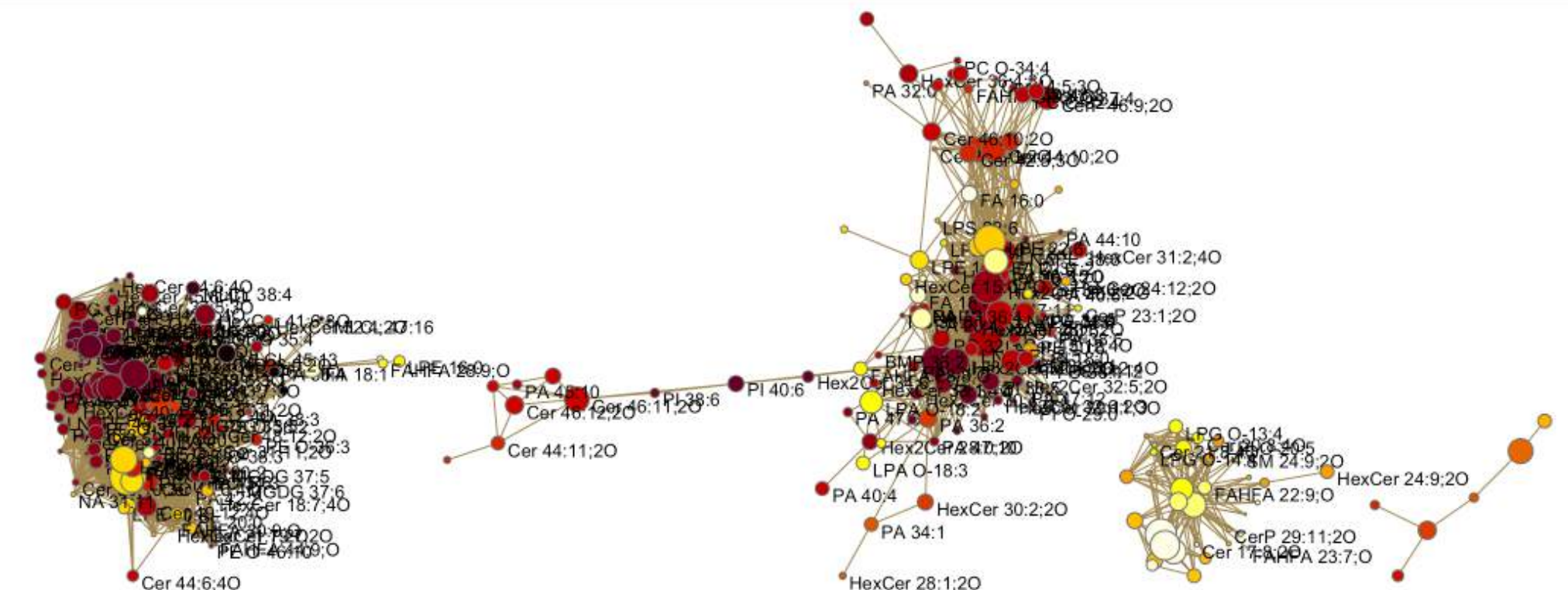
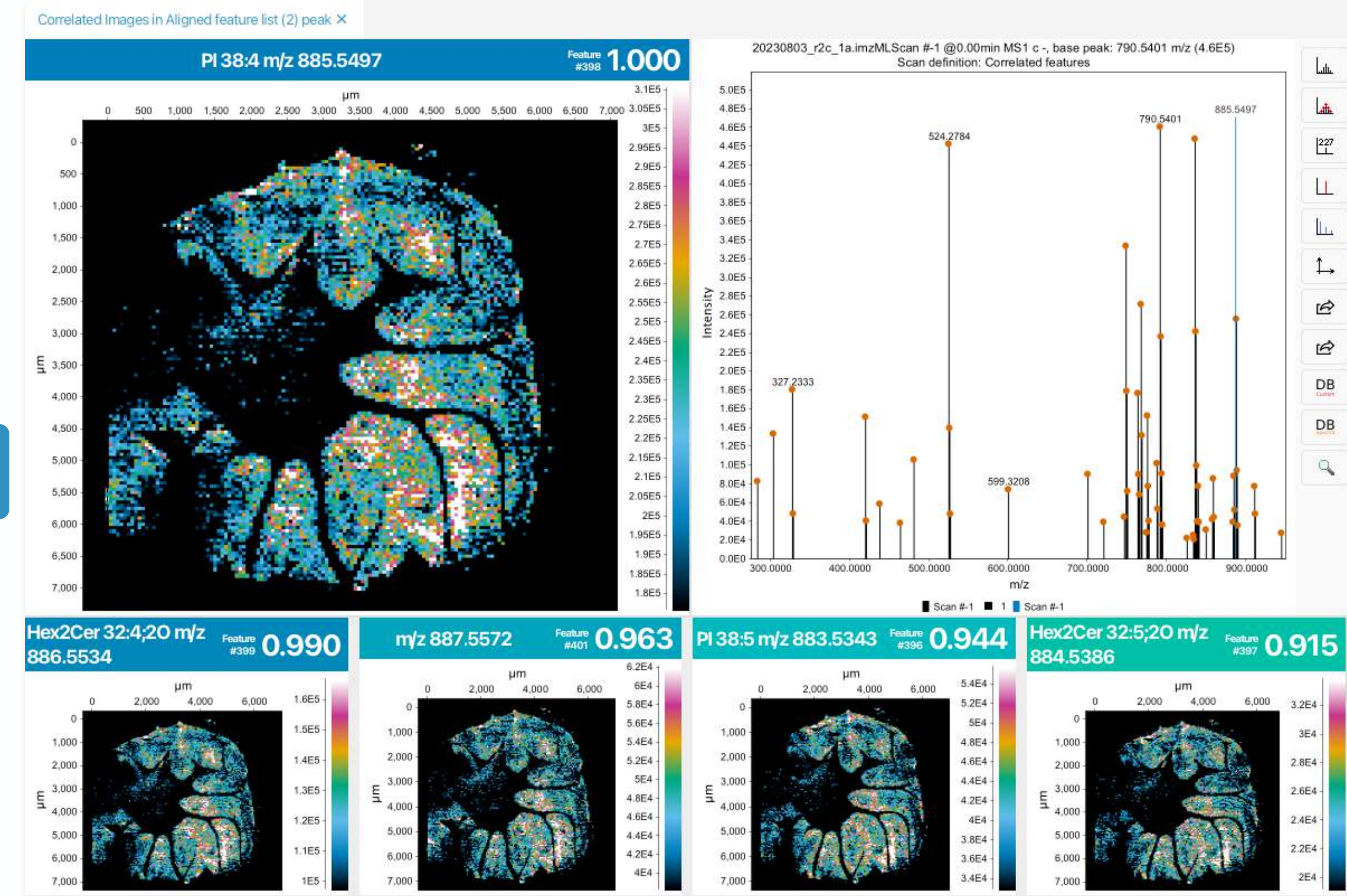
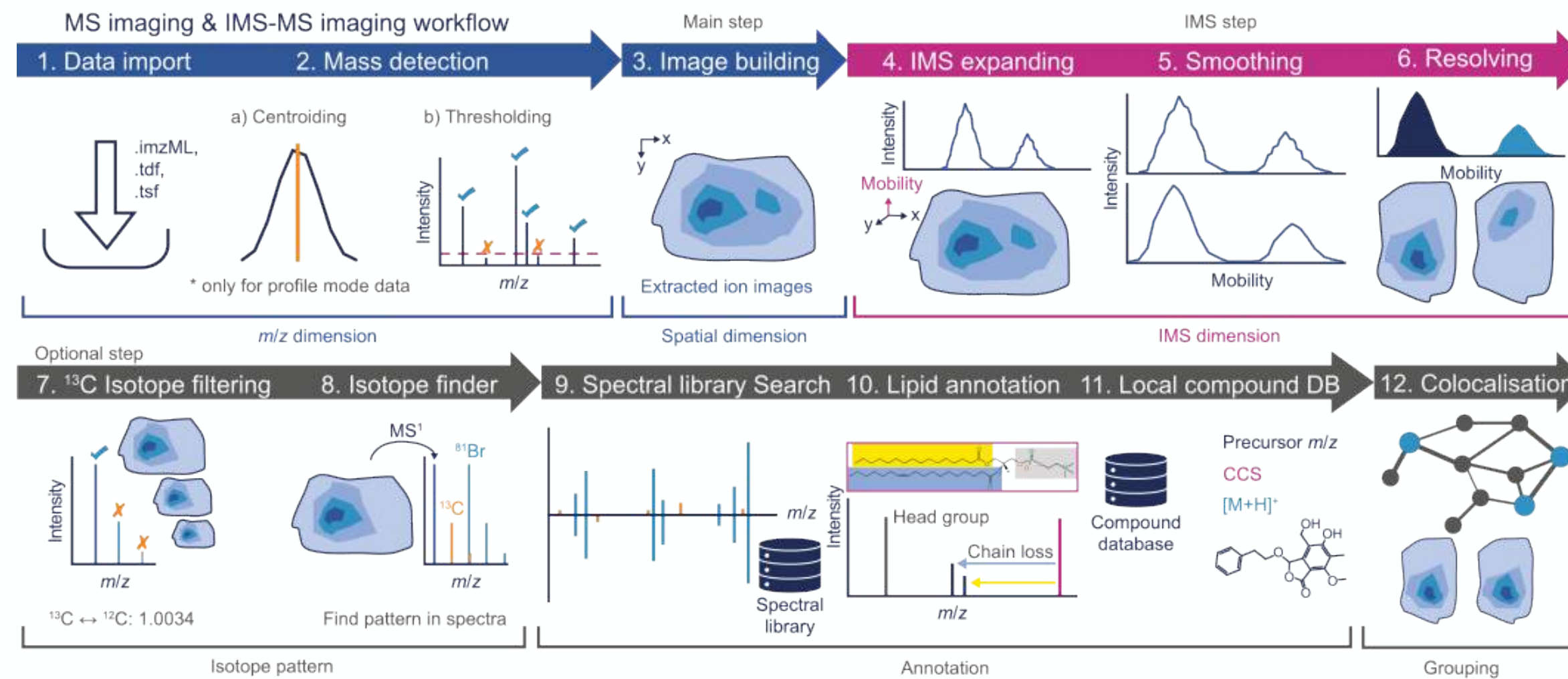
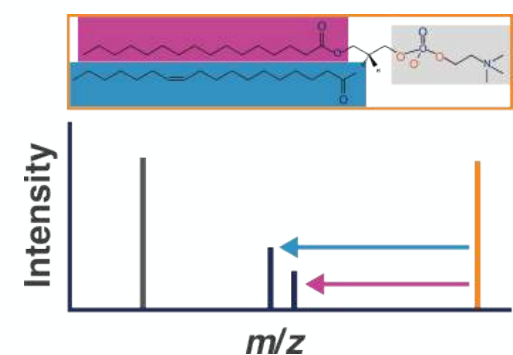


Image Co-localisation





Rule-based lipid annotation

Pre-defined lipid classes and fragmentation rules

LIPID MAPS hierarchy

All

Clear

▼ ☒ Glycerophospholipids

▶ ☐ Phosphatidylcholine

▶ ☐ Glycerophosphoethanolamines

▶ ☐ Glycerophosphoserines

▶ ☐ Glycerophosphoglycerols

▼ ☒ Glycerophosphoinositols

☒ PI Diacylglycerophosphoinositols

☐ PI Alkylacylglycerophosphoinositols

☐ PI Dialkylglycerophosphoinositols

☐ LPI Monoacylglycerophosphoinositols

☐ LPI Monoalkylglycerophosphoinositols

▶ ☐ Glycerophosphates

▶ ☐ Cardiolipin

▼ ☐ Sphingolipids

▶ ☐ Sphingoid bases

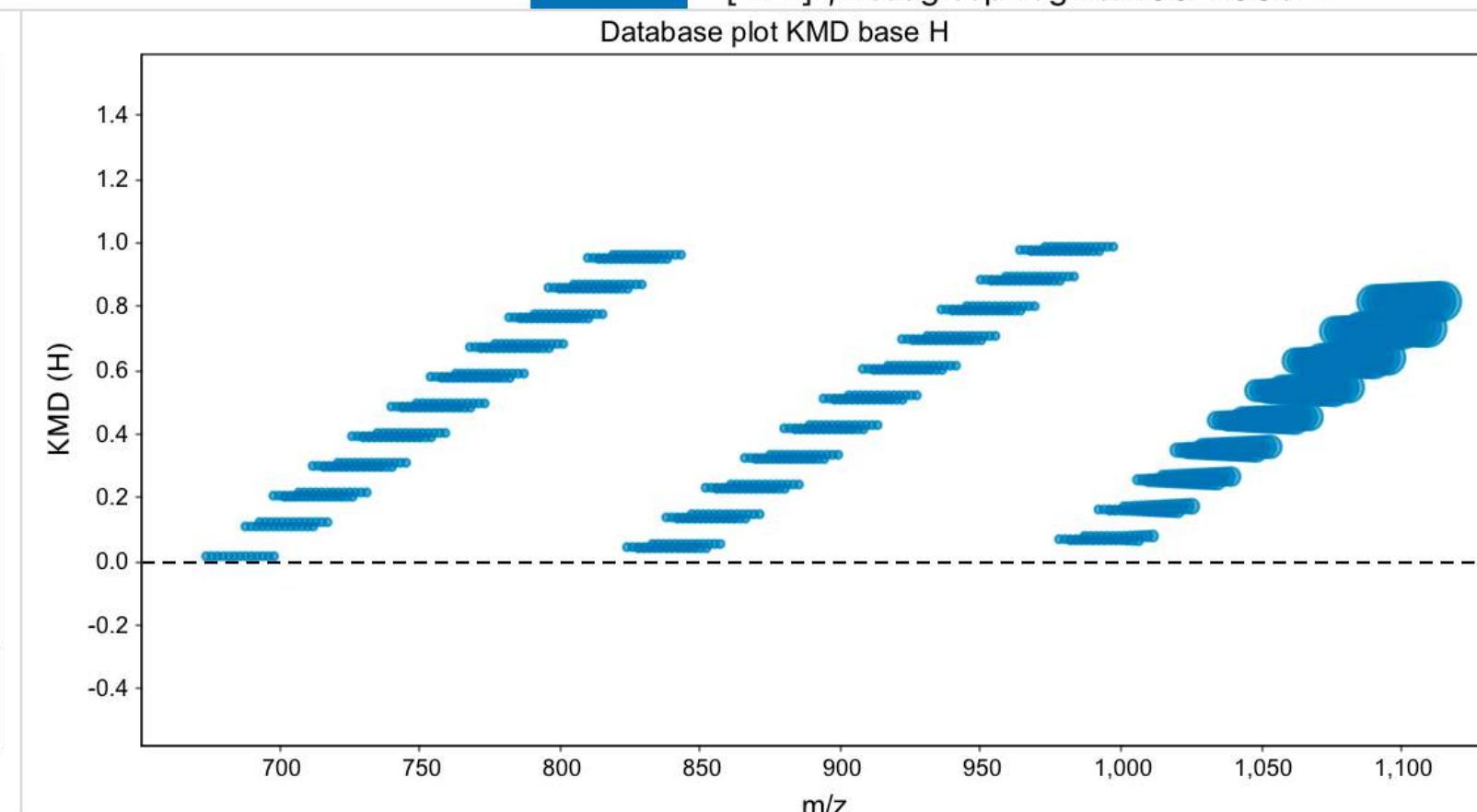
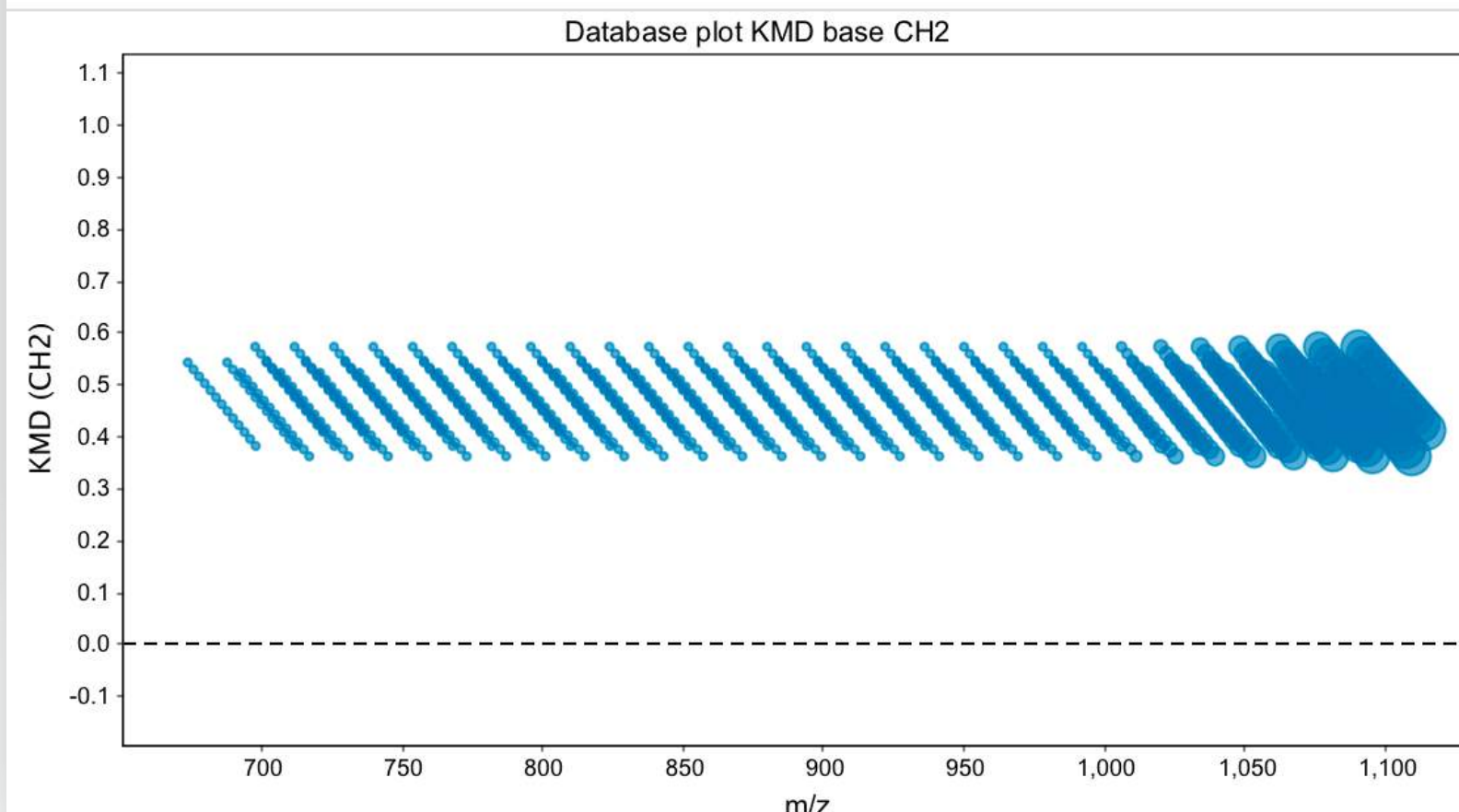
▶ ☐ Ceramides

▶ ☐ Neutral glycosphingolipids

In-silico lipid library

Status: No interference ☒ Possible interference ☐ Isomeric interference ☐

ID	Lipid class	Molecular formula	Abbreviation	Exact mass	Info	Sta...	MS/MS fragments
1	Diacylglycerophosphoinositols	C33H63O13P	PI 24:0	[M-H] ⁻ 697.3934 [M+Na] ⁺ 721.3898 [M+NH4] ⁺ 716.4345		☒	[M+NH4] ⁺ , Headgroup fragment n1 C6H16O9PN [M+Na] ⁺ , Headgroup fragment n1 C6H13O9PNa [M+Na] ⁺ , Headgroup fragment C6H13O9PNa+ [M-H] ⁻ , Headgroup fragment C9H16O10P- [M-H] ⁻ , Headgroup fragment C9H14O9P- [M-H] ⁻ , Headgroup fragment C6H12O9P- [M-H] ⁻ , Headgroup fragment C6H10O8P- [M-H] ⁻ , Headgroup fragment C6H8O7P- [M-H] ⁻ , Headgroup fragment C3H6O5P- [M-H] ⁻ , Headgroup fragment H2PO4- [M-H] ⁻ , Headgroup fragment PO3- [M-H] ⁻ , Headgroup fragment n1 C6H12O5 [M-H] ⁻ , Acylchain fragment [M-H] ⁻ , Acylchain fragment n1 [M-H] ⁻ , Acylchain minus formula fragment n1 H2O [M-H] ⁻ , Acylchain plus formula fragment n1 C6H12O5
2	Diacylglycerophosphoinositols	C33H61O13P	PI 24:1	[M-H] ⁻ 695.3777 [M+Na] ⁺ 719.3742 [M+NH4] ⁺ 714.4188		☐	[M+NH4] ⁺ , Headgroup fragment n1 C6H16O9PN [M+Na] ⁺ , Headgroup fragment n1 C6H13O9PNa [M+Na] ⁺ , Headgroup fragment C6H13O9PNa+ [M-H] ⁻ , Headgroup fragment C9H16O10P- [M-H] ⁻ , Headgroup fragment C9H14O9P- [M-H] ⁻ , Headgroup fragment C6H12O9P- [M-H] ⁻ , Headgroup fragment C6H10O8P-



Rule-based lipid annotation

Create custom lipid classes and fragmentation rules

Define lipid hierarchy, name and backbone formula

Lipid category

Glycerophospholipids

Lipid main class

Glycerophosphates

Custom lipid class name

4-Acyl Diphosphoryl Lipid A (1 sec)

Custom lipid class abbreviation

4-Acyl DPLA (1 sec)

Lipid backbone molecular formula

C12H24O15P2

Add...

Clear

Amid mono hydroxy chain

Amid mono hydroxy chain

Acyl mono hydroxy chain

Acyl chain

Add...

Import...

Export...

Remove selected

Clear

[M-H]⁻, Acylchain fragment n1

[M-H]⁻, Headgroup fragment n1 H3PO4

[M+H]⁺, Precursor

[M-H]⁻, Precursor

[M+Na]⁺, Precursor

[M+Na]⁺, Headgroup fragment n1 H3PO4

[M+H]⁺, Headgroup fragment n1 H3PO4

Preset lipid class

OK

Cancel

Lipid class summary

Species level:

4-Acyl DPLA (1 sec) 66:1;3O

Molecular species level:

4-Acyl DPLA (1 sec) 18:0_14:0;O_16:1;O_18:0;O

Molecular formula:

C78H148N2O22P2

Neutral mass:

1526.9999

Number of chains:

4

Ion notations:

[M+Na]⁺ 1549.9891

[M-H]⁻ 1525.9926

[M+H]⁺ 1528.0072

Simulate lipid fragments

	Chains	Carbons	DBEs
#1	16	1	
#2	18	0	
#3	14	0	
#4	18	0	

In-silico fragments (random intensities)

[M+Na]⁺

[M-H]⁻

[M+H]⁺

Add side chains

Add fragmentation rules

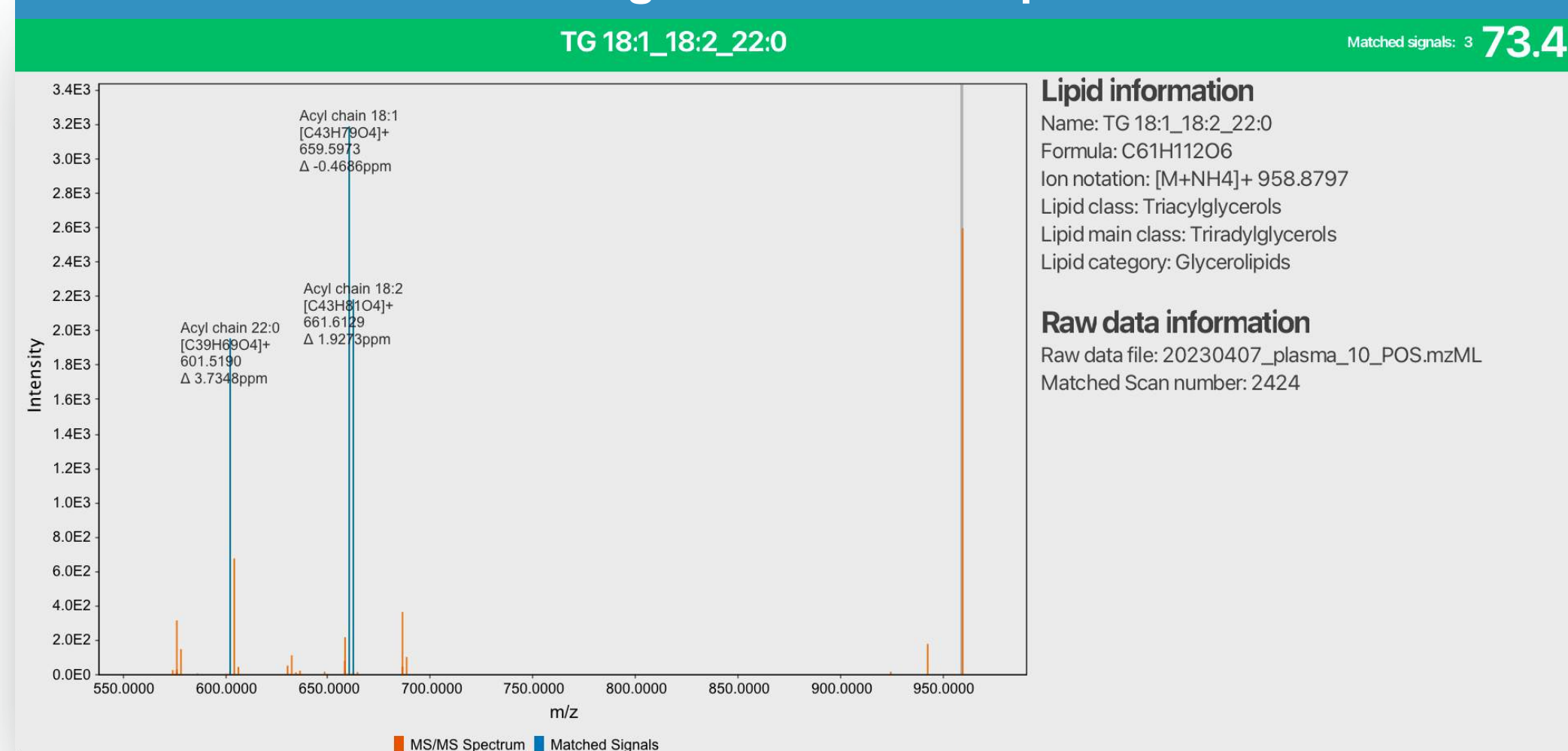
Simulate example lipid species

Investigate in-silico spectra

Lipid annotation quality control

Variety of tools to check and validate lipid annotations

Matched signals in MS/MS spectrum

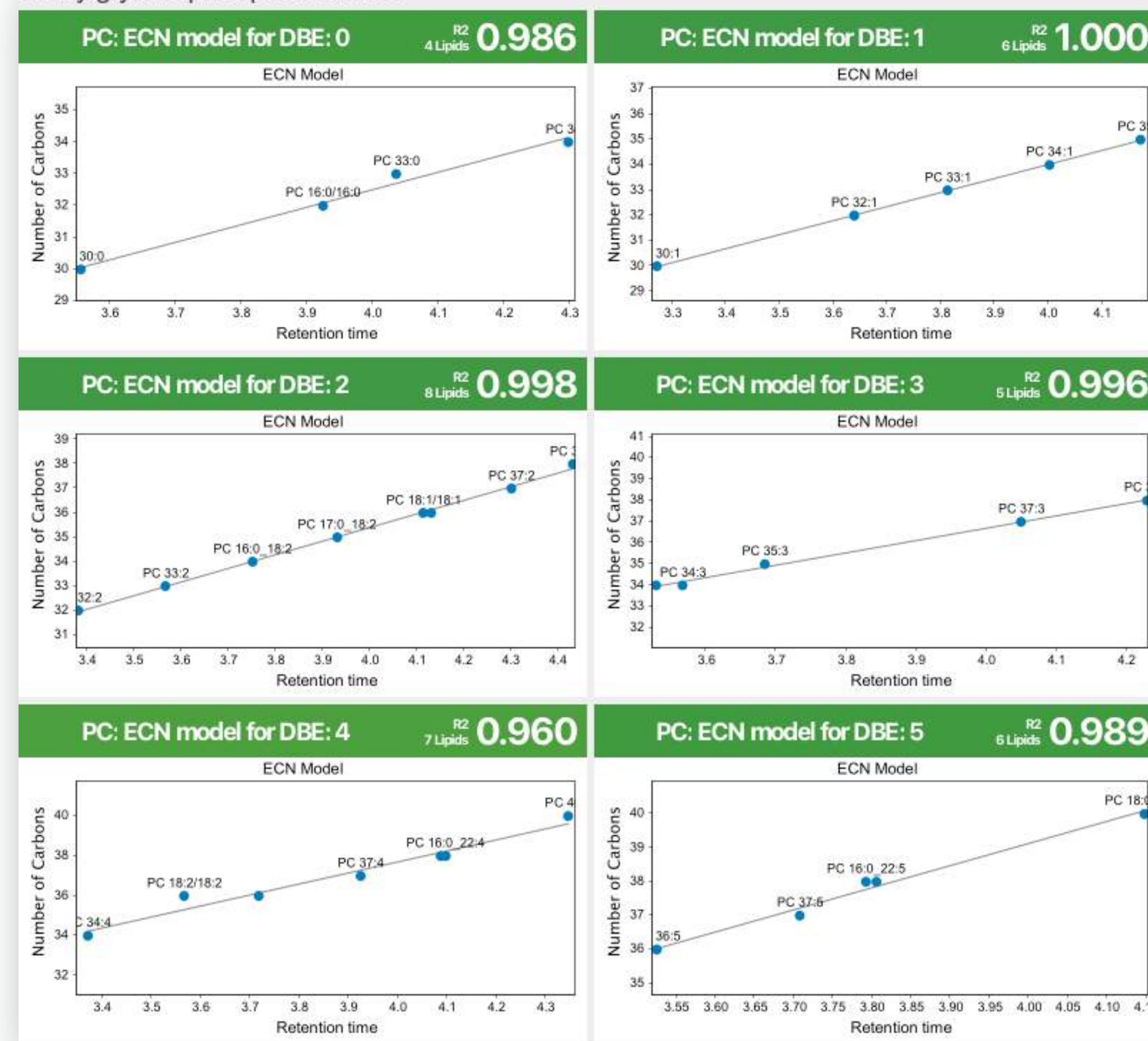


Kendrick mass plots



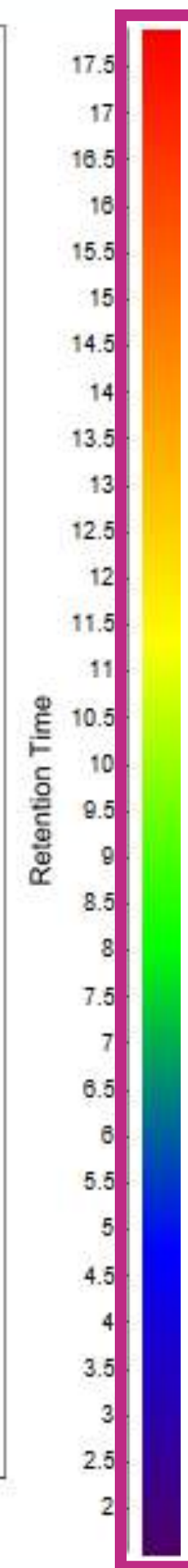
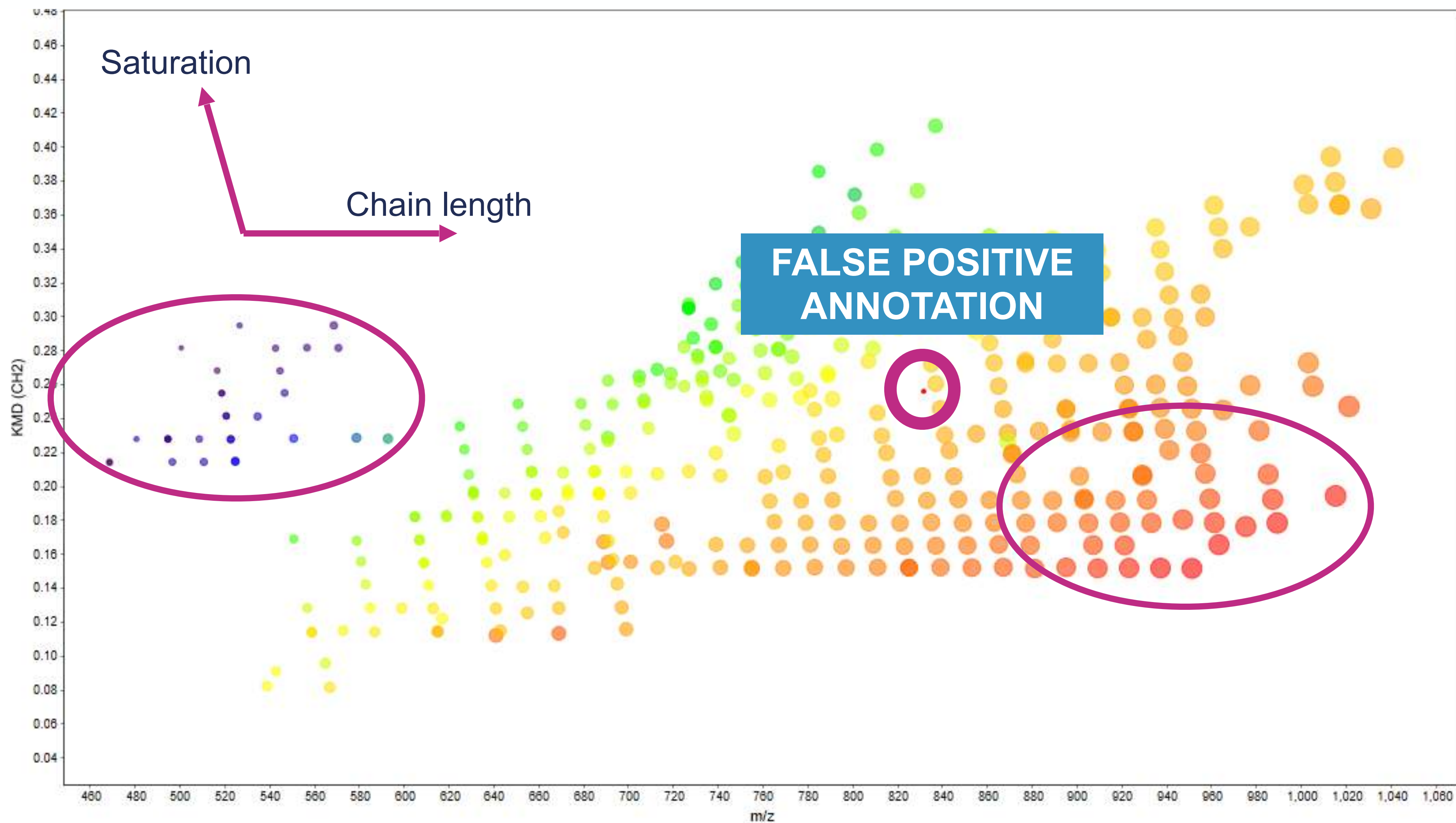
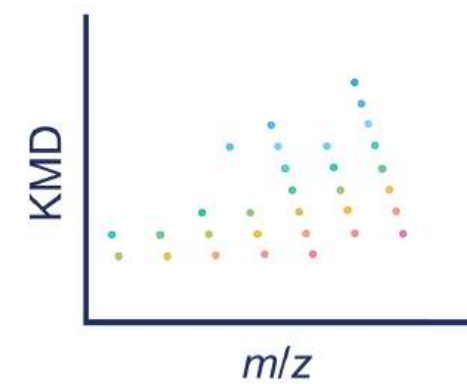
ECN plots

Diacylglycerophosphocholines



Kendrick Mass Defect plot

Lipids on horizontal lines



Color shows RT,
for elution pattern

Bubble size shows
mobility, small lyso
lipids first, large
TGs last

Equivalent carbon number models

Check retention time trends (RP chromatography)

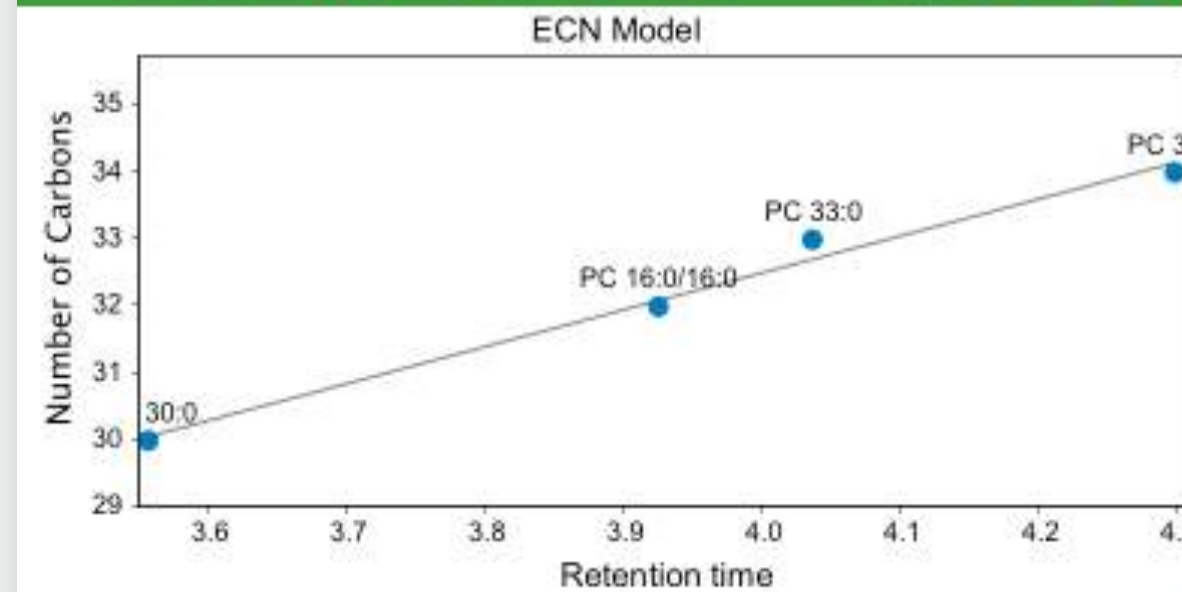
- Group lipids of **same class** with **same number of double bond equivalents** with **number of carbons**.
- In RP chromatography, higher retention on column if more CH₂ in side-chains.
- Almost linear trend

ECN plots

Diacylglycerophosphocholines

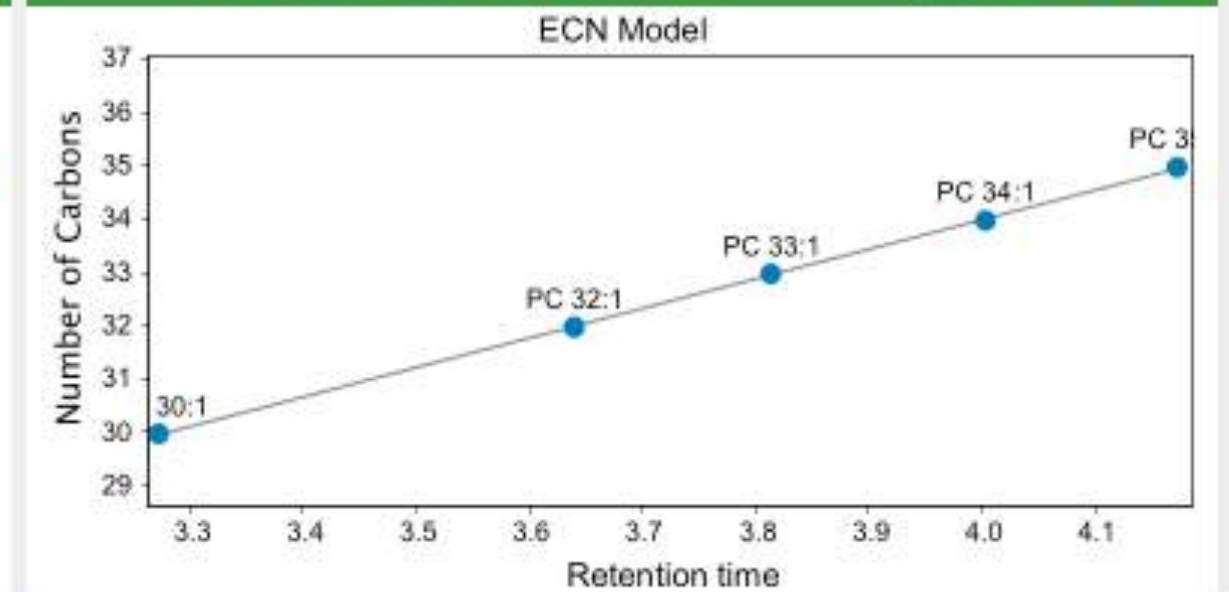
PC: ECN model for DBE: 0

R² 4 Lipids **0.986**



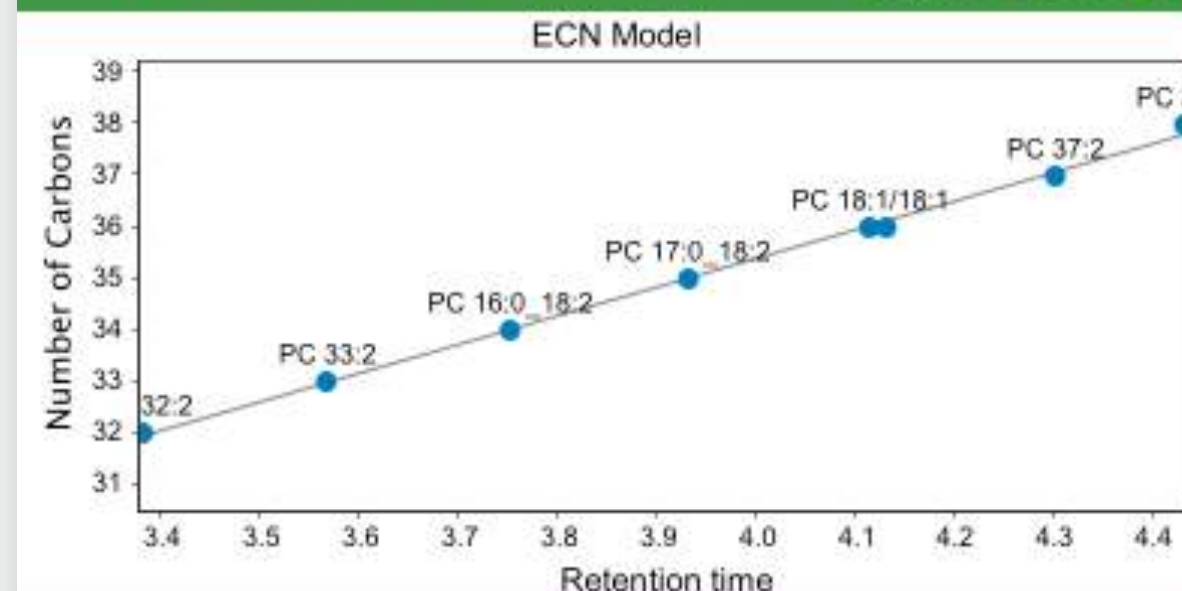
PC: ECN model for DBE: 1

R² 6 Lipids **1.000**



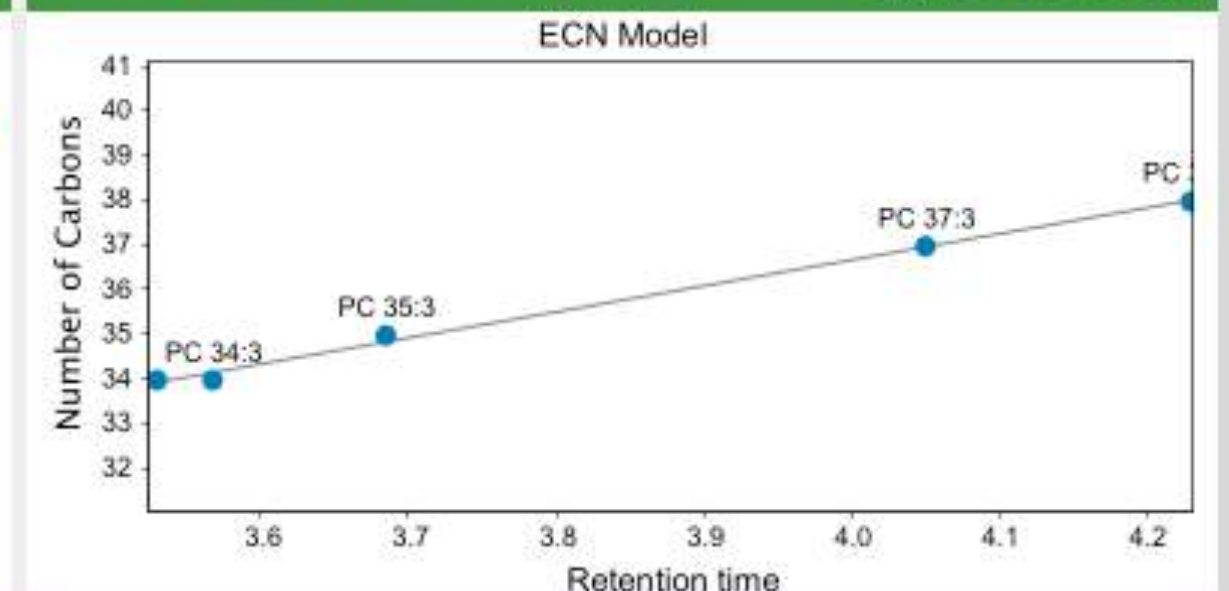
PC: ECN model for DBE: 2

R² 8 Lipids **0.998**



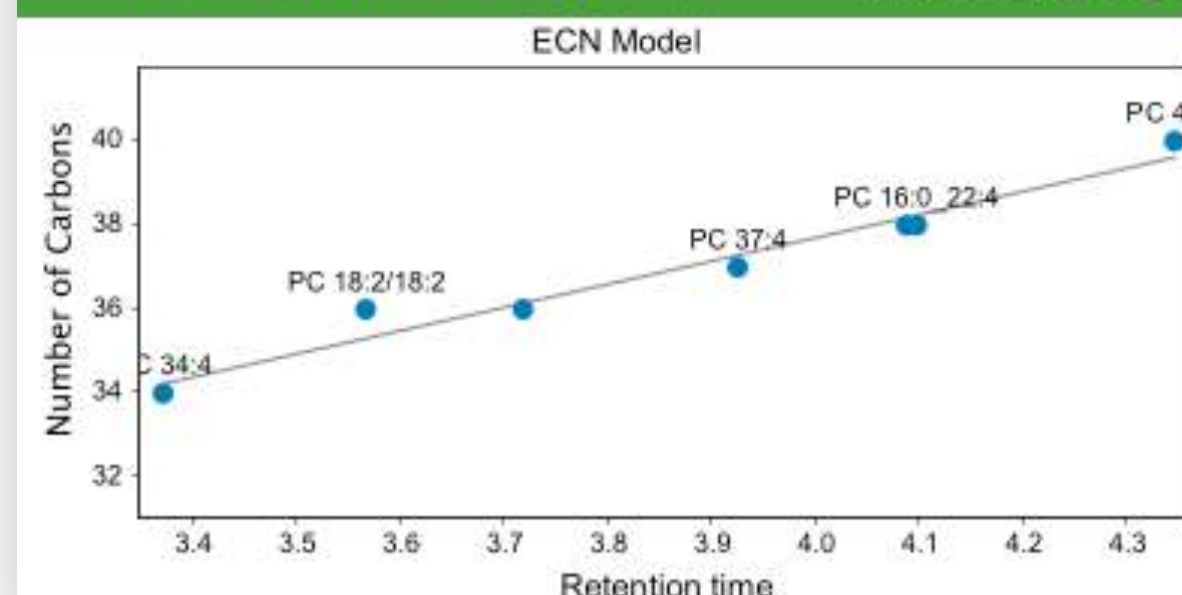
PC: ECN model for DBE: 3

R² 5 Lipids **0.996**



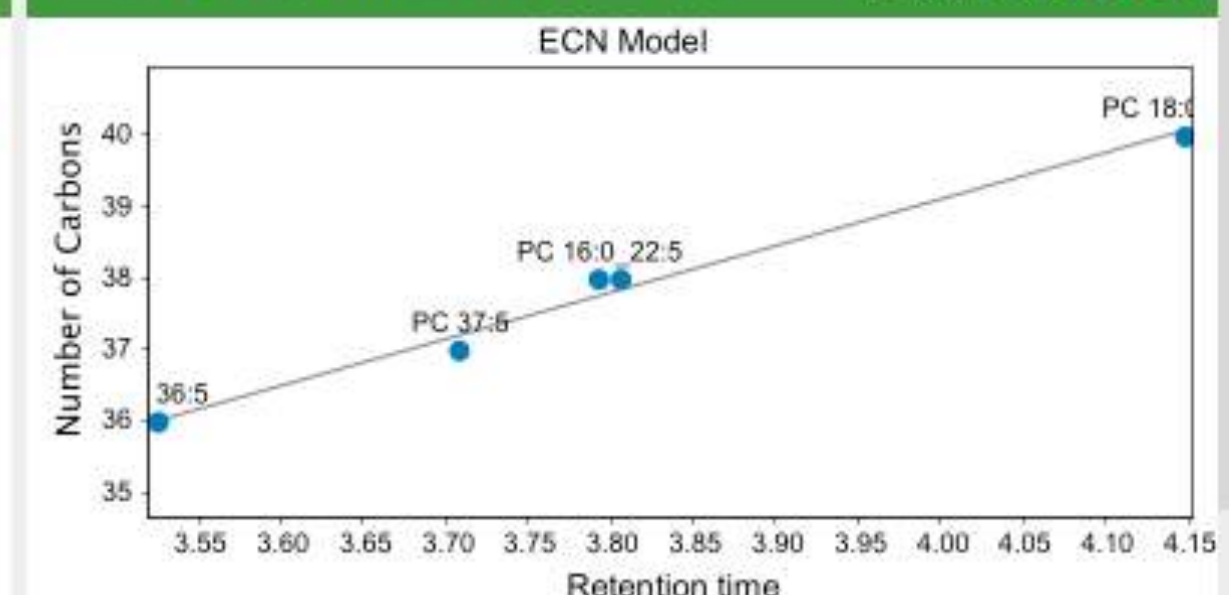
PC: ECN model for DBE: 4

R² 7 Lipids **0.960**



PC: ECN model for DBE: 5

R² 6 Lipids **0.989**



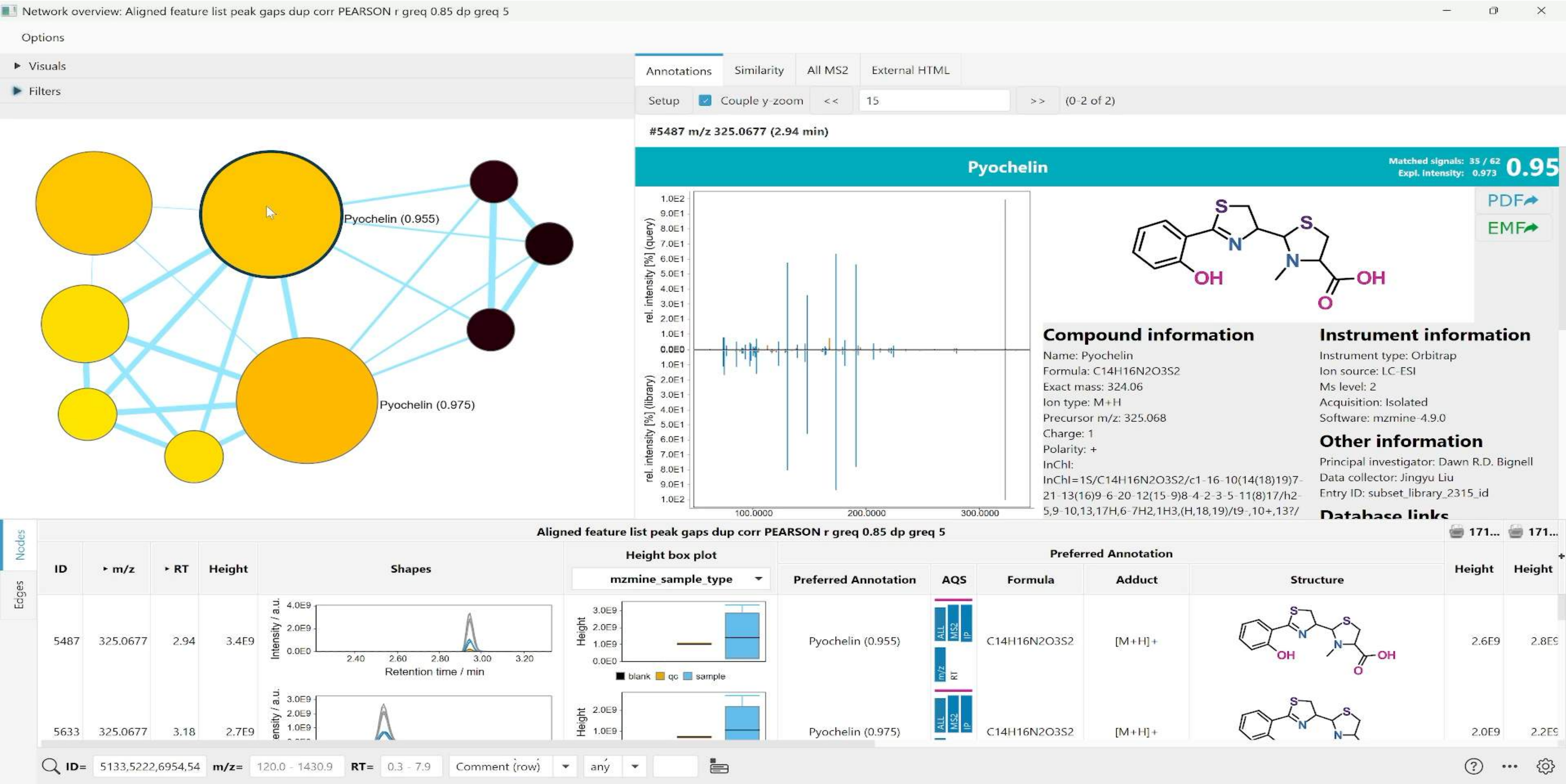
Statistics Dashboard



Statistics: Region of Interest



Interactive Molecular Networks



Interactive Molecular Networks

mzmine 4.9.0

ProjectRaw data methodsFeature detectionFeature list methodsVisualizationmzwizardToolsOther detectorsMRMWindowsUsersWorkspacesHelp

MS data filesFeature lists

Welcome to mzmine xFeature Table xStatistics dashboard x

Aligned feature list peak gaps dup corr PEARSON r greq 0.85 dp greq 5

ID	m/z	RT	Height	Shapes	Height box plot	Preferred Annotation	AQS	Formula	Adduct	Structure
2246	295.1647	1.06	1.0E8			Spectral Match to Tyr-Leu from NIST14 (0.935)		C15H22N2O4	[M+H] ⁺	
2545	237.1232	1.15	1.4E7			Phe-ala (0.925)		C12H16N2O3	[M+H] ⁺	

Q ID= 1,5-6 m/z= 120.0 - 1430.9 RT= 0.3 - 7.9 Formula > >

[7:41:10 PM]: EDGE colors

Focus nodesCenterFull network

Phe-ala

Matched signals: 4 / 16
Expl. intensity: 0.944 0.92

Compound information

Name: Phe-ala
Formula: C12H16N2O3
Exact mass: 236.116
Ion type: [M+H]⁺
Precursor m/z: 237.123
Charge: 1
Polarity: +
InChI:
InChI=1S/C12H16N2O3/c1-8(12(16)17)14-11(15)10(13)7-9-5-3-2-4-6-9/h2-6,8,10H,7,13H2,1H3,(H,14,15)(H,16,17)/t8-,10-/m0/s1
InChI key: MIDZLCFIAINOQN-WPRPVWTQSA-N
SMILES:

Instrument information

Instrument type: Orbitrap
Ion source: ESI
Ms level: 2
Acquisition: Crude
Software: mzmine 4.9.0

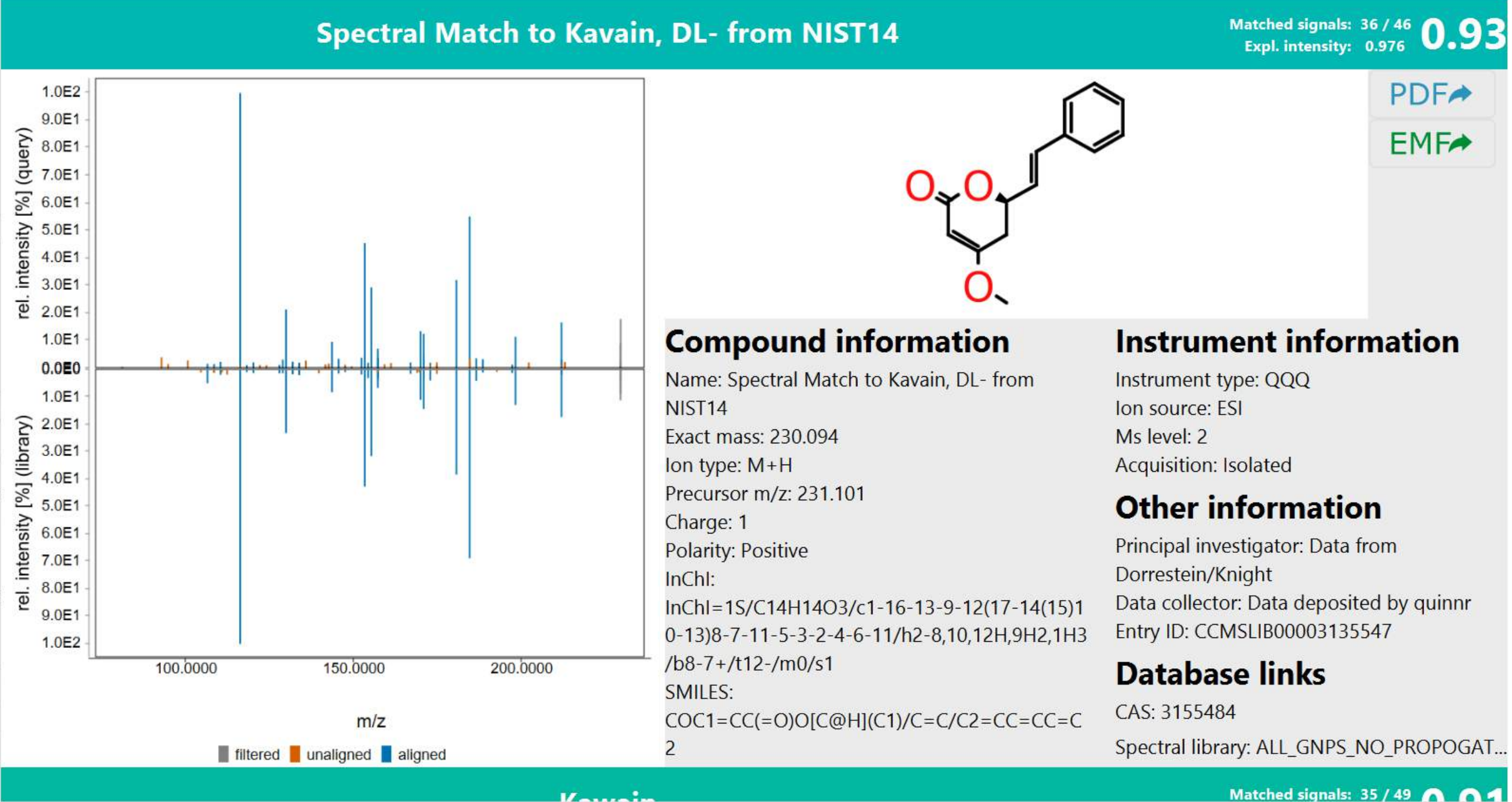
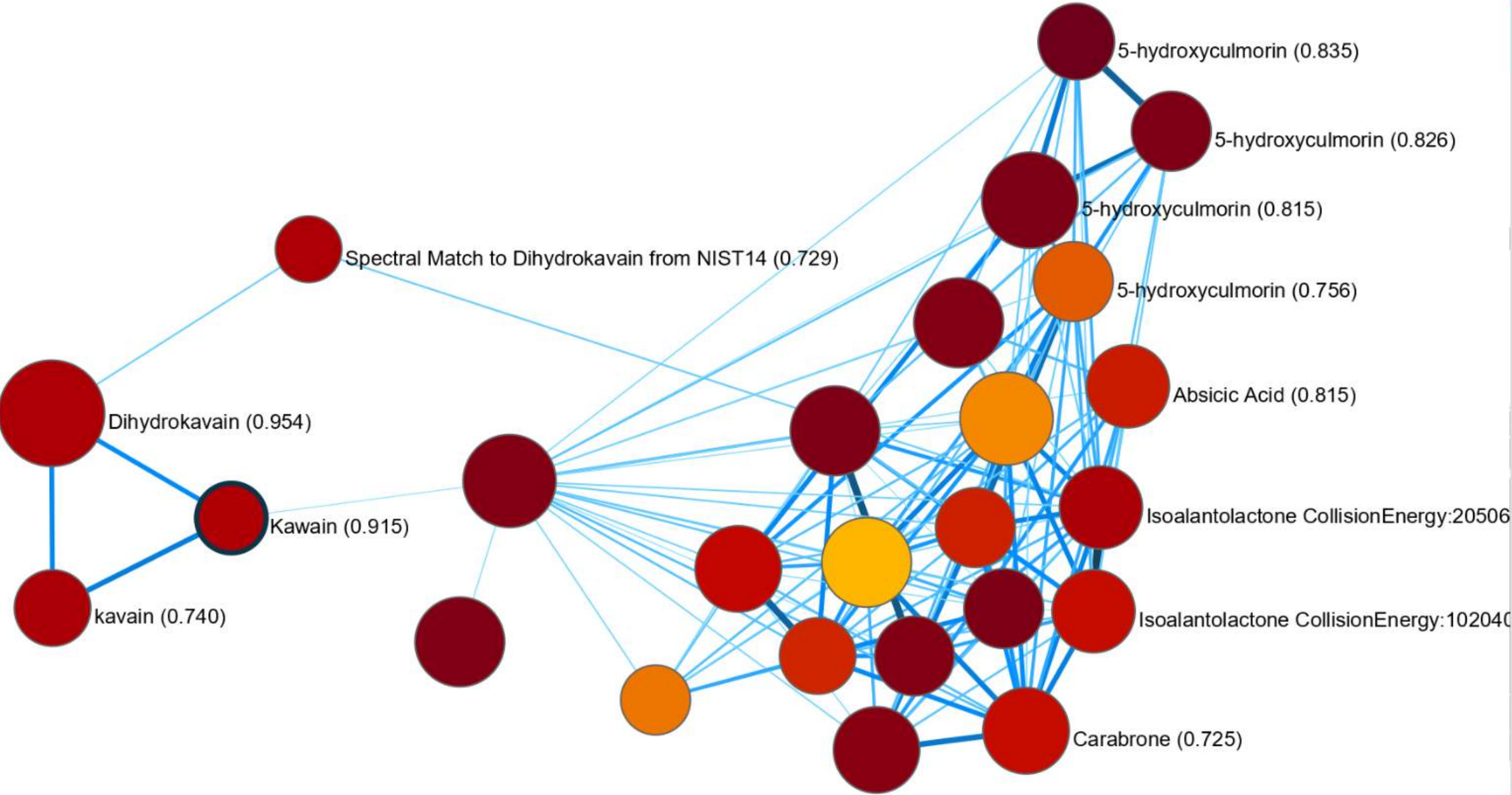
Other information

Principal investigator: Tomas Pluskal
Data collector: Corinna Brungs
Entry ID: subset_library_1942_id

Database links

Spectral library: spectral_library_subset.json

Interactive Network Dashboard



Nodes		Aligned feature list peak gaps dup corr PEARSON r greq 0.85 dp greq 5												
Edges	ID	m/z	RT	Height	Charge	Shapes	Height box plot	Area box plot	Spectral match					
							ATTRIBUTE_Sample_Area	ATTRIBUTE_Sample_Location	Spectral match	Similarity	Adduct	Formula	Structure	
	6502	233.1169	8.52	2.4E7	1				Dihydrokavain (0.954)	0.954	[M+H] ⁺			
	6504	215.1065	8.53	1.3E6	1				Spectral Match to Dihydrokavain from NIST14 (0.729)	0.729	M+H-H2O			
	6532	231.1013	8.57	2.4E6	1				kavain (0.740)	0.740	[M+H] ⁺	C14H14O3		

Q ID= 6083,6548,7492,60

m/z= 152.9 - 1444.4

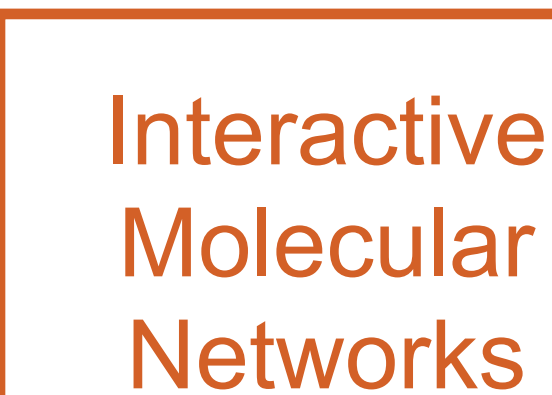
RT= 0.8 - 13.0

Comment (row)

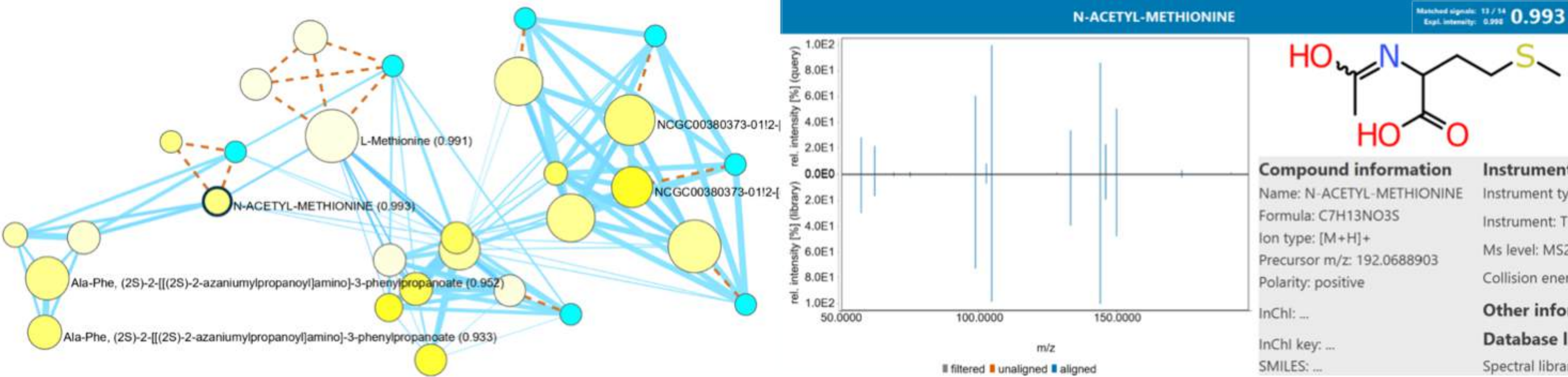
any

?

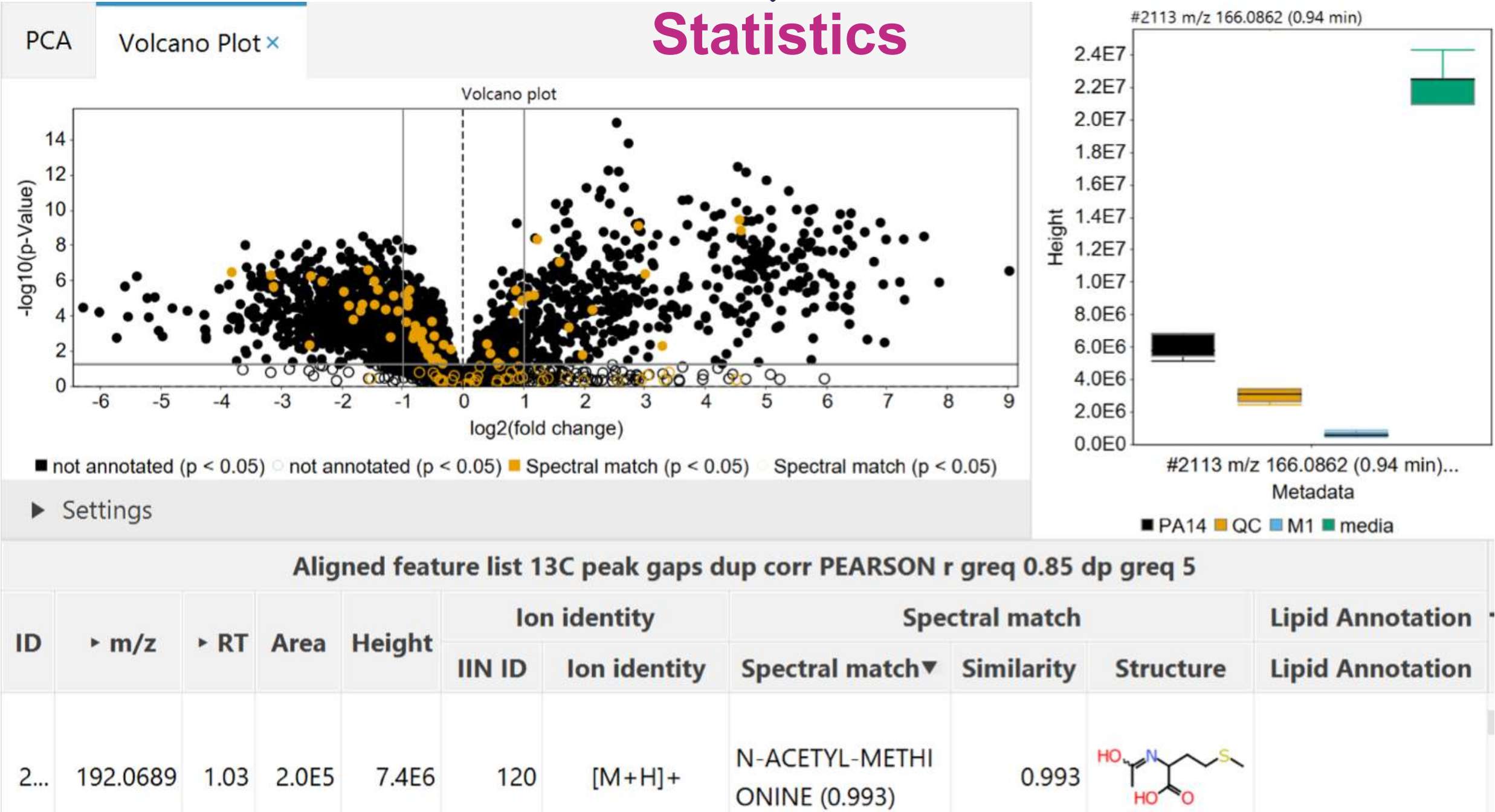
...



Linked Statistics Dashboard & Molecular Networking



Interactive Networking Spectral Library Match



Molecular Networking in MZmine

MS data files

Feature lists

Libraries

skin_05_720_UB.mzML eics sm r deiso

skin_05_720_UB.mzML eics sm r

skin_05_720_UB.mzML eics sm

skin_05_720_UB.mzML eics

▼ skin_04_600_UB.mzML

skin_04_600_UB.mzML eics sm r deiso

skin_04_600_UB.mzML eics sm r

skin_04_600_UB.mzML eics sm

skin_04_600_UB.mzML eics

▼ skin_03_480_UB.mzML

skin_03_480_UB.mzML eics sm r deiso

skin_03_480_UB.mzML eics sm r

skin_03_480_UB.mzML eics sm

skin_03_480_UB.mzML eics

▼ bld_plt2_08_480_1.mzML

bld_plt2_08_480_1.mzML eics sm r deiso

bld_plt2_08_480_1.mzML eics sm r

bld_plt2_08_480_1.mzML eics sm

bld_plt2_08_480_1.mzML eics

▼ bld_plt2_03_720_1.mzML

bld_plt2_03_720_1.mzML eics sm r deiso

bld_plt2_03_720_1.mzML eics sm r

bld_plt2_03_720_1.mzML eics sm

bld_plt2_03_720_1.mzML eics

▼ bld_plt1_01_120_1.mzML

bld_plt1_01_120_1.mzML eics sm r deiso

bld_plt1_01_120_1.mzML eics sm r

bld_plt1_01_120_1.mzML eics sm

bld_plt1_01_120_1.mzML eics

▼ Aligned feature lists

Aligned feature list peak gaps dup corr

Aligned feature list peak gaps dup

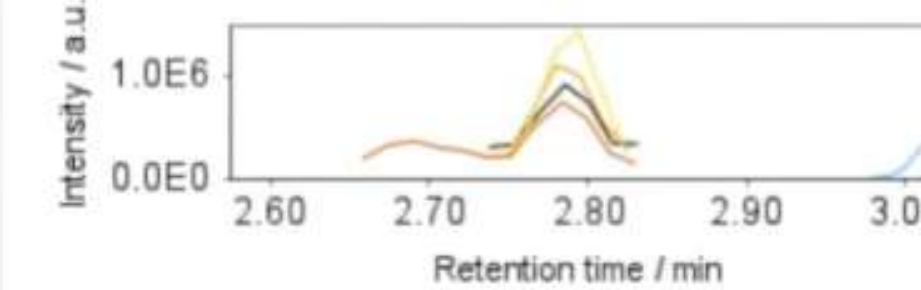
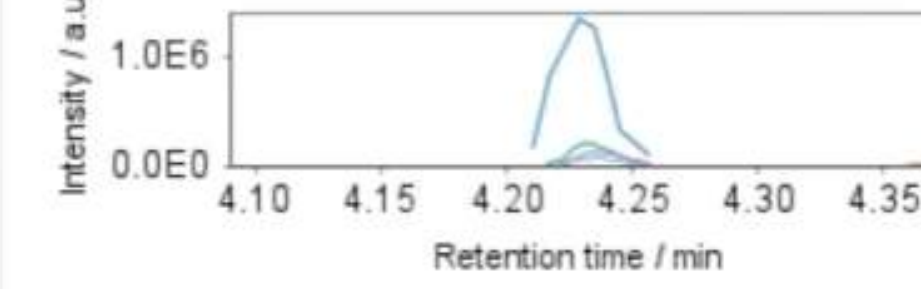
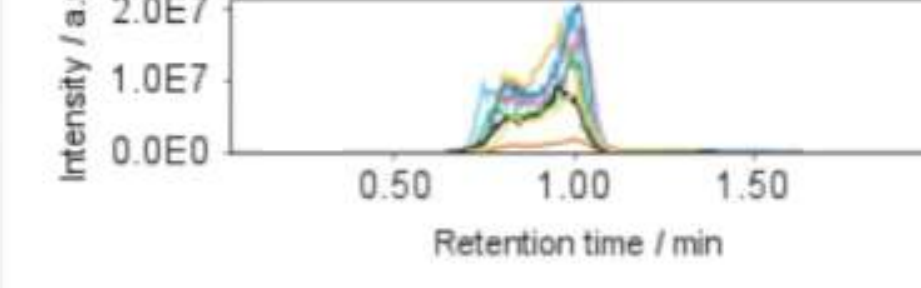
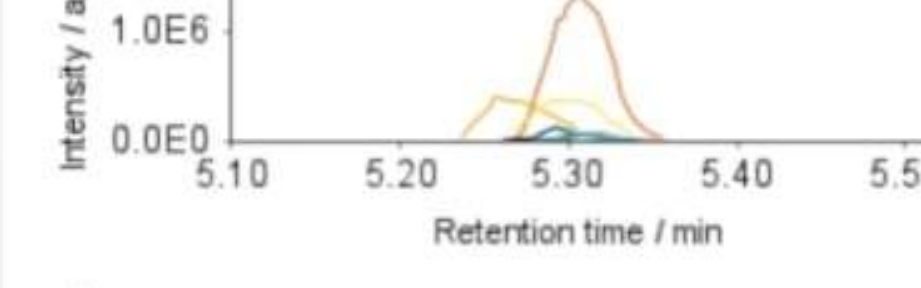
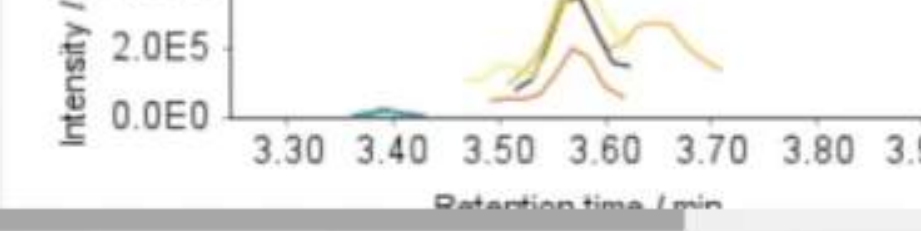
Aligned feature list peak gaps

Aligned feature list peak

Aligned feature list

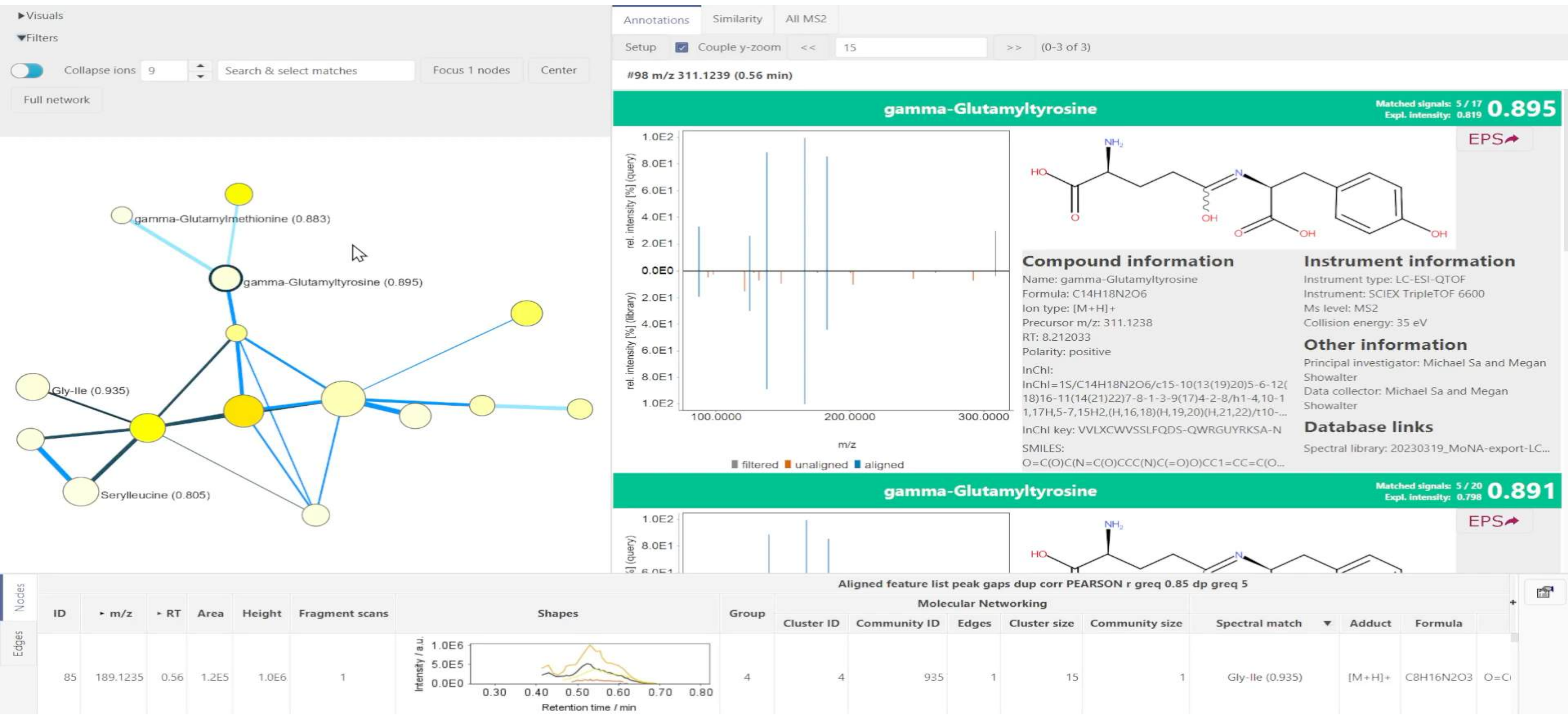
Feature Table X

Aligned feature list peak gaps dup corr PEARSON r greq 0

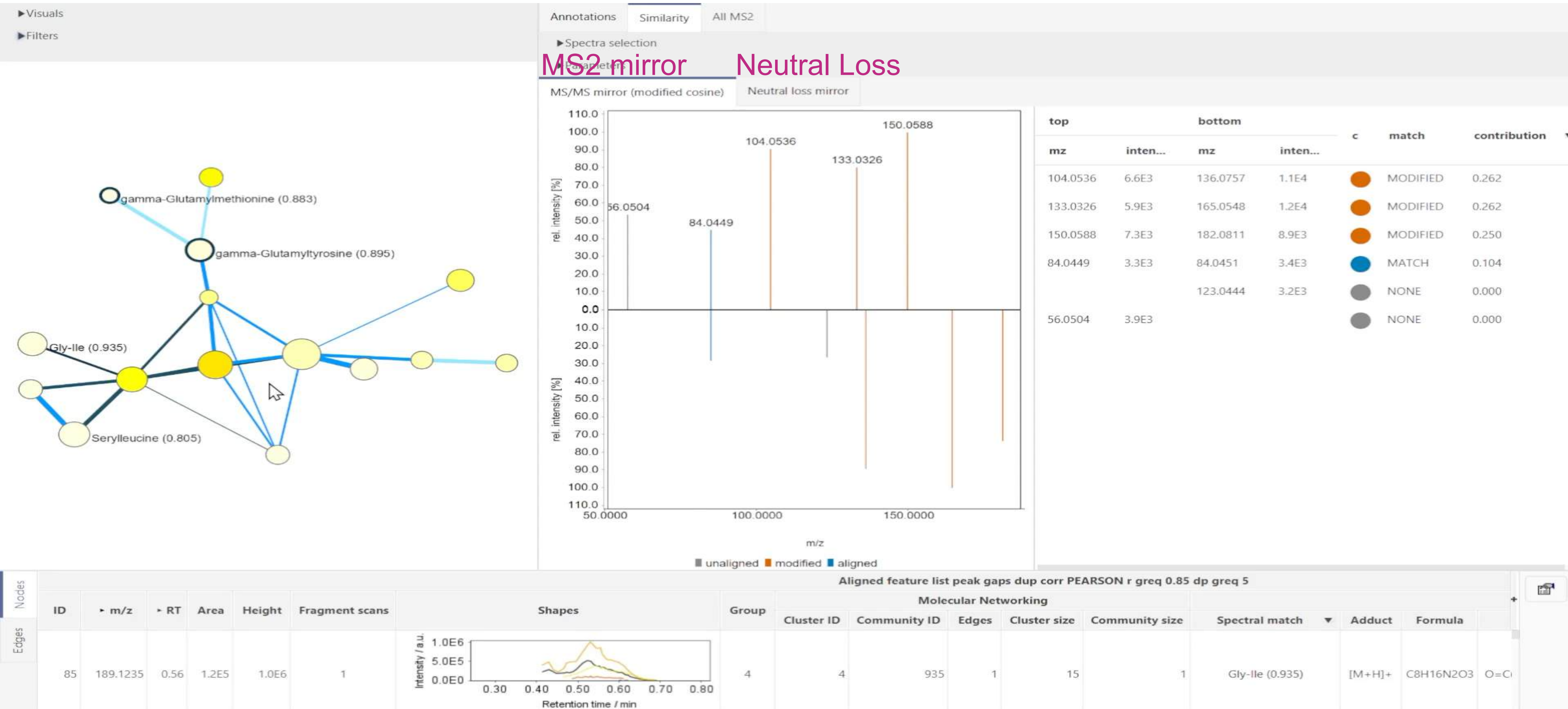
ID	m/z	RT	Area	Height	Fragment scans	Shapes	Group	Spectral mat...	Adduct	Formula	SMILES	Precursor m/z	Neu
592	153.0548	2.40	4.2E5	4.5E6	4		86	Vanillin (0.867)	[M+H] ⁺	C8H8O3	<chem>O=CC1=CC=C(O)C(OC)=C1</chem>	153.0545	
1241	169.0497	2.91	6.4E4	1.4E6	1		86	VANILLIC ACID ...	[M+H] ⁺	C8H8O4	<chem>O=C(O)C1=CC=C(O)C(OC)=C1</chem>	169.0495	
3171	426.3216	4.29	3.4E4	1.3E6	1		121	URSOCHOLIC ...	[M+NH] ⁺	C24H40O5	<chem>O=C(O)CCC(C)C1CCC2C3C=CC(=C2)C(=C1)C</chem>	426.3214	
180	205.0973	0.99	3.3E6	2.0E7	19		73	TRYPTOPHAN (...)	[M+H] ⁺	C11H12N...	<chem>O=C(O)C(N)CC1=CNC=CC=C1</chem>	205.0972	
4218	327.0784	5.30	5.6E4	1.3E6	2		133	Triphenyl phos...	[M+H] ⁺	C18H15O...	<chem>O=P(OC1=CC=CC=C1)(OC1=CC=CC=C1)OC1=CC=CC=C1</chem>	327.0781	
4338	399.2508	5.55	1.1E5	2.6E6	2			Tri(butoxyethyl)...	[M+H] ⁺	C18H39O...	<chem>CCCCOCCOP(=O)(OCCOCC)OCCOCC</chem>	399.2506	
2391	151.1119	3.48	4.6E4	4.3E5	3		86	Thymol (0.840)	[M+H] ⁺	C10H14O	<chem>OC1=CC(=CC=C1C(C)C)C</chem>	151.1118	

ID: m/z: 150.0629 - 1370.9621 RT: 0.43 - 8.79 ID :

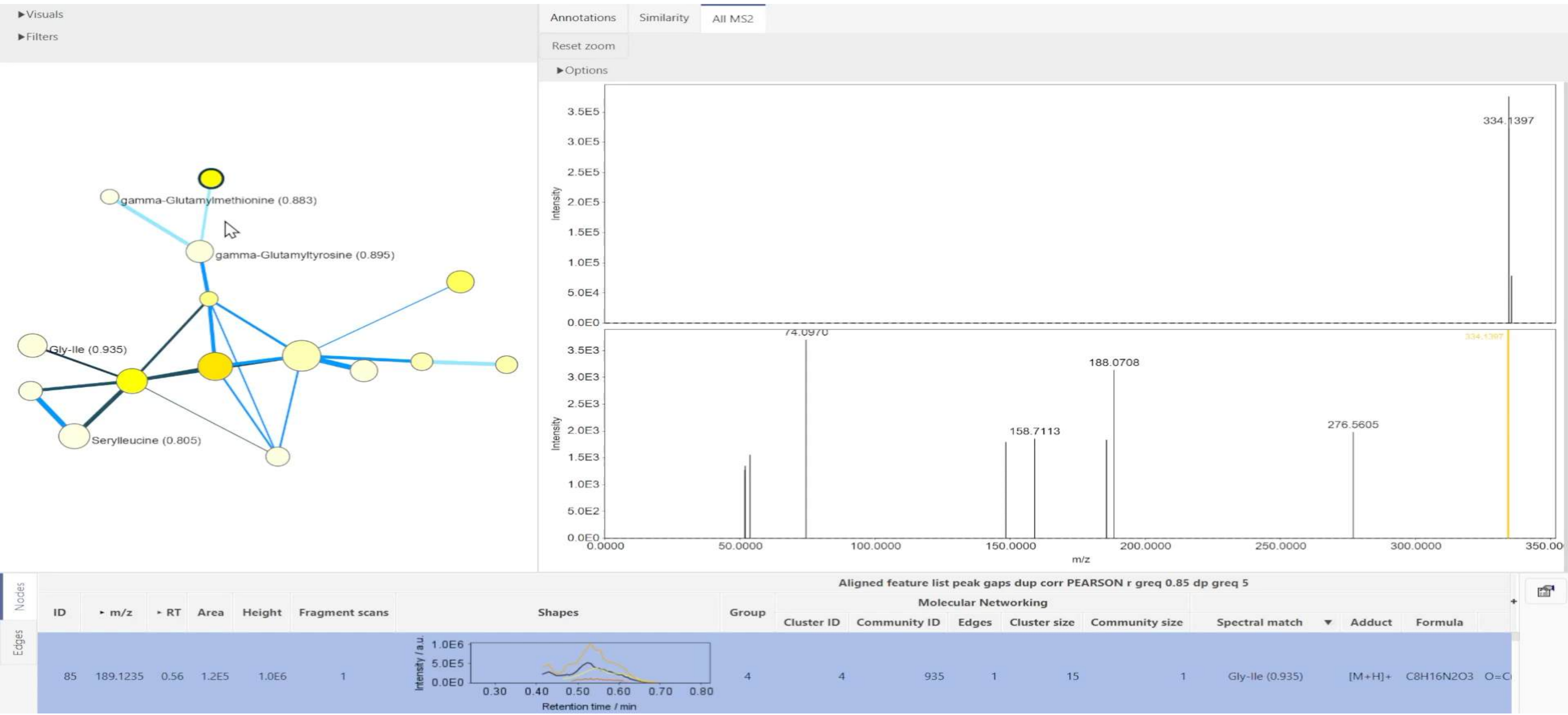
Molecular Networking – Annotations (click)



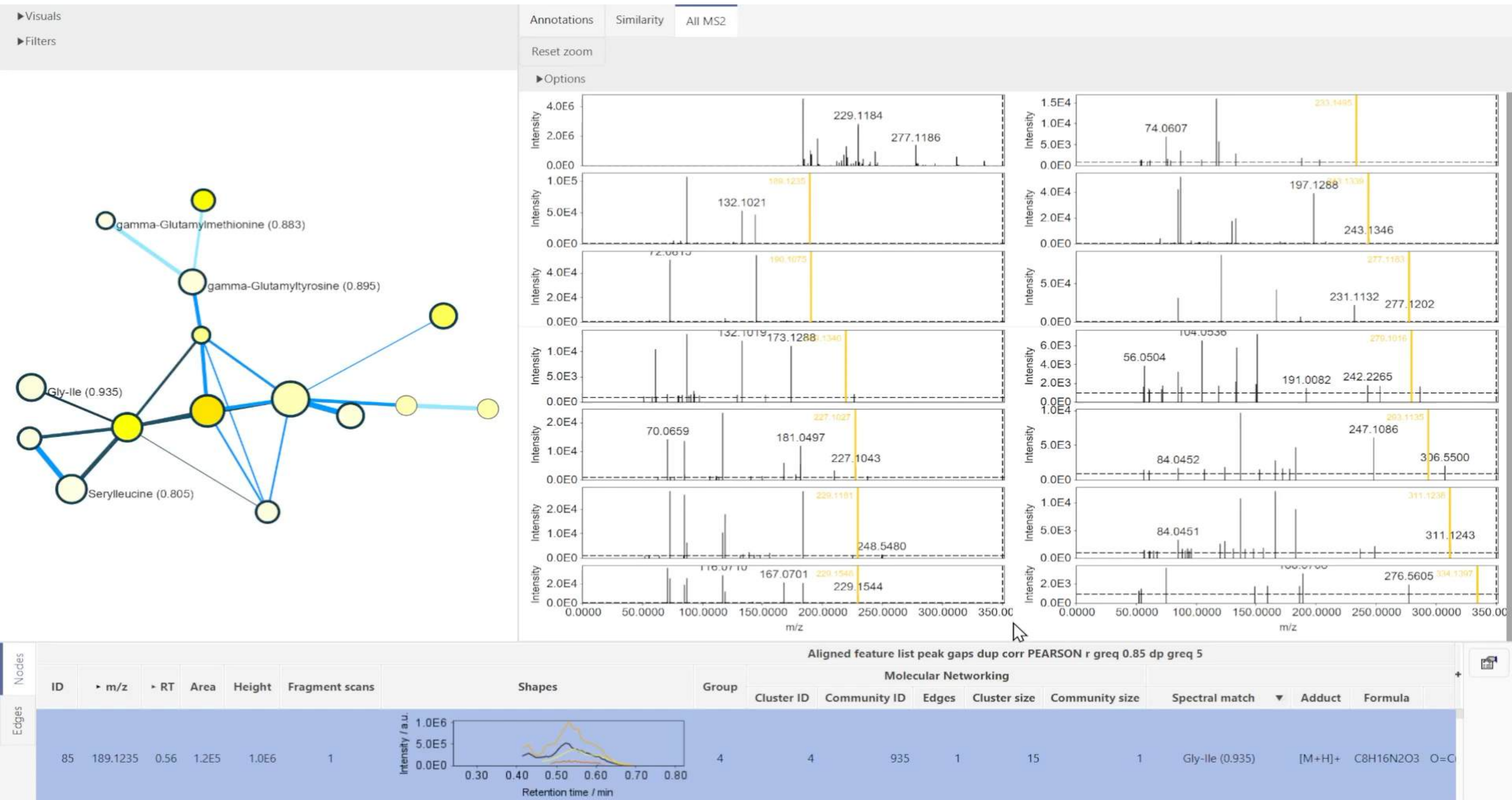
Molecular Networking – Mirror Match + Neutral Loss Mirror



Molecular Networking – All MS²



Molecular Networking – All MS² Same Fragments



Molecular Networking – Full Network

[illegible]

Molecular Networking – Annotation Search

[illegible]

Molecular Networking – Ion Identity Networks (orange)

Visuals

Filters

Collapse ions 3

Search & select matches

Focus 1 nodes

Center

Full network

#655 m/z 271.0319 (2.44 min)

Sulfamethizole, 4-amino-N-(5-methyl-1,3,4-thiadiazol-2-yl)benzenesulfonamide
Matched signals: 13 / 19
Expl. intensity: 0.885

Compound information

Name: Sulfamethizole,
4-amino-N-(5-methyl-1,3,4-thiadiazol-2-yl)benzenesulfonamide
Formula: C₉H₁₀N₄O₂S₂
Mol. weight: 270.0245
Exact mass: 270.0245
Ion type: [M+H]⁺
Precursor m/z: 271.0318
Polarity: positive

Instrument information

Instrument type: LC-ESI-QTOF
Instrument: Bruker maXis Impact
Ion source: ESI
Resolution: 35000
Ms level: MS2
Collision energy: Ramp 20.6-30.8 eV

Other information

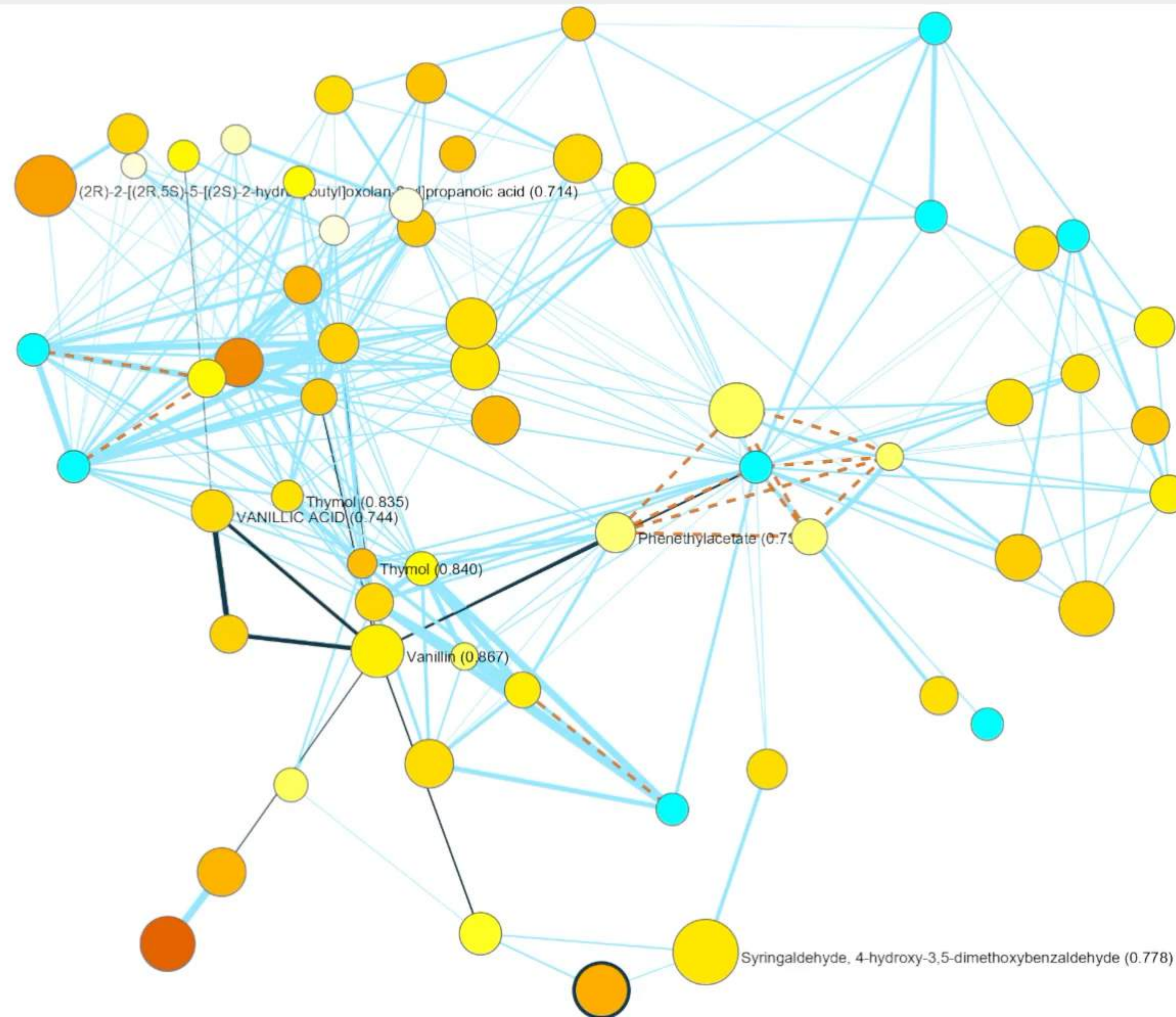
Principal investigator: Nikiforos Alygizakis, ...

Nodes

Edges

Aligned feature list peak gaps dup corr PEARSON r greq 0.85 dp greq 5

ID	m/z	RT	Area	Height	Fragment scans	Shapes	Group	IIN ID	Ion identity	Size	Neutral mass	Cluster ID	Community ID	Edges	Cluster size	Community size
655	271.0319	2.44	1.3E6	1.9E7	11		157	62	[M+H] ⁺	5	270.0247	18	404	1	4	4
654	293.0138	2.43	7.4E4	1.2E6	0		157	62	[M+Na] ⁺	5	270.0247					

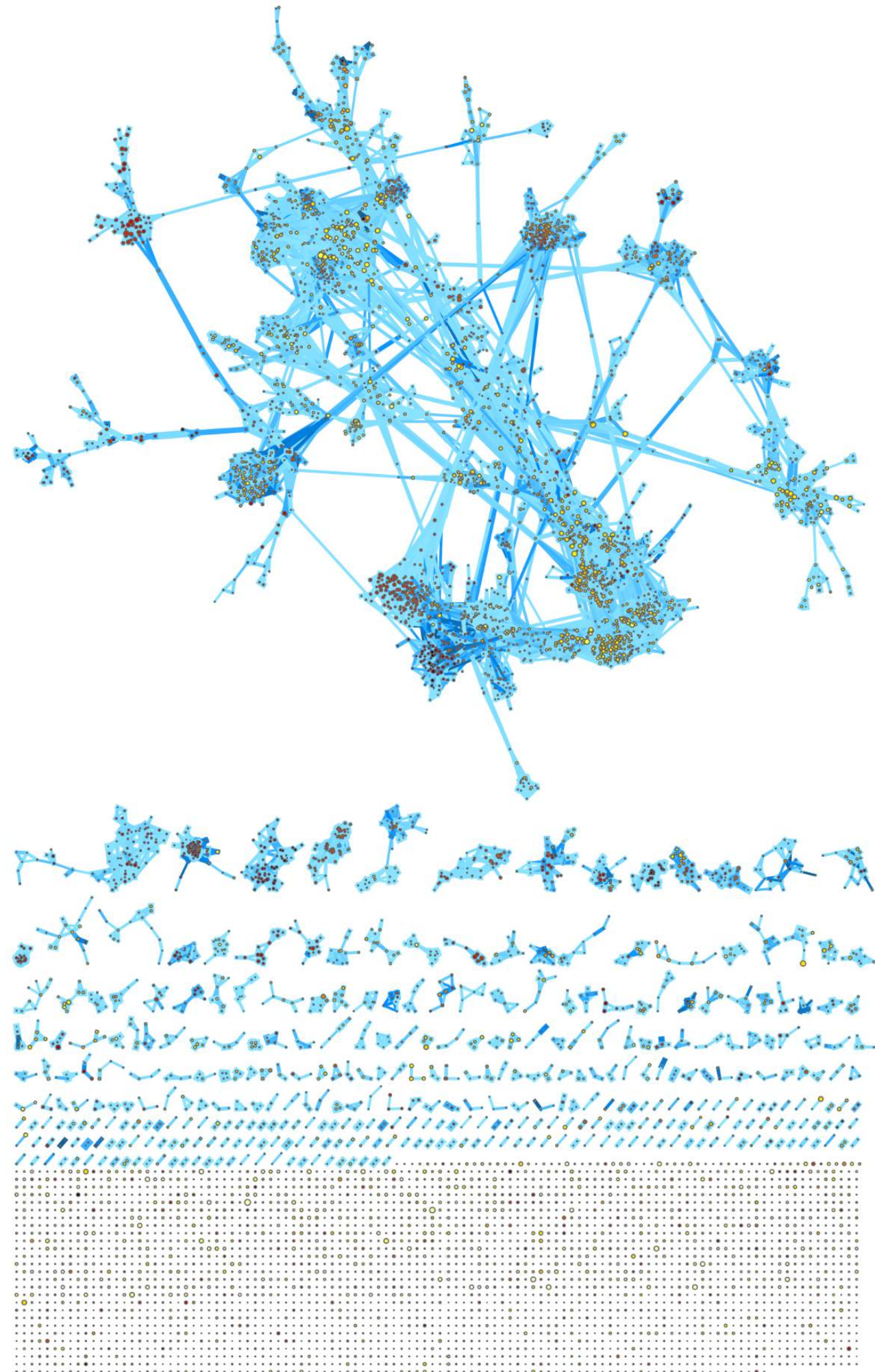


Integration of ML Models

- Direct hosting in mzmine (download latest button)
- Export PyTorch model in TorchScript (JIT trace)
- Easy to use: No Python & virtual environment setup

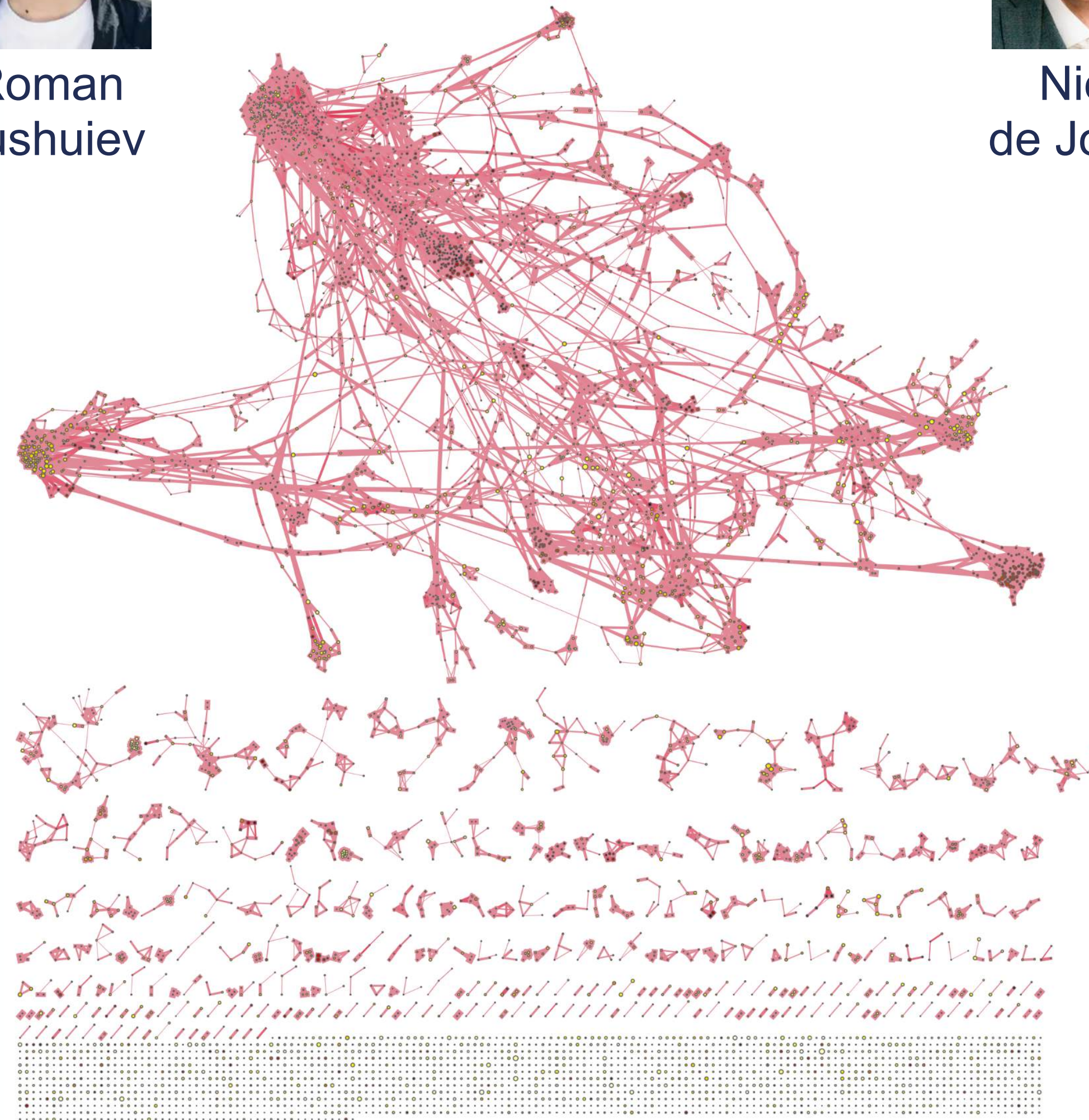
ML Examples – Spectral Embedding/Networking

Modified Cosine



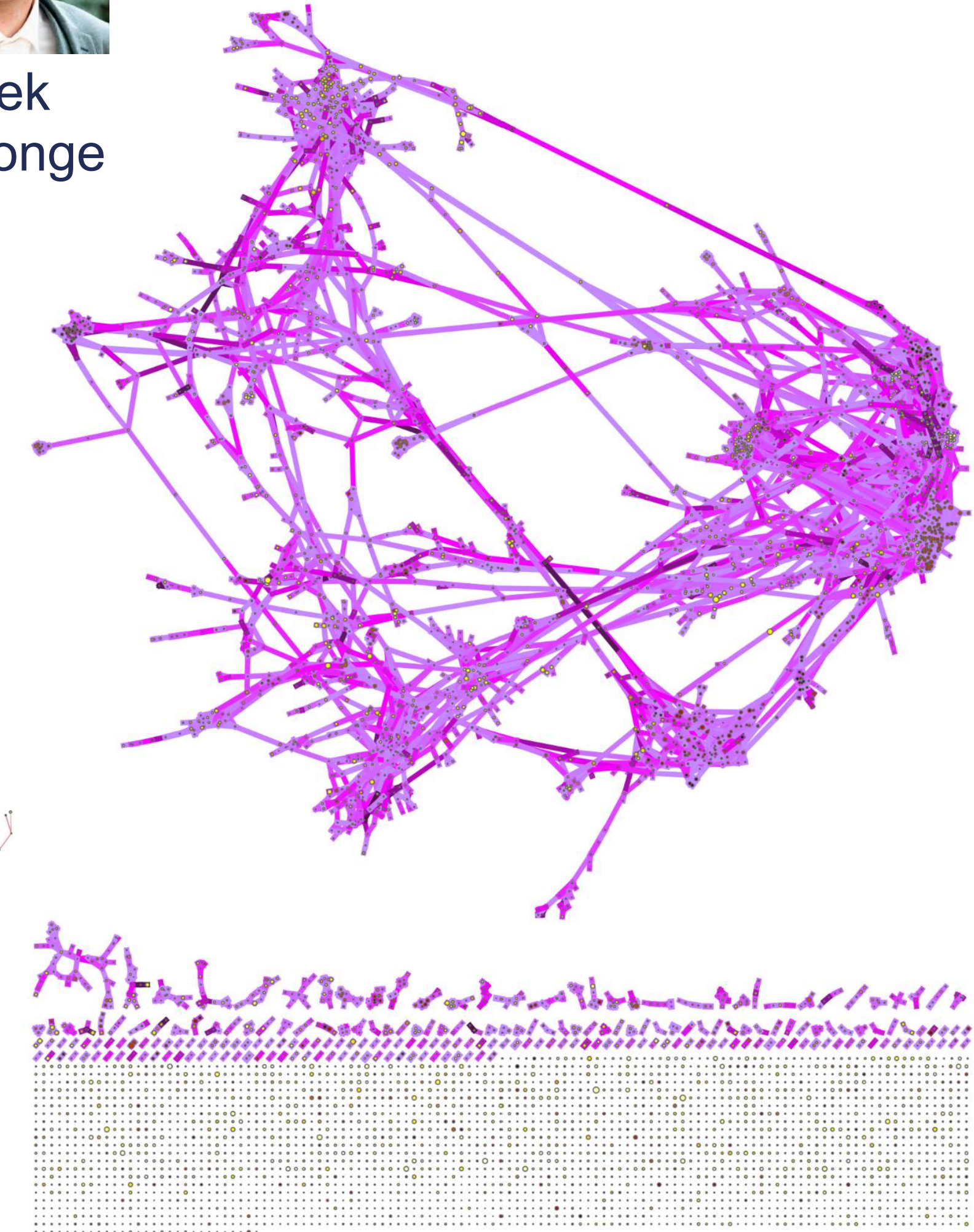
Roman
Bushuiev

DreaMS



Niek
de Jonge

MS2Deepscore

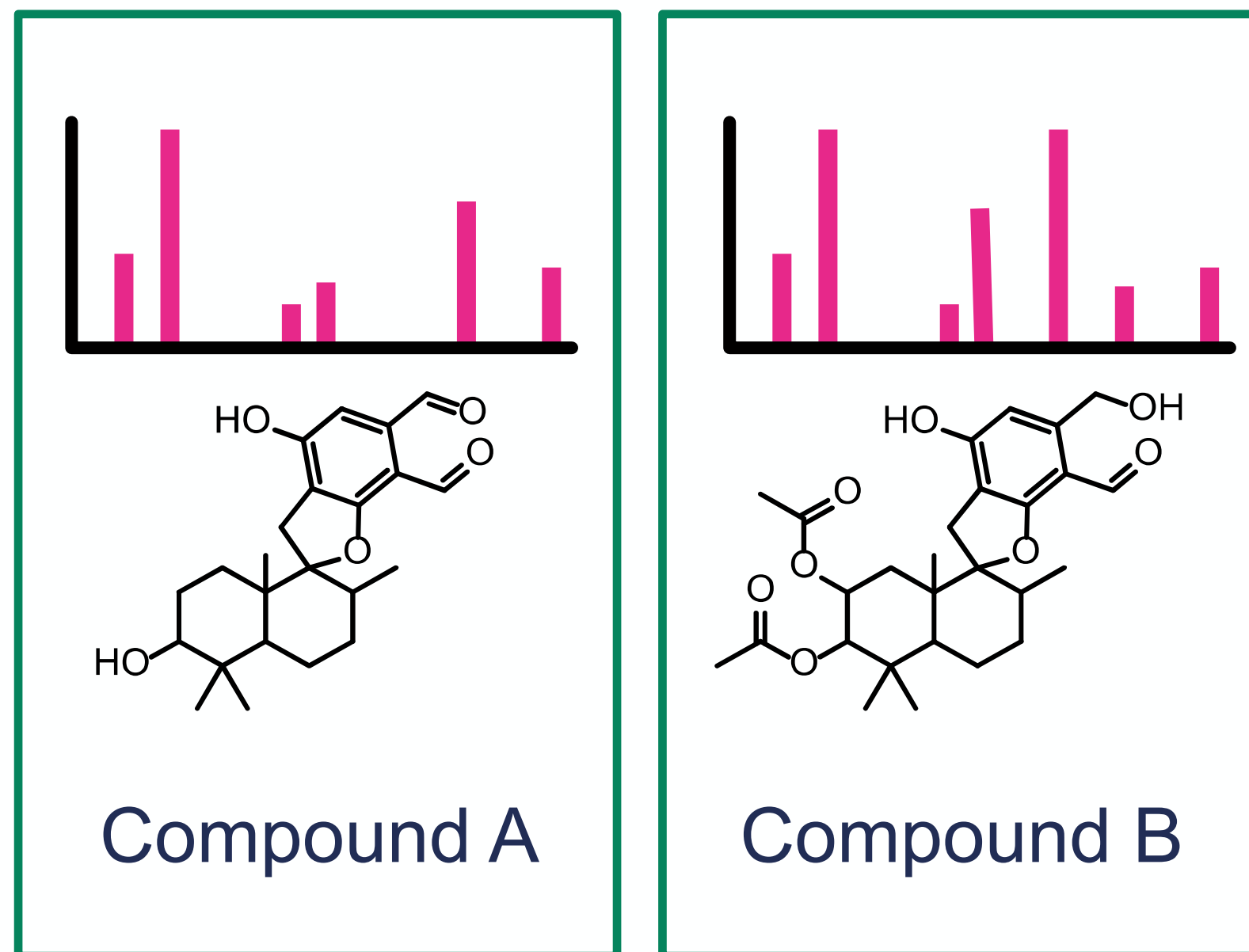


PyTorch model, dreams-docs.readthedocs.io

PyTorch model, github.com/matchms/ms2deepscore

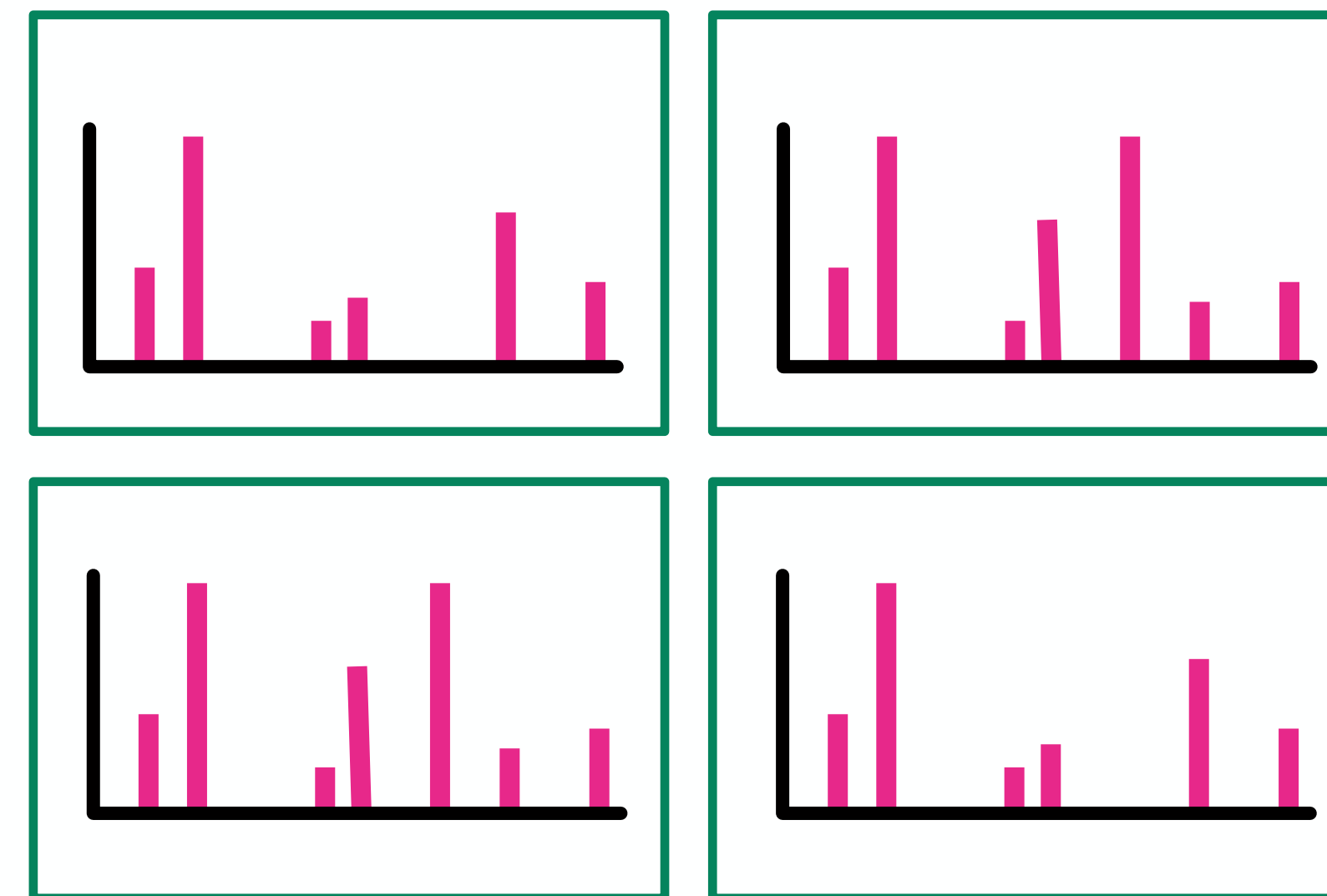
Beyond Modification Aware Matches – Machine Learning

Supervised Learning



Labeled Data
Compound-Spectral Pairs

Self-Supervised Learning



Unlabeled Data
Labeled Data for Fine Tuning



Chemical similarity prediction

DreaMS Model
24M unique spectra

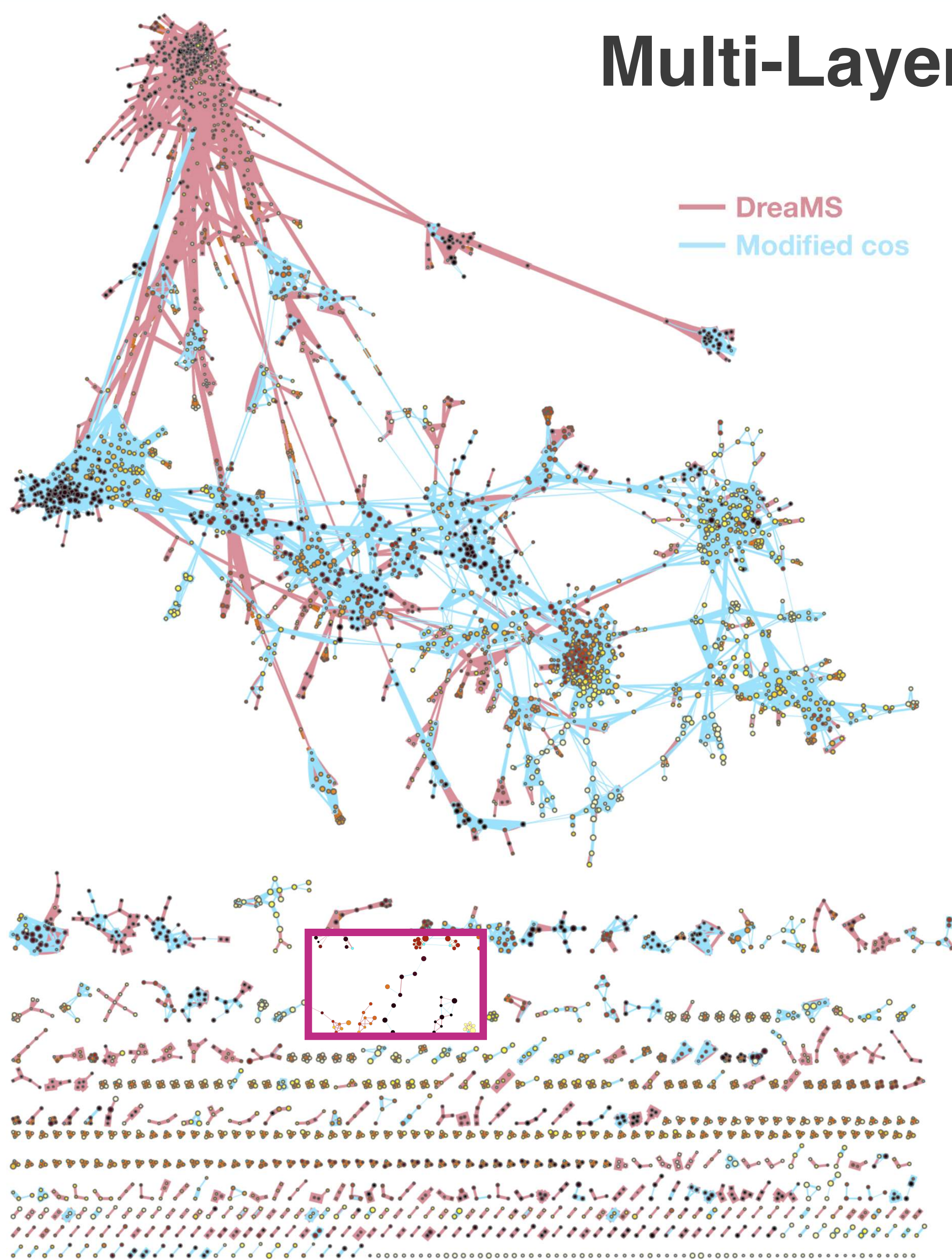


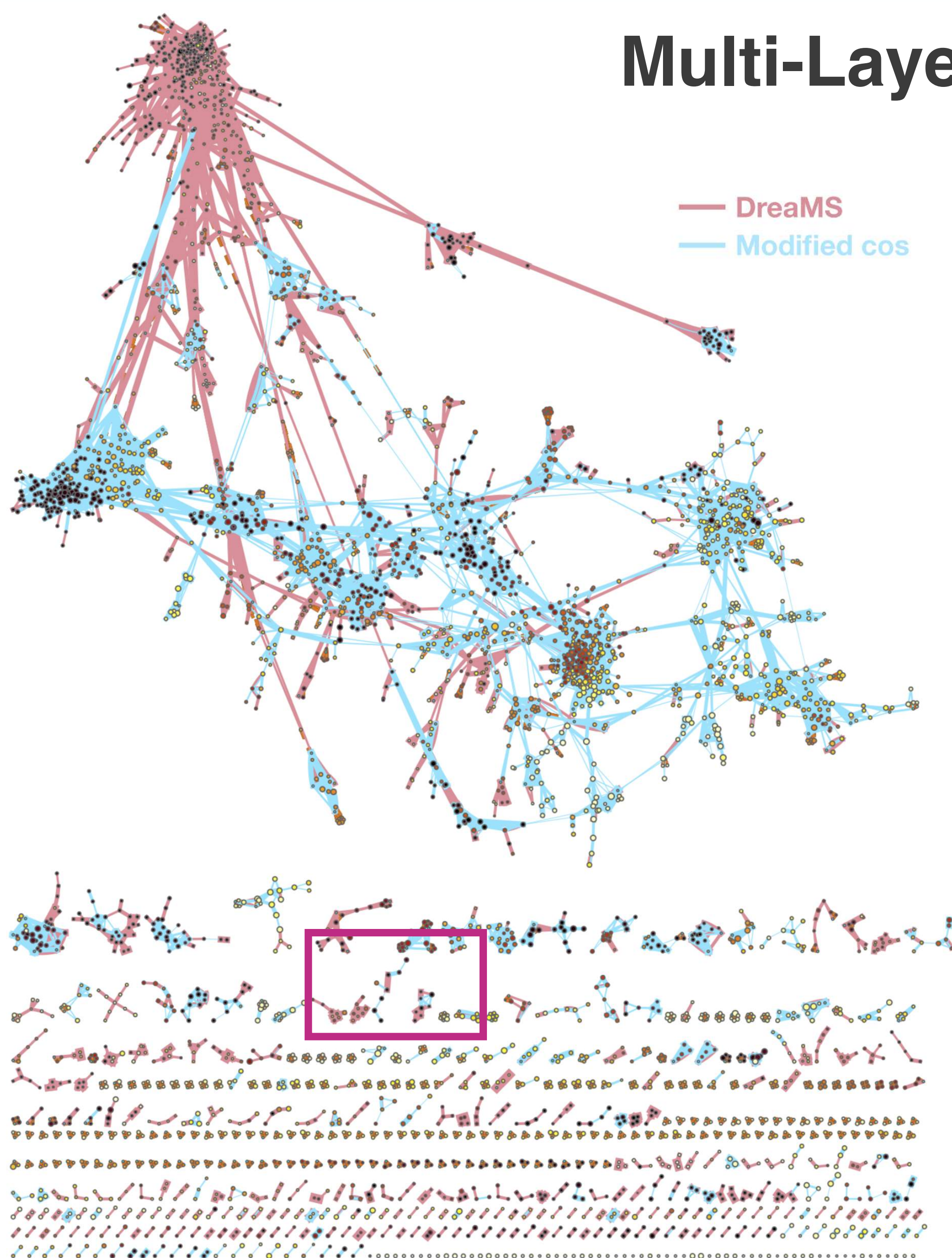
Niek de Jonge



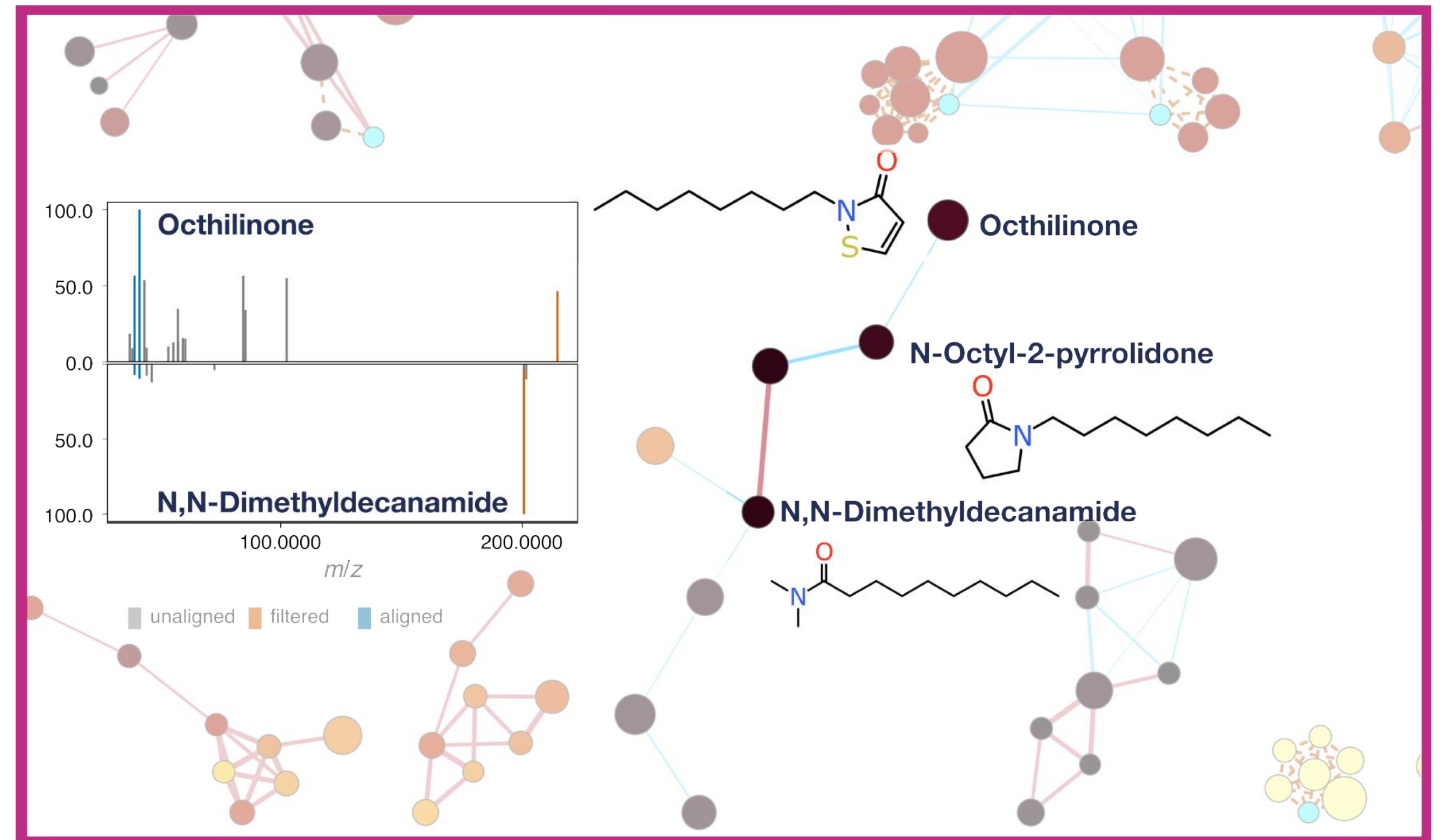
Roman Bushuiev

Multi-Layer Networks in mzmine



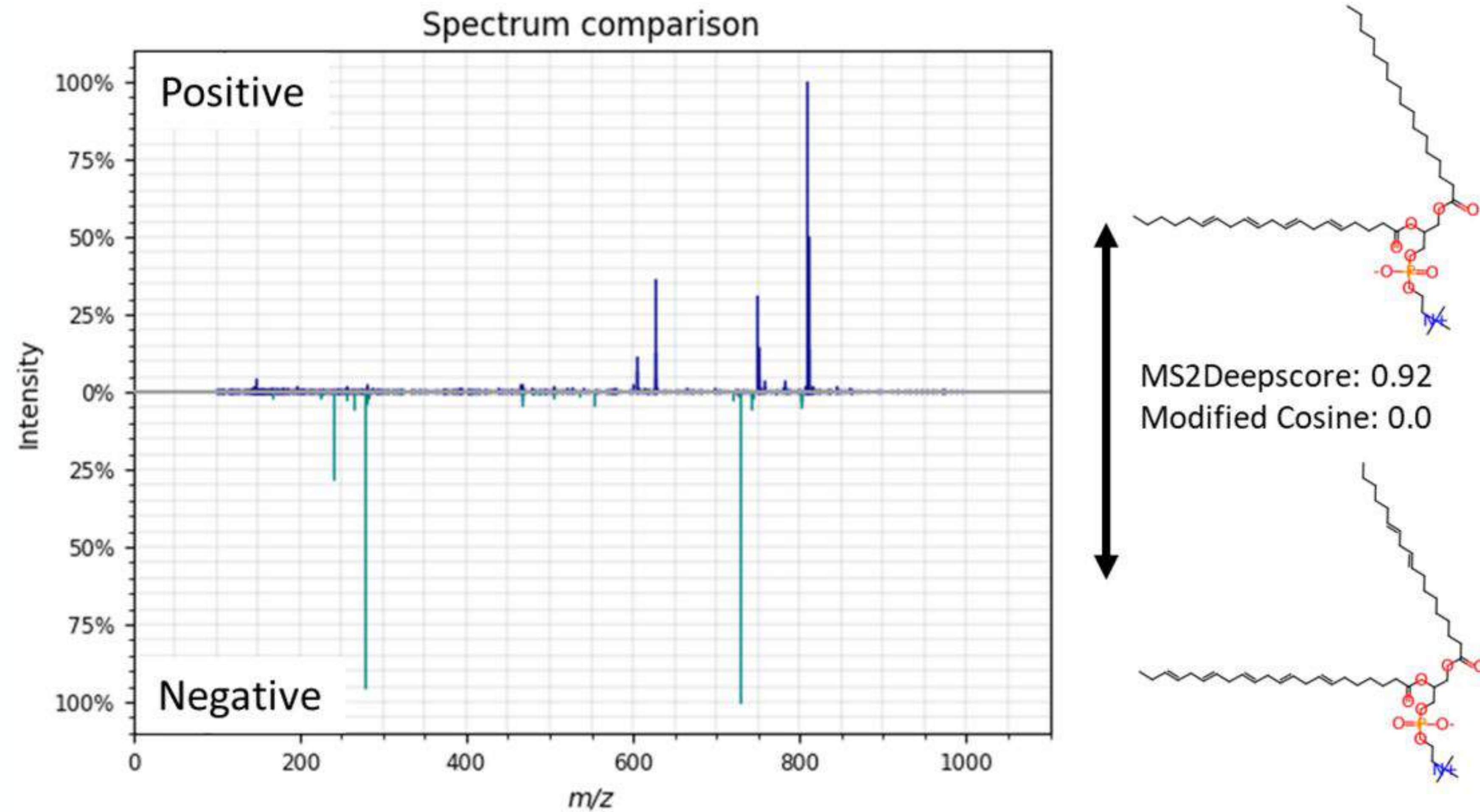


Modified cosine ~ single modifications



Deep learning model DreaMS clusters analogs with complex modifications

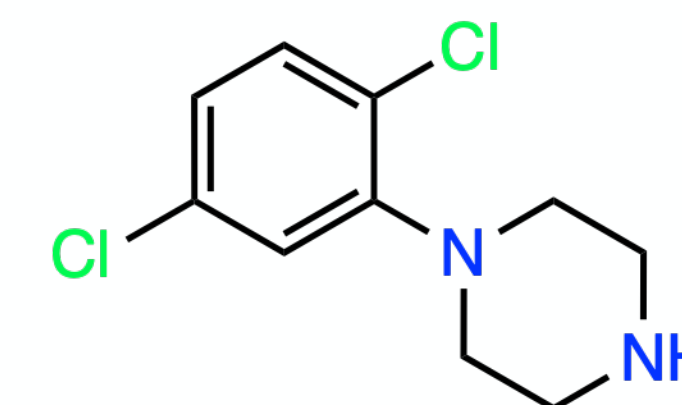
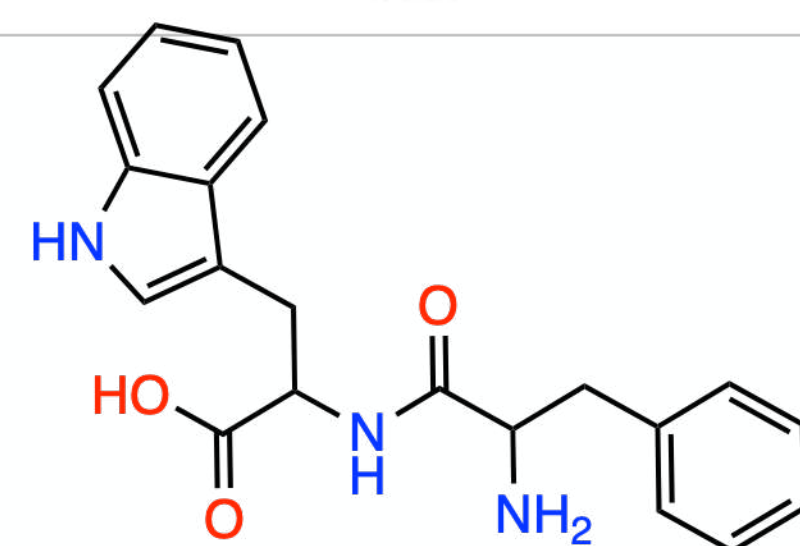
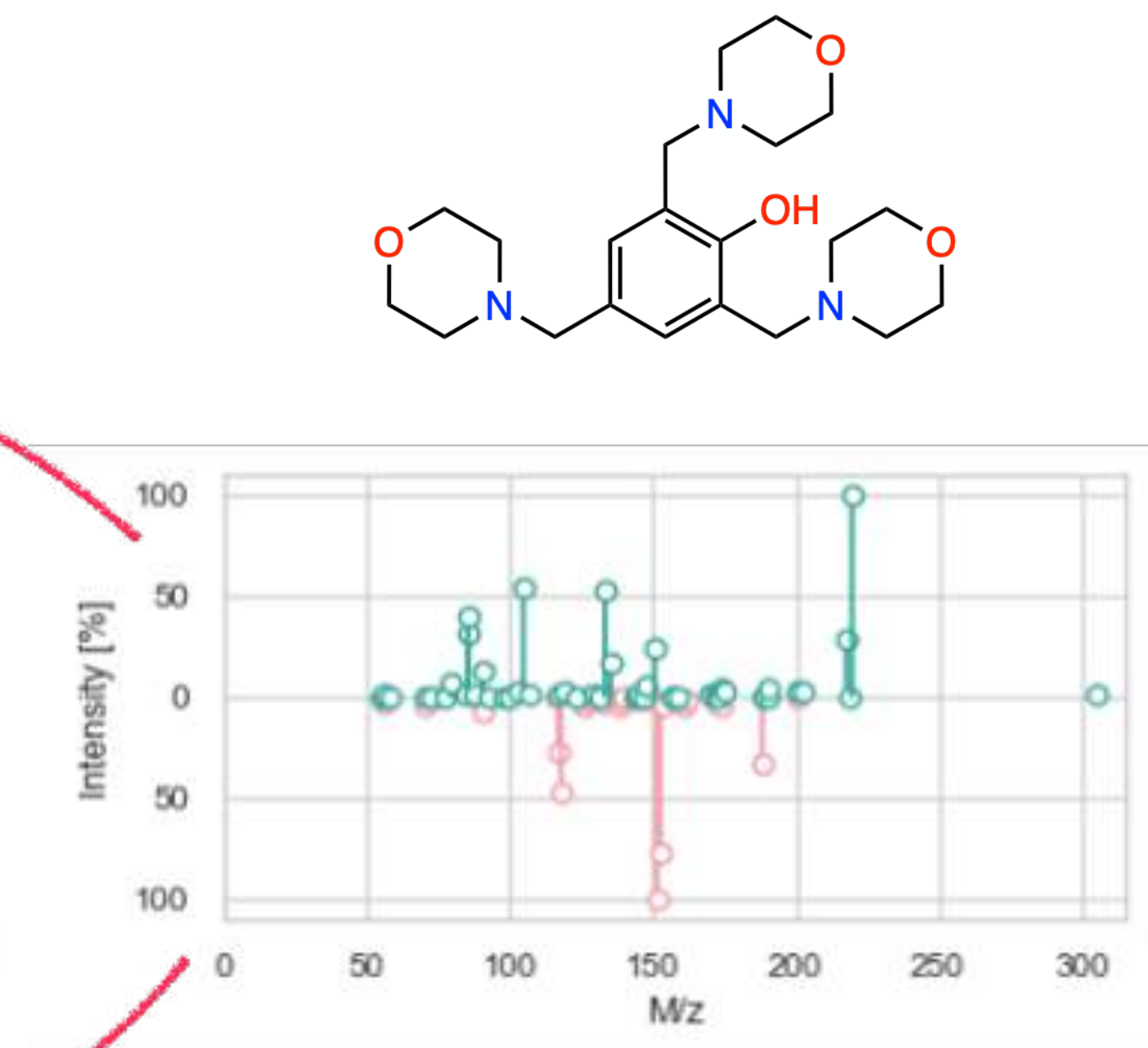
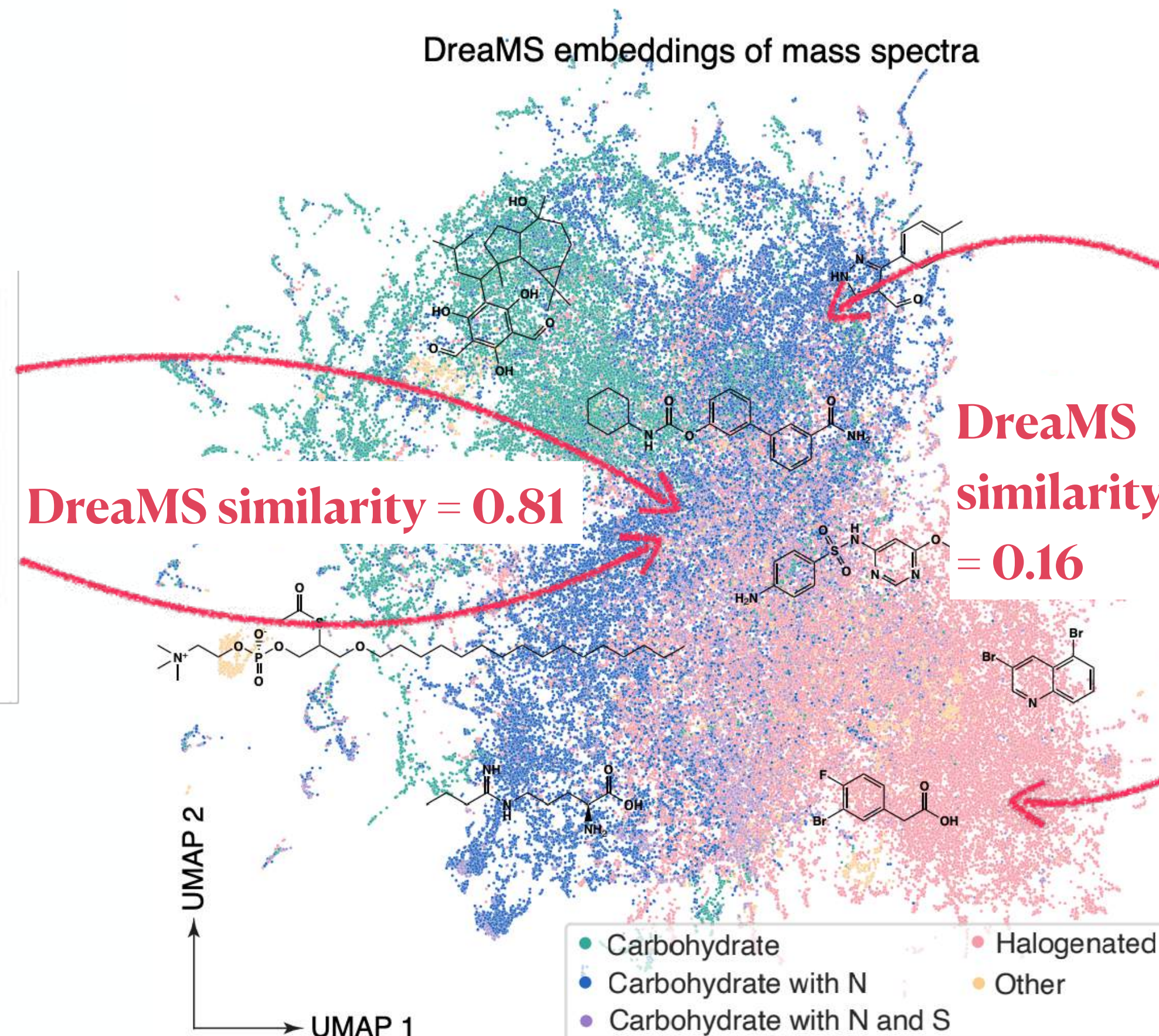
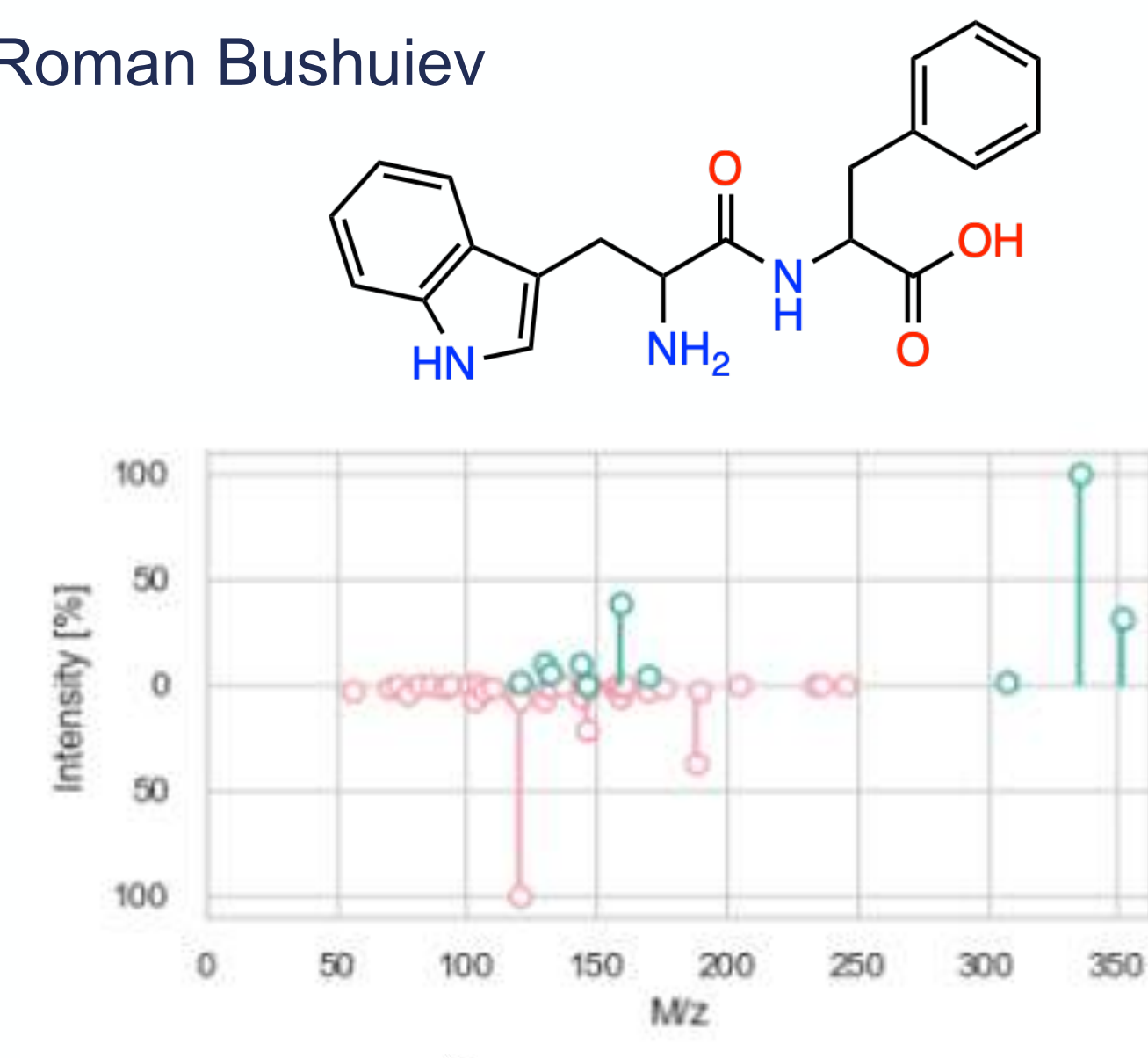
MS2DeepScore Bridging Polarities





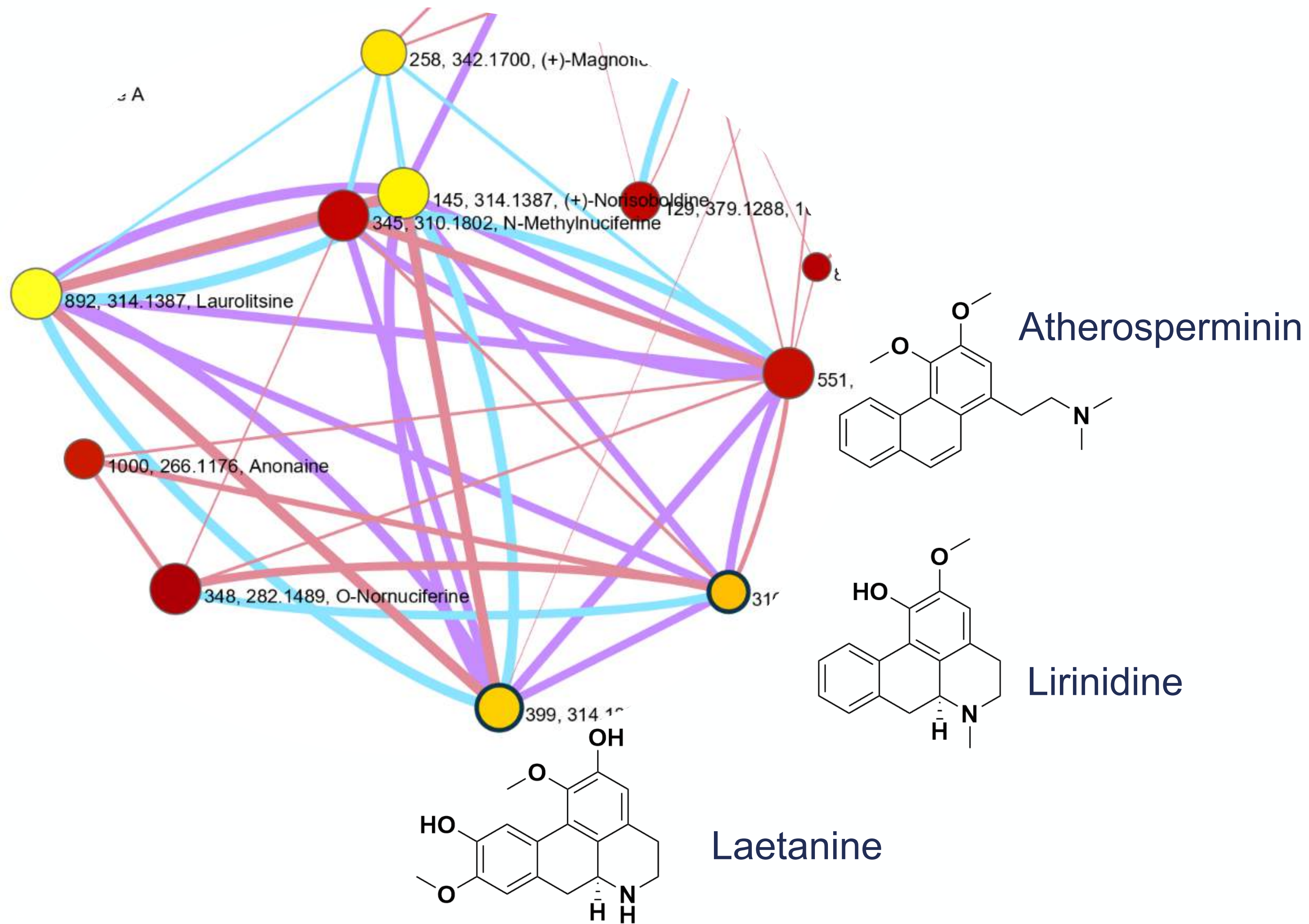
Roman Bushuiev

DreaMS Spectral Similarity



Similar DreaMS representations = similar molecular structures

DreaMS

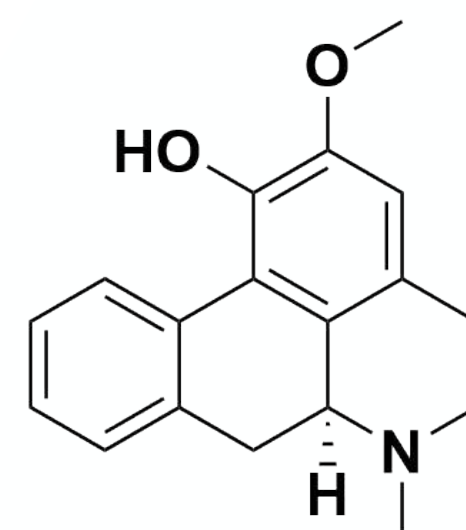
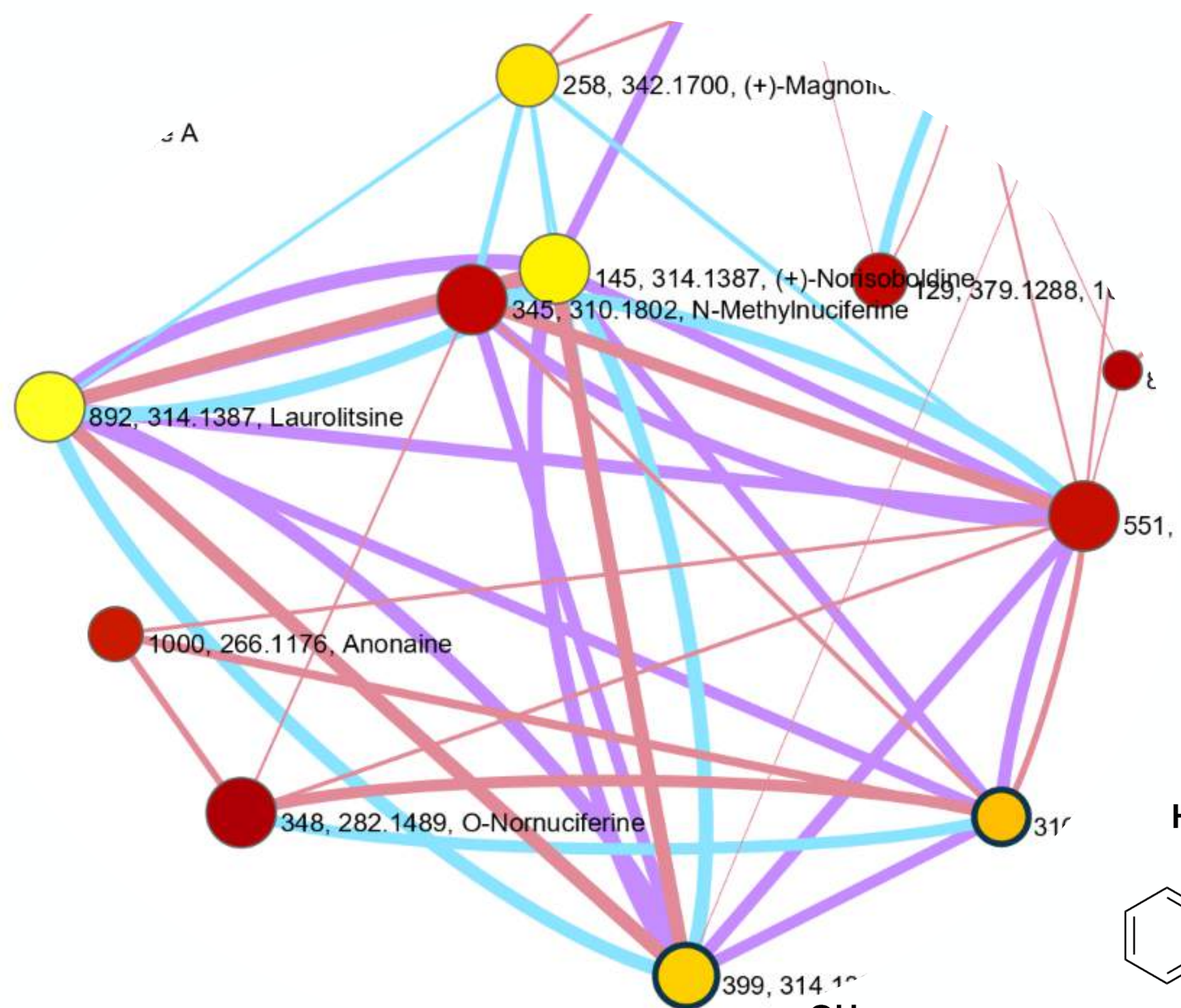


Application to a Library Subset

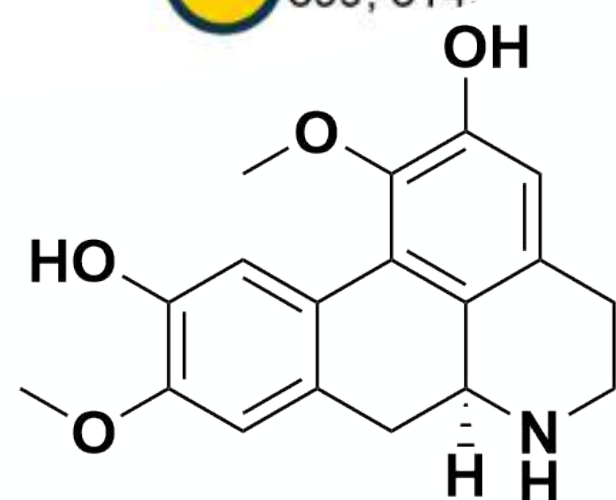
Modified cosine

MS2DeepScore

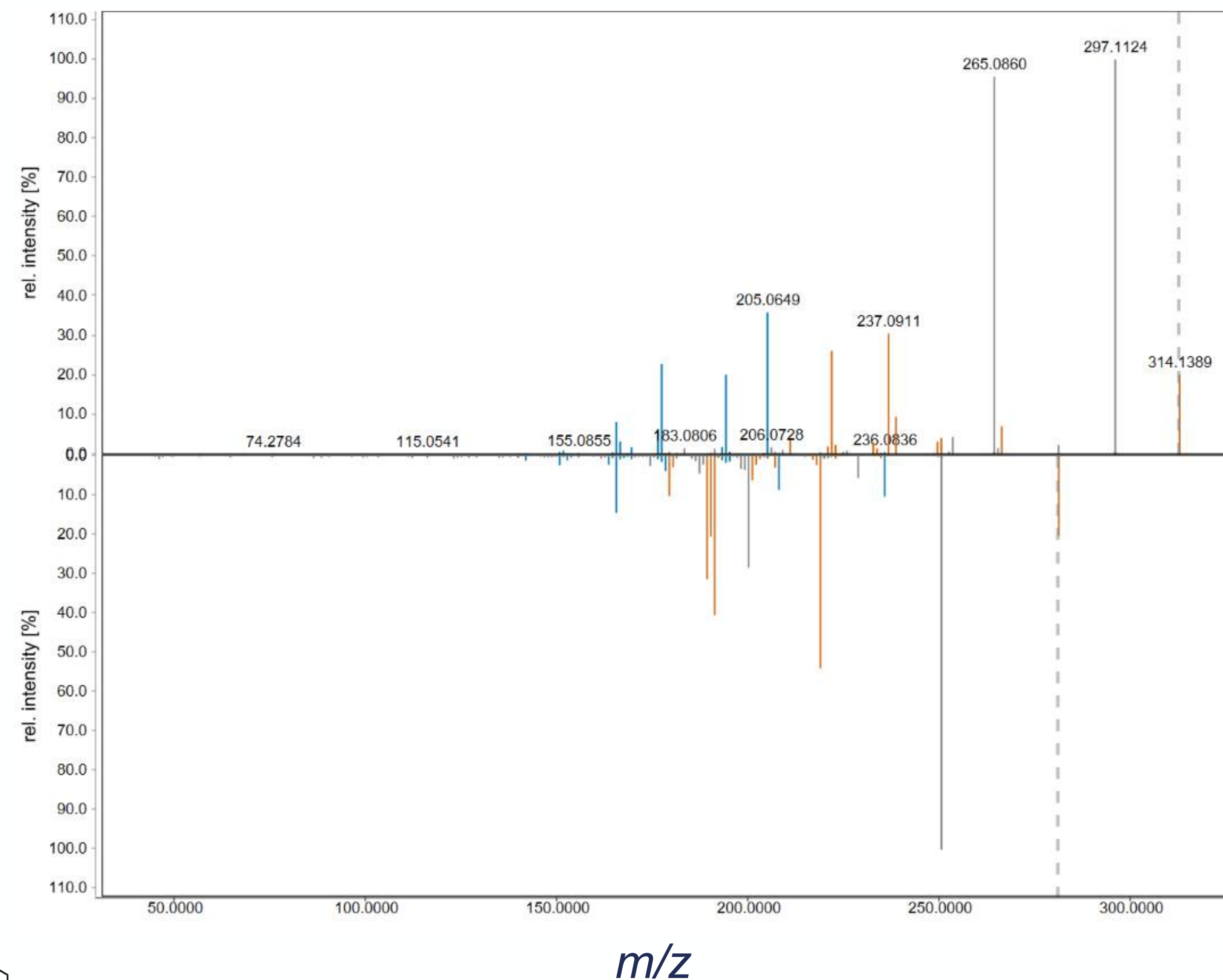
DreaMS



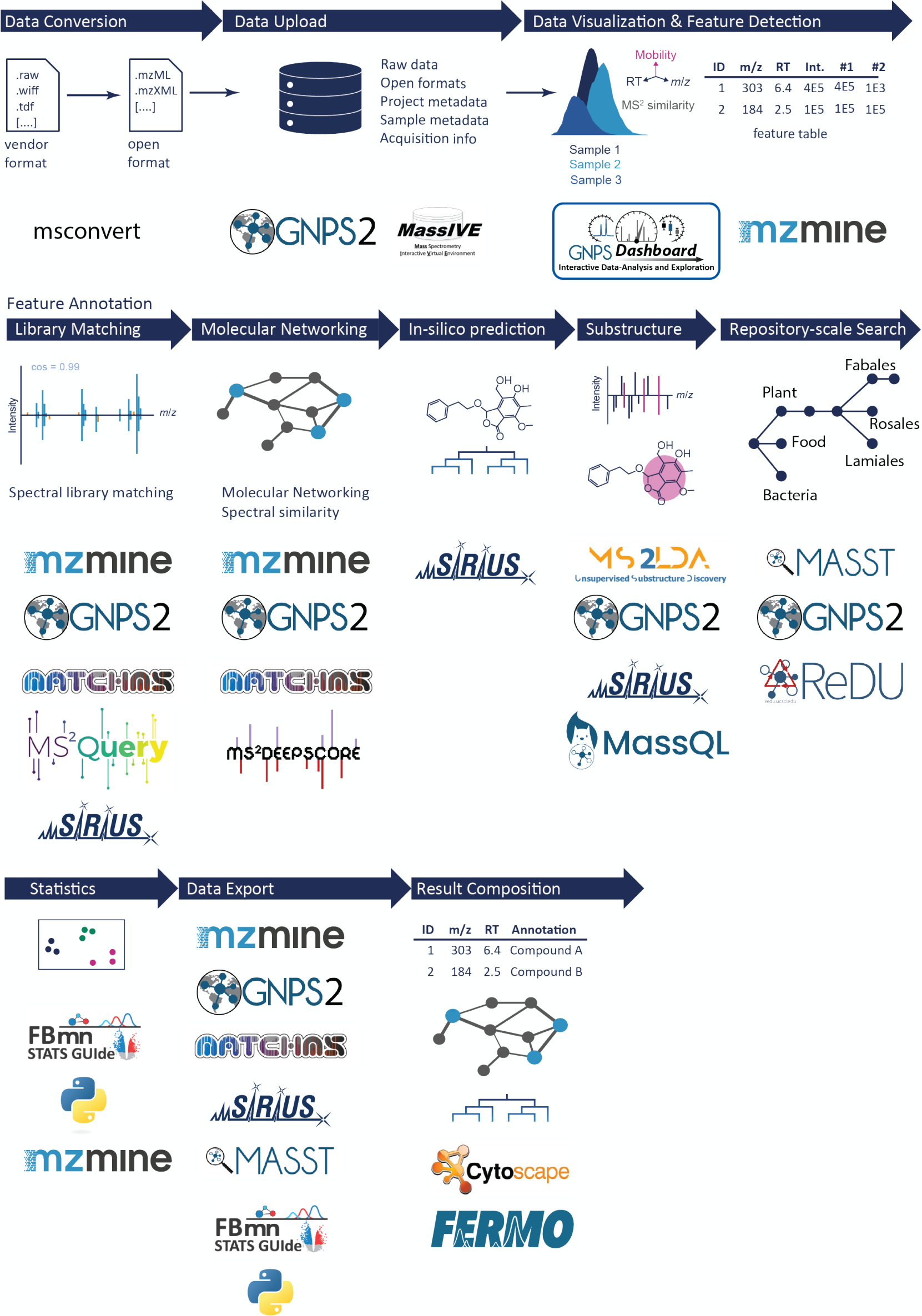
Lirinidine



Laetanine

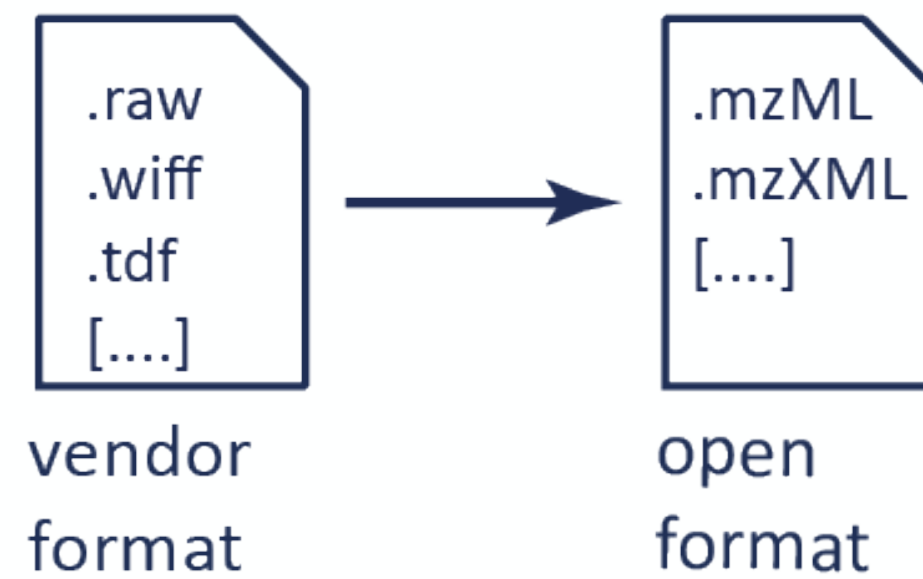


Processing Pipeline



Processing Pipeline

Data Conversion



msconvert

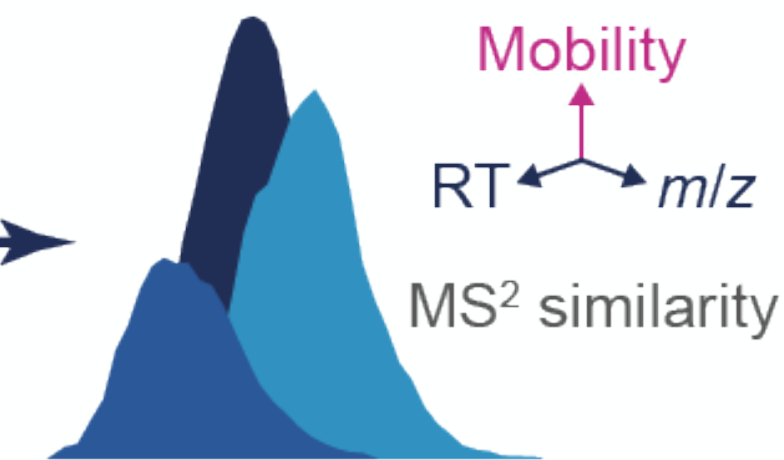
Data Upload



Raw data
Open formats
Project metadata
Sample metadata
Acquisition info

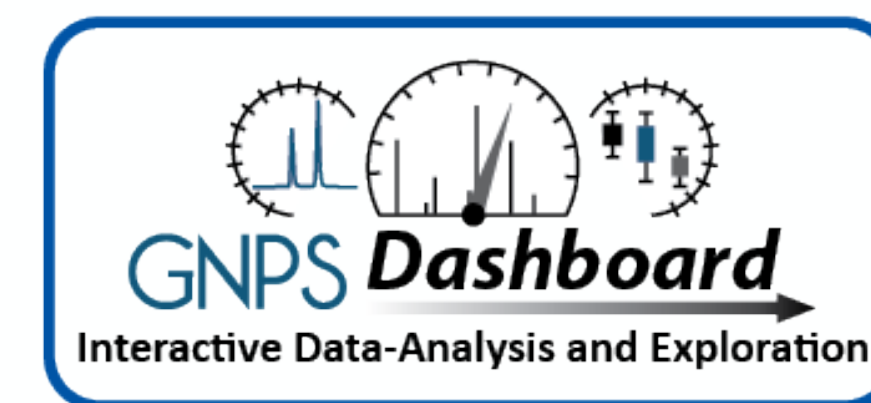


Data Visualization & Feature Detection



ID	m/z	RT	Int.	#1	#2
1	303	6.4	4E5	4E5	1E3
2	184	2.5	1E5	1E5	1E5

feature table

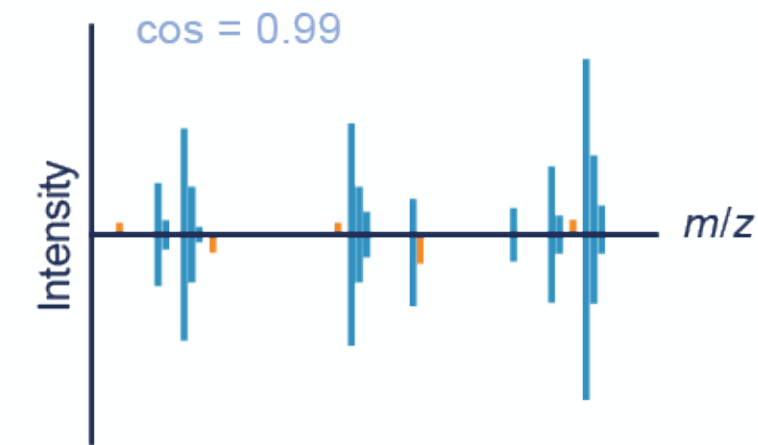


mzmine

Processing Pipeline

Feature Annotation

Library Matching



Spectral library matching

mzmine

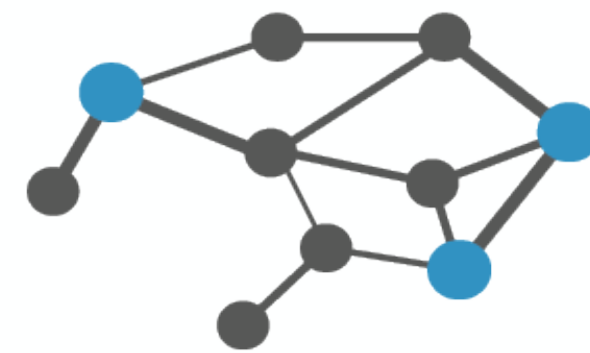
GNPS2

MATCH.ms

MS²Query

SIRUS

Molecular Networking



Molecular Networking
Spectral similarity

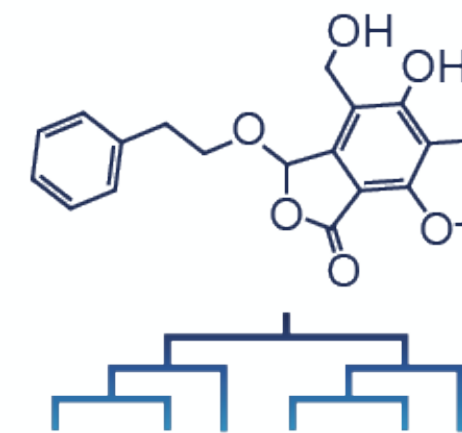
mzmine

GNPS2

MATCH.ms

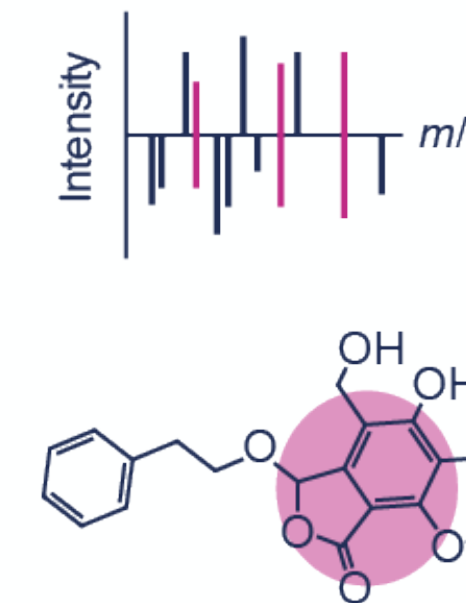
MS²DEEPCORE

In-silico prediction



SIRUS

Substructure



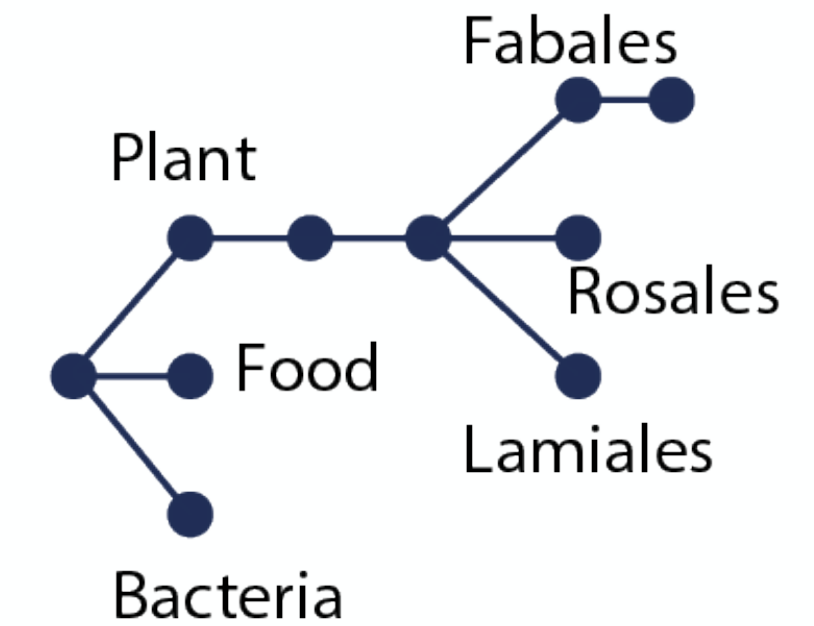
MS2DA
Unsupervised Substructure Discovery

GNPS2

SIRUS

MassQL

Repository-scale Search

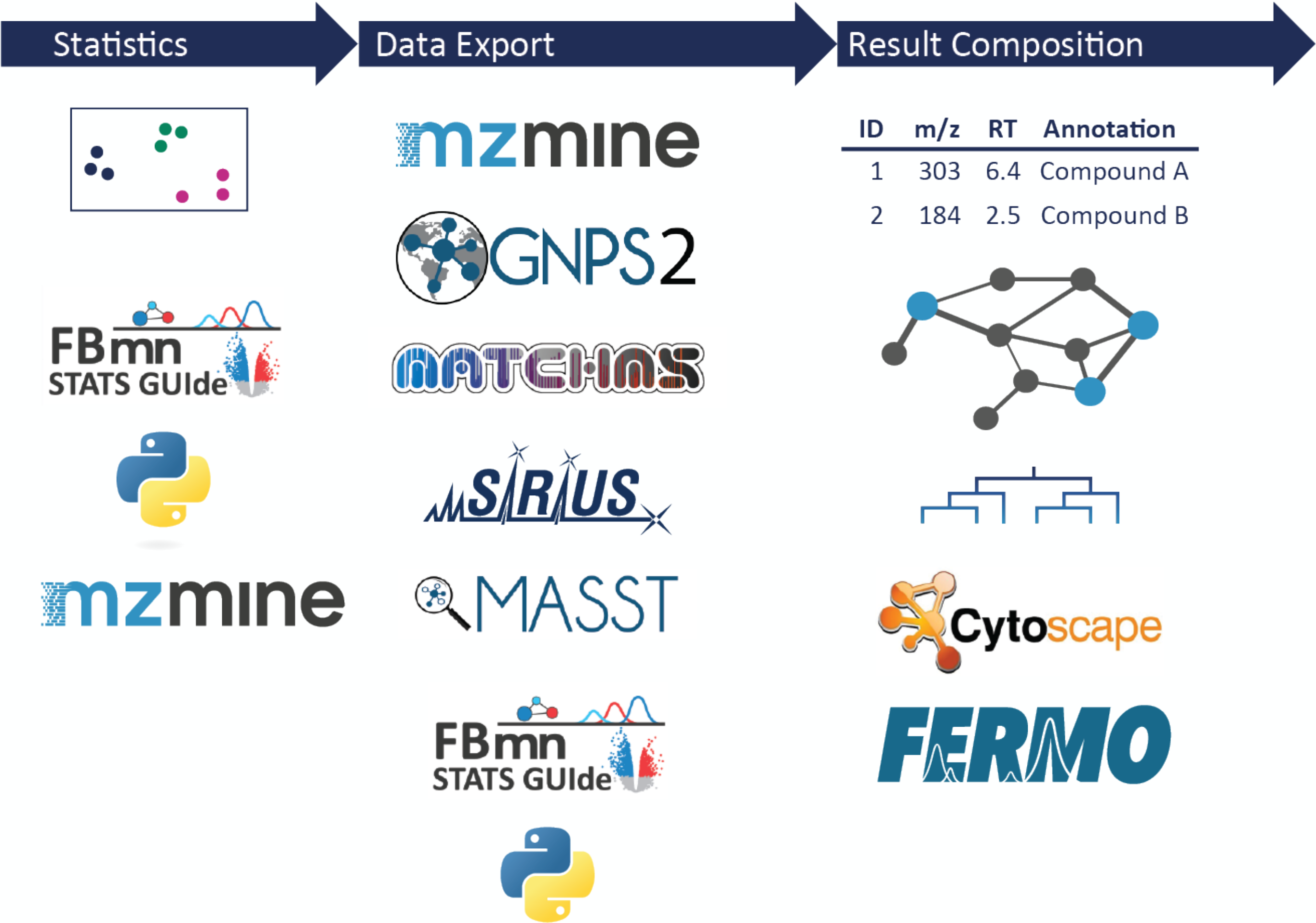


MASST

GNPS2

ReDU

Processing Pipeline



MS Data Formats and Data Conversion

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MS Data Formats



Vendor/Instrument	Data format
Thermo	.raw (file)
Bruker (timsTOF)	.d (.tdf/.baf)
Bruker (impact/maXis)	.d (.baf)
Waters	.raw (folder)
Agilent	.d (.bin)
Shimadzu	.lcd
SCIEX	.wiff, .wiff2

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Open data formats	Application
mzML	General purpose MS data
imzML	MS imaging
mzData, mzXML	Outdated – use mzML if possible

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Converters to mzML	Formats supported
<u>ProteoWizard mzsconvert</u>	Most formats supported
<u>ThermoRawFileParser</u>	Specific for Thermo files
<u>Waters data_convert/data_connect</u>	Specific for Waters files (only non-IM)